Bayesian Structural Inference for Dynamic Crowdsourcing Contests

Linsheng Zhuang
Center for Intelligent Computing
Great Bay University, China
zhuangls@gbu.edu.cn

Jussi Keppo
NUS Business School & Institute of Operations Research and Analytics
National University of Singapore, Singapore
keppo@nus.edu.sg

Zhi Chen
NUS Business School
National University of Singapore, Singapore
bizczhi@nus.edu.sg

Online crowdsourcing contests generate rich dynamic behavior, but their core strategic state is latent: contestants react to noisy public leaderboard feedback rather than true relative performance. We develop a tractable dynamic game with Bayesian filtering and a corresponding structural estimation framework to recover effort costs, innovation risk, and feedback precision. With symmetric effort costs, sufficiently low innovation risk and effort costs, and the stated conditions on initial uncertainty and organizer preferences, a finite-precision design strictly outperforms every full-disclosure design, even when prizes and deadlines are reoptimized and precision has no direct cost. A closed-form leading-order effort policy links feedback precision to the range of perceived performance gaps over which prize incentives sustain effort, enabling tractable Bayesian estimation. For validation, we use a path-wise holdout design: private-leaderboard path values are excluded from the estimation likelihood and subsequently used for diagnostic tests, although ex post private-score information is used to define the focal rivalry. Across 73 Kaggle contests, higher estimated innovation risk is associated with greater private-path volatility, while higher estimated feedback precision is associated with smaller public–private gaps. These relationships also appear in time-split diagnostics and are interpreted as descriptive associations conditional on ex post focal-pair selection. Persistent differences across contests in score scale, evaluation metric, or submission frequency remain possible common drivers.

Key words: Bayesian Statistics, Structural Estimation, Crowdsourcing Contest, Dynamic Contest

1. Introduction

Crowdsourcing contests are widely used by firms to source solutions to complex problems from external contributors. Platforms such as Kaggle[1], Topcoder[2], DrivenData[3], and AIcrowd[4] host thousands of these contests, attracting data scientists, engineers, and developers to compete by submitting solutions to real-world problems. On Kaggle, for instance, a typical contest spans several weeks, during which participants compete for monetary prizes by repeatedly submitting solutions, observing their relative ranking on a public leaderboard, and adjusting their strategies in response to evolving feedback and the observed actions of competitors.

The central empirical challenge in these online crowdsourcing contests is that the true competitive state is unobserved during the contest. Participants observe noisy public leaderboard updates rather than true relative performance, and they choose effort based on filtered beliefs rather than on the latent state itself. The econometrician faces an analogous partial-observability problem during estimation, making it difficult to recover strategic behavior and incentives from contest-phase data alone.

Motivated by this identification problem, we develop a dynamic contest model and structural Bayesian estimation framework tailored to settings such as Kaggle, where contestants choose continuous effort while learning from noisy public leaderboard signals. Applying the framework to the Meta-Kaggle dataset (Risdal and Bozsolik 2022), we recover latent competition states and structural primitives from contest-phase public information and evaluate them using post-contest private-leaderboard paths excluded from the likelihood.

To illustrate the framework in greater detail, Figure 1 depicts the key components and their interactions in a typical Kaggle contest. One notable feature is the existence of two leaderboards: a *private* leaderboard, which is hidden until the contest ends and ultimately determines the final ranking and prize allocation, and a *public* leaderboard, which is visible to all participants during the contest.[5] Leaderboard displays are contestant-specific and update asynchronously. After observing the public leaderboard, contestants update their beliefs and form a perceived state of the competition. Based on these beliefs, contestants choose effort according to the model's Markov perfect equilibrium (MPE) strategies. These effort levels shape submission intensity. In our empirical representation, each submission updates the submitting contestant's score, while the opponent's most

[1] Kaggle: `https://www.kaggle.com`

[2] Topcoder: `https://www.topcoder.com`

[3] DrivenData: `https://www.drivendata.org`

[4] AIcrowd: `https://www.aicrowd.com`

[5] The public leaderboard provides interim feedback, while the hidden private leaderboard limits direct adaptation to the final evaluation sample (see also Blum and Hardt 2015). In our model, uncertainty about final rankings can sustain competition; public–private disagreement alone does not identify overfitting.

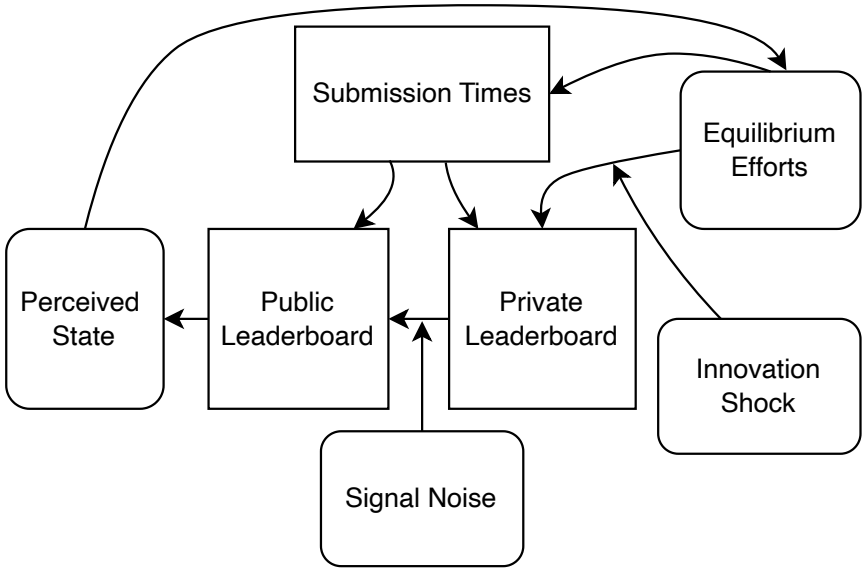


Figure 1 Structure of a Typical Kaggle Contest

Note: This figure illustrates the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles and rounded rectangles represent observed data and latent states, respectively. The private leaderboard is visible only to the contest organizer during the contest and is revealed to all participants only after it ends. The diagram separates public feedback during the contest from private-score evaluation after it ends. In the model, public signals and past efforts form common information; in the empirical application, submission activity proxies for effort. The reconstructed latest-submission score paths need not coincide with the displayed best-so-far leaderboard or each contestant's information set (Section 3).

recently submitted score is carried forward. In the model, latent performance evolves according to the real-time effort level and an innovation shock.

Our Bayesian analysis links submission times and public leaderboard trajectories to the underlying contest dynamics, with innovation risk and feedback precision as the main empirical targets. To assess whether the estimated model captures meaningful contest primitives, we use private-leaderboard paths to evaluate two predictions illustrated in Figure 1: higher estimated innovation risk should be associated with greater private-path volatility, whereas higher estimated signal precision should be associated with smaller public-private leaderboard gaps. Although private-leaderboard path values are excluded from estimation, reconstructed private-score rankings are used to select the focal pair. We therefore interpret the validation exercises as path-wise holdout diagnostics conditional on ex post pair selection. We also conduct time-split exercises that relate primitives estimated from early public contest data to private-path outcomes in the later contest period. This reduces overlap in the estimation and diagnostic windows, but does not rule out persistent contest-level score-scale, metric, or activity differences as common drivers. Section 6.3 also compares out-of-sample submission-count predictions with nonstrategic benchmarks.

Empirically, we carefully select 73 high-quality contests from the Meta-Kaggle database and preprocess the data to ensure consistency across estimation methods. Conditional on the focal-pair selection rule, we estimate contest-specific innovation risk and signal precision using only public contest-phase information. The diagnostics show positive cross-contest associations in the predicted directions: higher estimated innovation risk coincides with greater private-path volatility, and higher estimated signal precision with smaller public–private gaps. The associations also appear in time-split specifications, but should not be read as isolating incremental structural information from persistent contest-level score scale, metric type, or submission activity. We also solve the nonlinear game numerically to compare joint adjustments to prizes, deadlines, and feedback precision that preserve baseline expected solution quality under alternative delay costs.

This paper makes three main contributions. First, we develop a dynamic game-theoretic model and a Bayesian structural framework for contests with noisy public feedback and latent performance dynamics. Relative to contest models centered on discrete submit-or-stop decisions, our approach models continuous effort accumulation and makes belief updating from noisy signals structurally central. Building on the equilibrium structure of Ryvkin (2022), we model the winner as determined by the private leaderboard, hidden until the contest ends, so that the informativeness of public feedback shapes equilibrium effort incentives, while retaining a closed-form leading-order effort policy for structural estimation. Second, under symmetric effort costs and the initial-uncertainty and organizer-preference conditions in Proposition 1, we identify a region of low innovation risk and low effort costs in which a finite-precision design yields a higher organizer payoff than every full-disclosure design, even when prizes and deadlines are jointly optimized and precision carries no direct cost. By preserving uncertainty about the winner, noisy feedback can sustain effort that a substantial disclosed lead would otherwise discourage. Third, we use private-leaderboard path values, which are excluded from the likelihood, to construct post-estimation validation targets. These diagnostics are consistent with interpreting the estimated parameters as innovation risk and signal precision. Evidence on out-of-sample submission-count prediction favors the theoretically derived effort rule over a more flexible alternative. Comparisons with simpler benchmarks are mixed: a constant-rate model performs better, while results against a deadline-dependent model are inconclusive (Section 6.3).

1.1. Related Literature

This study is most closely related to the following strands of literature.

Contest Design. A rich literature on *static* contest design (Ales et al. 2017) investigates how to structure competitive environments to elicit performance, typically through a contest success function mapping effort to winning probabilities (Skaperdas 1996).[6] Because our setting is dynamic

[6] There are exceptions, such as beauty or barrier contests.

with repeated feedback, this static branch serves mainly as background for the mechanism-design questions we build on, including prize allocation, participation rules, and information disclosure (Moldovanu and Sela 2001, Ales et al. 2021, Bergemann et al. 2022).

While these studies offer valuable insights into incentive design, they largely overlook how participants dynamically adjust their efforts in response to evolving competition. In contrast, this paper focuses on *dynamic* contests, where strategic decisions unfold over time. Two main modeling approaches have been used to capture such dynamic strategic interactions:

One strand of the literature models dynamic contests as multistage processes, where participants exert effort over time while facing uncertainty about intermediate progress. This structure facilitates the analysis of the mid-contest policy instruments available to the contest designer, such as feedback disclosure and screening mechanisms. In two-stage contests, interim feedback shapes subsequent effort choices (Aoyagi 2010, Mihm and Schlapp 2019). Participants exert effort in both stages, and the final output reflects the cumulative impact of early performance and subsequent improvement. Similarly, Bimpikis et al. (2019) study a two-stage model in which agents exert continuous effort to complete Stage A (potentially infeasible) and Stage B (feasible), with breakthroughs governed by Poisson processes. In all these models, the contest designer may strategically disclose interim feedback. Another example is Khorasani et al. (2023), who analyze a two-stage contest in which solvers invest exploratory effort to generate viable ideas, with only those passing an imperfect screening mechanism proceeding to execution.

The second approach, more closely related to our work, treats output as a stochastic process whose drift depends on the participant's effort. This line of research originates from the tug-of-war framework, first formalized by Harris and Vickers (1987) as a one-dimensional simplification of a multi-stage R&D race. Subsequent studies build on this foundation by modeling each player's progress as a Brownian motion. Budd et al. (1993), for example, analyze a duopoly innovation race where the state evolves as a Brownian motion driven by the difference in effort, and characterize an approximate equilibrium. Moscarini and Smith (2011) advance this approach by modeling the game state as the output gap between players and derive closed-form equilibrium strategies under pure-strategy play. Ryvkin (2022) adapts this framework to a dynamic contest with a fixed deadline, providing a closed-form solution for the Markov perfect equilibrium.

The precision of public feedback is itself a design instrument in data-science contests. Huang et al. (2024) show that intermediate training–test splits can encourage high-dispersion modeling approaches. We instead hold innovation risk fixed and study how public-feedback precision shapes effort.

This paper studies online crowdsourcing contests, which typically feature fixed deadlines, repeated submissions, and public real-time feedback, all governed by platform-specific rules and

evaluation metrics. Hence, our analysis builds on the model by Ryvkin (2022), whose simplifications yield a closed-form equilibrium that makes structural estimation in a dynamic setting analytically and computationally tractable. We depart from their full-information setting by introducing a noisy public signal and Bayesian filtering, thereby bringing partial observability into a tractable dynamic contest model.

Structural Estimation of Contests. Beyond theoretical modeling, a growing body of research employs structural estimation techniques to recover key parameters of contest environments using observed data.

In the structural estimation of static contests, when a contest success function (CSF) is explicitly specified, the primary goal is to estimate its functional form or key parameters (Hwang 2012, Kang 2016, Huang and He 2021). For beauty contest auctions with unobserved allocation rules, equilibrium first-order conditions can be used to recover bidders' costs. For example, Yoganarasimhan (2016) proposes a two-step semiparametric estimator, using an EM-like algorithm in the first step, to recover both unobserved auction heterogeneity and bidder cost distributions. Related auction studies use equilibrium restrictions to recover bidders' valuation distributions nonparametrically (Guerre et al. 2000, Shakhgildyan 2025).

Most dynamic structural estimation exercises rely on the Markov Perfect Equilibrium (MPE) of a dynamic game. However, when the model is calibrated to closely fit real-world data, the MPE typically lacks a closed-form solution. For example, Jiang et al. (2022) develop a finite-horizon dynamic game with discrete time, state, and action spaces to model creators' strategic behavior in response to feedback during online logo design contests. Using data from 810 contests, they implement a two-step maximum likelihood estimation (MLE) procedure under the assumption of ex-ante homogeneous contests and players: the first step estimates action-specific state transition probabilities, while the second recovers utility parameters.

Specific to online crowdsourcing environments, Zhang et al. (2019) build a dynamic structural model to study how participants in crowdsourcing contests learn and choose contests over time. Their model shows that competing against superstars accelerates learning by providing more informative feedback. Using Bayesian updating and a hierarchical model, they estimate how individuals trade off monetary rewards, learning opportunities, and participation costs, based on data from Topcoder.

Most closely related to our work, Lemus and Marshall (2021) structurally estimate contest behavior using Kaggle data. They propose a finite-horizon dynamic game in which, after each submission, players decide whether to continue or exit, essentially solving an optimal stopping problem. Players can only make this decision immediately after completing a submission, which arrives at random intervals and incurs a privately drawn cost. Their estimation proceeds in two stages: the first uses

maximum likelihood and the EM algorithm to estimate primitive parameters such as the distribution of player types and scores; the second applies a conditional choice probability (CCP)-based estimator to identify the cost distribution. Our model differs in two key respects: (1) we link submission intensity to current effort rather than model submissions as discrete costly actions; and (2) we explicitly model belief updating from noisy signals as a structural channel that shapes both behavior and identification.

The structure of the paper is as follows: Section 2 introduces our theoretical model of online crowdsourcing contests. Section 3 presents the Bayesian estimation framework. In Section 4, we describe the data-generating process and apply the Bayesian framework to synthetic data. Section 5 applies the framework to real-world Kaggle data and evaluates the estimated primitives using private-leaderboard path values excluded from the likelihood. A replication package (code and implementation details) will be provided with the paper.

2. The Model

Two players, i and j, compete for a prize $\theta > 0$ in a contest. The winner receives the prize and the loser receives zero. Following Ryvkin (2022), the contest starts at time zero and ends at a fixed deadline $T > 0$. At each $t \in [0, T)$, each player $k \in \{i, j\}$ chooses effort $q_{k,t} \geq 0$ and incurs an instantaneous quadratic cost $C_k(q_{k,t}) = c_k q_{k,t}^2/2$, where $c_k > 0$ and a lower c_k corresponds to higher ability. Let y_t denote the *latent private-score gap* between players i and j at time t, evolving as

$$dy_t = (q_{i,t} - q_{j,t})dt + \sigma dW_t \tag{1}$$

where W_t is a Brownian motion and $\sigma > 0$ measures the innovation risk. For given efforts, a larger σ increases the uncertainty in performance changes.

The actual contest is equipped with a submission system in which contestants submit solutions asynchronously over time. Each submission generates new public and private scores for the submitting contestant, while the opponent's most recently submitted scores are carried forward when constructing the corresponding leaderboard gaps. For tractability, we approximate this irregular sequence of submission-triggered updates by a continuous-time public signal of the latent private-score gap y_t. The feasible precision parameter satisfies $\lambda > 0$, and the signal evolves by

$$dZ_t = y_t dt + \frac{dB_t}{\sqrt{\lambda}} \tag{2}$$

where B_t is a standard Brownian motion independent of W_t. A larger λ implies less noise and more informative feedback; in the empirical mapping, this corresponds to a more informative public leaderboard signal about latent relative performance. This continuous-time signal is an idealized analogue of the stream of public leaderboard updates observed after submissions.

The model parameters are common knowledge. Public submission activity provides a proxy for effort; the theoretical model idealizes this indirect observation by treating past efforts as observable to both players. Neither player has additional private information about y_t. We study public Markov equilibria. Let I_t denote the common history of public signals and past efforts, and define the posterior mean and variance by $\tilde{y}_t := \mathbb{E}(y_t \mid I_t)$ and $S_t := \mathbb{E}[(y_t - \tilde{y}_t)^2 \mid I_t]$. Initially, $y_0 \mid I_0 \sim \mathcal{N}(\tilde{y}_0, S_0)$, where $\tilde{y}_0$ is publicly known and $S_0 > 0$ is fixed independently of λ. The initial filtering error is independent of future increments of W and B. Under standard linear-Gaussian assumptions, the Kalman-Bucy filter (Section 1.2 of Bensoussan 1992; see also Frogerais et al. 2012, Barrau and Bonnabel 2017) implies

$$d\tilde{y}_t = (q_{i,t} - q_{j,t})dt + \sqrt{\lambda} S_t \, d\widehat{B}_t \tag{3}$$

$$\frac{dS_t}{dt} = \sigma^2 - \lambda S_t^2 \tag{4}$$

$d\widehat{B}_t := \sqrt{\lambda}(dZ_t - \tilde{y}_t dt)$ is the innovation Brownian motion, so equation (3) is equivalent to the usual Kalman–Bucy representation. Hence, the conditional distribution satisfies $y_t \mid I_t \sim \mathcal{N}(\tilde{y}_t, S_t)$, so the posterior state is fully characterized by $(\tilde{y}_t, S_t)$. The steady-state value associated with equation (4) is $\bar{S} \equiv \sigma/\sqrt{\lambda}$. Given S_0, equation (4) determines a deterministic path S_t, so $(\tilde{y}_t, t)$ remains a sufficient strategic state. Appendix B.3 reports the solution.

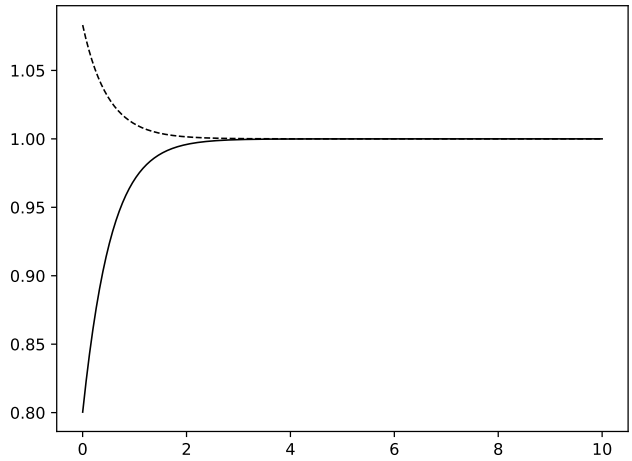


Figure 2 Convergence of the posterior variance S_t to steady state.

Note: With $\lambda = \sigma = 1$, the solid curve starts below the steady state ($S_0 < 1$) and the dashed curve starts above it ($S_0 > 1$). Both converge to $\bar{S} = 1$. The horizontal axis is time t; the vertical axis is posterior variance S_t. Variance falls when learning dominates new innovation ($\lambda S_t^2 > \sigma^2$) and rises in the reverse case. The theoretical model allows either initial condition; steady-state initialization is a restriction used in the empirical estimation.

The sign of y_T determines the winner. Let I_T denote the common information just before the terminal private scores are revealed. Since $y_T \mid I_T \sim \mathcal{N}(\tilde{y}_T, S_T)$, the expected terminal prizes are

$\Pi_i(\tilde{y}_T) = \theta\Phi(\tilde{y}_T/\sqrt{S_T})$ and $\Pi_j(\tilde{y}_T) = \theta\Phi(-\tilde{y}_T/\sqrt{S_T})$, where Φ is the standard normal cumulative distribution function. Let $m_k(\tilde{y},t)$ denote player k's equilibrium Markov effort strategy. Taking player j's strategy $q_{j,\tau} = m_j(\tilde{y}_\tau,\tau)$ as given, player i's value is

$$V^i(\tilde{y},t) = \sup_{\{q_{i,\tau}\geq 0\}_{\tau\in[t,T)}} \mathbb{E}\left[\Pi_i(\tilde{y}_T) - \int_t^T C_i(q_{i,\tau})d\tau \,\middle|\, \tilde{y}_t = \tilde{y}\right],$$

The supremum is over nonnegative effort processes predictable with respect to I_τ and satisfying $\mathbb{E}\int_t^T q_{i,\tau}^2 d\tau < \infty$. The optimization is subject to equations (3)–(4). Player j's problem is obtained by replacing $\tilde{y}$ with $-\tilde{y}$ and exchanging the player labels and cost parameters. The corresponding Hamilton-Jacobi-Bellman (HJB) equation for player i is

$$0 = \max_{q_{i,t}\geq 0}\left[-\frac{c_i q_{i,t}^2}{2} + V_y^i\cdot(q_{i,t} - m_j(\tilde{y},t)) + V_t^i + \frac{V_{yy}^i}{2}\lambda S_t^2\right].$$

Here, subscripts y denote derivatives with respect to the posterior mean $\tilde{y}$. For an interior equilibrium, substituting the first-order conditions $q_{i,t} = V_y^i/c_i$ and $q_{j,t} = -V_y^j/c_j$ yields the system

$$\frac{1}{2c_i}(V_y^i)^2 + \frac{1}{c_j}V_y^iV_y^j + V_t^i + V_{yy}^i\frac{\lambda S_t^2}{2} = 0$$
$$\frac{1}{2c_j}(V_y^j)^2 + \frac{1}{c_i}V_y^jV_y^i + V_t^j + V_{yy}^j\frac{\lambda S_t^2}{2} = 0$$

subject to boundary conditions $V^i(-\infty,t) = 0$, $V^i(+\infty,t) = \theta$, $V^j(-\infty,t) = \theta$, $V^j(+\infty,t) = 0$, and terminal conditions $V^i(\tilde{y},T) = \Pi_i(\tilde{y})$ and $V^j(\tilde{y},T) = \Pi_j(\tilde{y})$ for all $\tilde{y}\in\mathbb{R}$. The following theorem establishes equilibrium existence and gives a leading-order effort formula for small incentive indices.

THEOREM 1 **(Equilibrium Existence and Leading-Order Effort)**. *For every finite $\lambda > 0$ and sufficiently small $w_i + w_j$, the game admits an interior Markov-perfect equilibrium with bounded, smooth effort policies. As $(w_i,w_j)\to(0,0)$, the effort policies of this equilibrium satisfy*

$$m_k(\tilde{y}_t,t) = \frac{\sigma^2 w_k}{\sqrt{Q_t}}\left\{\frac{1}{\sqrt{2\pi}}\exp\left[-\frac{\tilde{y}_t^2}{2Q_t}\right] + O(w_i+w_j)\right\}, \tag{5}$$

for $k\in\{i,j\}$, where $w_k := \theta/(\sigma^2 c_k)$ and $Q_t := S_t + \sigma^2(T-t)$. The remainder is uniform over $(\tilde{y}_t,t)\in\mathbb{R}\times[0,T)$, holding T,σ,λ, and S_0 fixed.

The quantity Q_t combines uncertainty about the current gap, S_t, with innovation risk over the remaining horizon, $\sigma^2(T-t)$. The theorem separates the nonlinear equilibrium from its leading-order approximation, denoted by $m_k^{(1)}$. The following comparative statics describe this approximation at a given perceived gap; they do not assert the same signs for the nonlinear equilibrium or for effort averaged over endogenous contest histories. Lemma 1 establishes equilibrium existence for arbitrary positive prizes at finite precision; the small-incentive condition here concerns the approximation. The uniform remainder does not guarantee small percentage errors where effort is close to zero.

Prize, ability, and competitive position. At a given gap and Q_t, doubling the prize doubles $m_k^{(1)}$, while doubling the player's effort-cost parameter halves it. Effort is highest at a perceived tie and falls with the absolute gap: a marginal improvement matters most when it can change the winner. With equal costs, the perceived leader and laggard exert the same leading-order effort at a given state. With unequal costs, their effort ratio is $m_i^{(1)}/m_j^{(1)} = c_j/c_i$, regardless of who leads. This symmetry belongs to the approximation; nonlinear strategic responses can differ. Neither policy implies a finite gap at which a contestant stops completely: at finite precision, effort remains positive at every finite gap, although the leading-order policy approaches zero as the gap grows.

Feedback and uncertainty. Differentiating the leading-order policy while holding the perceived gap fixed gives

$$\frac{\partial \log m_k^{(1)}}{\partial Q_t} = \frac{\tilde{y}_t^2 - Q_t}{2Q_t^2}.$$

More uncertainty lowers effort in a close contest ($|\tilde{y}_t| < \sqrt{Q_t}$), because a small performance improvement is less decisive, but raises effort in a lopsided contest ($|\tilde{y}_t| > \sqrt{Q_t}$), because the outcome is less settled. Holding S_0 and other primitives fixed, higher precision lowers S_t for $t > 0$ and hence reverses these two effects. It concentrates effort around close perceived races. Higher initial uncertainty or innovation risk raises Q_t and broadens the range of gaps over which contestants remain engaged. These effects explain why neither more disclosure nor more innovation risk has an unconditionally positive effect on effort.

Time remaining. For a fixed deadline, $dQ_t/dt = -\lambda S_t^2 < 0$. Thus, holding the perceived gap fixed, approaching the deadline raises leading-order effort in a close race and lowers it in a sufficiently lopsided race. A contestant's realized effort need not follow either pattern because the perceived gap also changes. Extending the deadline at a fixed date instead adds σ^2 per unit of extension to Q_t, weakening immediate effort near a tie but strengthening it at large gaps. An extension also provides more time to work and innovate. The organizer must weigh these distinct effects against delay cost, rather than infer total output from the instantaneous effort response.

2.1. Joint Design of Prize, Deadline, and Feedback Precision

The performance gap is sufficient for the contestants' strategic problem, but the organizer values the best terminal performance. We therefore recover individual private performance scores. Let x_t^k denote contestant k's score, normalized so that $(x_0^i + x_0^j)/2 = 0$. The initial gap is $y_0 := x_0^i - x_0^j$, with $y_0 \mid I_0 \sim \mathcal{N}(\tilde{y}_0, S_0)$. Individual scores evolve according to

$$dx_t^k = q_{k,t}dt + \frac{\sigma}{\sqrt{2}}dW_t^k, \qquad k \in \{i, j\}, \tag{6}$$

where W^i and W^j are independent and $q_{k,t} = m_k(\tilde{y}_t, t)$ in equilibrium. With $W_t = (W_t^i - W_t^j)/\sqrt{2}$, the gap obeys equation (1). The organizer values the best terminal score,

$$\mathcal{B}(\tilde{y}_0, \theta, T, \lambda; S_0) := \mathbb{E}[\max\{x_T^i, x_T^j\} \mid I_0].$$

Even without effort, zero-mean innovation shocks can raise expected best quality through selection of the better terminal solution, giving $\mathcal{B}(0,0,T,\lambda;S_0)=\sqrt{(S_0+\sigma^2T)/(2\pi)}$. At zero duration, neither effort nor innovation affects scores, so $\mathcal{B}(0,\theta,0,\lambda;S_0)=\sqrt{S_0/(2\pi)}$.

The distinction between best quality and total effort matters economically. Since $\max\{x_T^i,x_T^j\}=(x_T^i+x_T^j)/2+|y_T|/2$, the organizer benefits from both average improvement and the opportunity to select the better solution. Two designs with the same expected total effort can therefore deliver different best-solution quality. The objective concerns the two terminal scores; it does not retain the best score ever attained along either path. Innovation continues even at zero effort in this model, and participation is fixed. Consequently, the zero-prize case below is a benchmark for the value of prize-induced effort, not a prediction of participation in an unpaid contest.

With the symmetric initial belief $\tilde{y}_0=0$, the organizer jointly chooses

$$\sup_{\theta,T\geq 0,\ \lambda\in(0,\infty]}\left\{u(\mathcal{B}(0,\theta,T,\lambda;S_0))-\theta-D(T)\right\}. \tag{7}$$

Here $u:[0,\infty)\to\mathbb{R}$ is twice continuously differentiable, strictly increasing, and concave. The delay cost $D:[0,\infty)\to[0,\infty)$ is continuously differentiable, strictly increasing, and convex, with $D(0)=D'(0)=0$ and $D'(T)/T$ nondecreasing for $T>0$. It is twice continuously differentiable near zero, with $0<D''(0)<\infty$. Short delays therefore have a locally quadratic cost. The design set requires a positive amount of public feedback. The endpoint $\lambda=\infty$ denotes the full-disclosure game: contestants observe the initial gap and its subsequent path, and expected output averages over the initial prior. For this benchmark, we use the symmetric Markov equilibrium constructed in the appendix.

PROPOSITION 1 **(Feedback Precision in Joint Design).** *Suppose $c_i=c_j=c>0$, $-u''(B)\geq\zeta u'(B)$ for all sufficiently large B and some $\zeta>0$, and S_0 is sufficiently large. There exist thresholds $c_F,\sigma_F>0$ with the following properties:*

(i) Finite feedback strictly dominates full disclosure. *For $0<c<c_F$ and $0<\sigma<\sigma_F$, some positive-prize, positive-deadline, finite-precision design strictly outperforms every full-disclosure design. Every joint optimizer satisfies $\theta^*>0$, $T^*>0$, and $\lambda^*<\infty$.*

(ii) Positive-prize full disclosure. *There exist $c>0$ and $\sigma>\sigma_F$ for which joint optima exist and satisfy $\theta^*>0$, $T^*>0$, and $\lambda^*=\infty$.*

(iii) Prize exit at high cost. *For every $\sigma>0$ and sufficiently large c, joint optima exist and satisfy $\theta^*=0$ and $T^*>0$. Precision is then payoff-irrelevant.*

The three instruments address different margins. A larger prize strengthens the reward for winning but costs the organizer dollar for dollar. More time permits additional effort and innovation but delays delivery. Precision determines how sharply contestants distinguish close from lopsided races and has no direct resource cost here. The concavity of u limits the value of additional expected

quality; the stronger tail-curvature condition in the proposition ensures that this marginal value eventually falls sufficiently quickly. The restrictions on D make short extensions inexpensive and prevent delay costs from growing too slowly. These assumptions help turn local incentive effects into comparisons over all prizes and deadlines.

More precise feedback can strengthen incentives by revealing that a contest is close, but revealing a substantial lead can discourage the trailing contestant from catching up and reduce the leader's pressure to improve. Finite precision can sustain competition by preserving uncertainty about the winner, at the cost of blurring the information that sharpens incentives in close contests.

Low innovation risk makes disclosed initial gaps difficult to overturn, while low effort costs make continued competition inexpensive to support through the prize. A short positive deadline allows effort to improve solution quality while limiting delay costs. This advantage persists even when full disclosure is allowed to reoptimize the prize and deadline.[7] With greater innovation risk, future shocks can preserve competitive uncertainty even under full disclosure; part (ii) establishes its joint optimality at some such parameter points. In part (iii), effort costs are sufficiently high that inducing effort no longer justifies prize expenditure. The optimal prize is therefore zero and precision is irrelevant, although a positive deadline remains worthwhile because innovation can improve expected best output without effort. Figure 3 illustrates the numerical arrangement of these regimes; the proposition gives sufficient conditions rather than a complete partition.

Large initial uncertainty makes the information policy consequential: full disclosure can immediately reveal a substantial lead, whereas finite feedback initially preserves uncertainty about that lead. The result does not say that increasing S_0 always makes less disclosure preferable. Nor does part (ii) establish a universal innovation-risk threshold above which full disclosure is optimal. The managerial implication is conditional: when effort is inexpensive and future innovations are unlikely to overturn disclosed gaps, preserving some uncertainty can improve output even after the prize and deadline are adjusted. When innovation itself keeps the race open, disclosure can be preferable. These are joint-design comparisons, not a claim that effort, optimal precision, or optimal prizes vary monotonically with every primitive.

3. Estimation Framework

This section links equilibrium effort to observed submission timing and public leaderboard updates to estimate the model's structural parameters. We specify the observation process and modeling assumptions, then construct an approximate Bayesian likelihood using the leading-order effort policy in Theorem 1.

[7] Part (i) does not assert that a joint optimum is attained; any attained joint optimum must use finite precision.

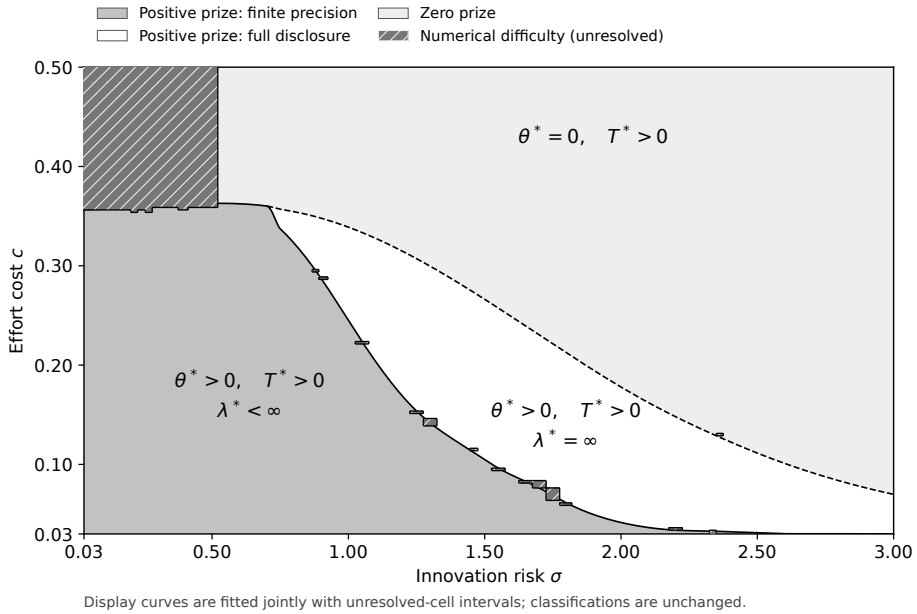


Figure 3 Numerical joint-design regions in the (c, σ) plane

Note: Medium gray denotes positive prize with finite precision, white positive prize with full disclosure, and light gray zero prize. Dark hatching marks 1,179 numerically unresolved grid points, including 926 with potentially competitive failed candidates. The curves are smoothed for display using the unresolved-cell intervals; smoothing does not resolve classification uncertainty. The search jointly varies the prize, deadline, and precision, with $S_0 = 1$, $u(B) = 3(1 - e^{-2B})$, $D(T) = T^2/2$, symmetric costs $c_i = c_j = c$, and initial perceived gap $\tilde{y}_0 = 0$. Both axes are linear and start at 0.03. Further details appear in Appendix B.1. The horizontal axis varies innovation risk σ and the vertical axis varies the common effort cost c (higher c means lower ability). The grid contains 14,553 points. Regions classify numerical design comparisons under this calibration, not empirically estimated regimes or analytically established boundary curves.

3.1. Data-Generating Process

For each Kaggle contest, the observable data from Meta-Kaggle can be grouped into three components.

The first component contains contest-level institutional features: contest duration, prize structure, and information disclosure rules. To match the two-player winner-take-all structure of the model, we restrict the empirical application to contests where a dominant top rivalry is plausible and monetary prize incentives can be mapped to an effective top-two winner-gain structure. The effective prize θ is observed rather than estimated. We express monetary payoffs in units of one hundred thousand U.S. dollars and use this unit for every contest.

The second component records submission events of player i, denoted by $\{\hat{t}_k^i\}_{k=1}^{N_i}$, where N_i is the total number of submissions. We model submissions as counting processes whose intensities depend on equilibrium effort and the preceding public history. The two contestants' submissions

are generated using independent Poisson randomness. To link observed submission timing to effort, we assume

$$\tau_i(t) = r \cdot m_i(\tilde{y}_t, t) \tag{8}$$

where $r > 0$ converts model effort into submission intensity; the same relationship applies to player j. This proportional relationship is an empirical measurement assumption rather than a result derived from the theoretical game. In estimation, r is treated as an auxiliary scale parameter rather than a contest primitive.

The third component contains *public* and *private* leaderboard series, denoted by $\hat{x}_t^{i(j)}$ and $x_t^{i(j)}$. In Kaggle contests, public and private scores are computed from different subsets of the test data. Public scores are displayed during the contest, whereas private scores are revealed only after it ends. We define the latent contemporaneous gap as $y_t := x_t^i - x_t^j$ and the empirical public gap as $\hat{y}_t := \hat{x}_t^i - \hat{x}_t^j$. We construct the empirical public gap by carrying forward each contestant's most recently submitted public score between updates. Because submissions are asynchronous, $\hat{y}_t$ need not compare contemporaneous submissions by the two contestants.

This reconstructed latest-submission gap need not coincide with the platform's displayed best-so-far public-score gap, nor does the data establish that every value in the reconstructed path belongs to the contestants' common information set. We therefore use it as a proxy for a common public signal, not as a direct record of the information observed by each contestant. The corresponding private latest-submission path can also differ from the best-so-far score that determines final ranking; its changes may include exploratory submissions that do not alter a contestant's retained best score.

The public gap provides a noisy empirical measure of the latent performance gap. At retained readout time t_k, let $h_k = t_k - t_{k-1}$, with $t_0 = 0$, denote the elapsed time since the preceding retained readout. In the empirical observation model, we assume a fresh Gaussian measurement error $\varepsilon_k \sim \mathcal{N}(0, 1/(\lambda h_k))$. Conditional on the preceding public history and the current update time, this error is independent of the latent state.[8] Substituting the latent state dynamics in equation (1) gives the approximate observation equation

$$\hat{y}_{t_k} \approx y_0 + \int_0^{t_k} [m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)]\, ds + \sigma W_{t_k} + \varepsilon_k. \tag{9}$$

Here, σW_{t_k} captures accumulated innovation noise, while ε_k captures the public-signal noise associated with the leaderboard update. Thus, larger λ implies more precise public feedback. The variance

[8] For a fixed interval of length h, equation (2) gives the average readout $(Z_t - Z_{t-h})/h = h^{-1}\int_{t-h}^{t} y_s\, ds + (B_t - B_{t-h})/(\sqrt{\lambda}h)$. The noise term has variance $1/(\lambda h)$. Replacing the interval-average latent gap by its endpoint value motivates the empirical observation model; it is not an exact consequence of sampling the continuous signal at endogenous submission times.

$1/(\lambda h_k)$ is a maintained measurement assumption motivated by an interval-averaged continuous signal; it is not established for individual submission scores. If submission-level discrepancy variance does not decline with elapsed time, the resulting precision estimates may partly reflect readout spacing. In addition, carrying forward an opponent's stale score can reuse the same public–private discrepancy across successive gaps, creating serial dependence that the fresh-error likelihood does not represent. The fixed-grid benchmark gives a related sensitivity check, but neither it nor the time split identifies the variance–spacing relationship or the effect of serial dependence; no direction of bias is established.

The realized initial gap is not observed during the contest. Following the initial belief in Section 2, we assume

$$y_0 \sim \mathcal{N}(\mu_0, S_0)\,, \qquad S_0 > 0. \tag{10}$$

We integrate over y_0 in the likelihood. The parameter μ_0 is the initial perceived gap, while S_0 measures uncertainty about the initial gap. Given (S_0, σ, λ), equation (4) determines the deterministic variance path S_t. The precision parameter λ enters both the observation system and the deterministic filtering-variance path.

The theoretical model uses a continuous public signal and therefore admits the continuous-time filtering equation in (3). The empirical leaderboard, however, is observed only at asynchronous submission times. An exact empirical implementation would therefore require a hybrid filter with continuous prediction between submissions and discrete measurement updates.

To retain the continuous-update representation, we carry forward the most recently observed public gap. The structural DGP uses the equilibrium policy m, whereas estimation uses its leading-order approximation $m^{(1)}$, defined in equation (14). Together with the deterministic variance path from equation (4), this gives the empirical state recursion

$$d\tilde{y}_t = \left[m_i^{(1)}(\tilde{y}_t, S_t, t) - m_j^{(1)}(\tilde{y}_t, S_t, t)\right] dt + \lambda S_t(\hat{y}_t - \tilde{y}_t)dt \tag{11}$$

with initial condition $\tilde{y}_0 = \mu_0$. The reconstructed public gap $\hat{y}_t$ is a piecewise-constant empirical path, not a pointwise observation of the theoretical continuous signal Z_t. Between retained updates it is carried forward without a new measurement, so its continuous use may overstate learning. The estimates are therefore conditional on this approximation. The fresh-error likelihood also abstracts from dependence induced when carried-forward opponent scores are reused in consecutive readouts. We do not estimate a correlated-error or stale-score model, so the implications of this approximation for separating λ from σ remain untested.

3.2. Modeling Considerations

This subsection discusses three practical modeling concerns that are important for interpretation: strategic submission timing, the two-player approximation, and the normality assumption. The goal is to clarify the scope of the baseline framework and connect each concern to the robustness analyses that follow.

Strategic submission timing. Submission-triggered leaderboard updates create delays in public feedback: no new leaderboard readout arrives during periods without submissions. Strategic timing or withholding can therefore affect the information available to contestants and complicate inference.

Our submission-intensity specification links submissions to effort but does not explicitly model strategic withholding. Strategic submission timing is an important concern in the literature. For example, Lemus and Marshall (2021) adopt a similar simplification by excluding such behavior from their model. Our estimates should therefore be interpreted for contests where active updating behavior dominates withholding behavior. The minimum-submission requirement in Section 5.1 excludes very sparse histories, but does not establish that strategic withholding is absent among the retained contestants.

Selection of Two Players. Another concern is the two-player approximation. Real contests involve many participants, so reducing the environment to two players changes strategic externalities and information structure. Our interpretation is that the model isolates the dominant top rivalry in contests where competition is concentrated near the frontier. In Section 6.1, we conduct a series of robustness checks. By varying the focal-pair definition, we assess whether the validation results depend on a particular choice of representative rivalry.

Gaussian approximation. The model relies on Gaussian signal and innovation shocks. Mapping bounded leaderboard scores into the model's real-valued state space therefore requires both a transformation and a maintained distributional approximation. The empirical transformation is defined in Section 5.1. Rescaling the transformed scores changes their units but does not determine whether their distribution is Gaussian. We therefore treat Gaussianity as a modeling assumption rather than an empirically established property of every contest.

3.3. Bayesian Estimation Framework

We construct an approximate Bayesian posterior for the general parameter vector $\Theta := (c_i, c_j, \sigma, \lambda, \mu_0, r, S_0)$ using the data introduced above. Specific estimation exercises may fix individual parameters or impose restrictions, such as the steady-state initialization $S_0 = \sigma/\sqrt{\lambda}$, and estimate only the remaining free parameters.

Reconstructed private-score rankings are used to select the focal pair. Conditional on this selection, estimation uses the pair's contest-phase submission counts and public leaderboard readouts.

The private leaderboard trajectory y_t, observed ex post in Meta-Kaggle, is reserved for post-estimation path validation.

We discretize time into M intervals of length Δ_{num}, with endpoints $s_\ell = \ell\Delta_{\text{num}}$. Let n_ℓ^i denote contestant i's submission count in $(s_{\ell-1}, s_\ell]$, so that $\sum_{\ell=1}^M n_\ell^i = N_i$. Holding the submission intensity at its interval-start value gives the submission likelihood contribution (and analogously for j)

$$\mathcal{L}_i(\Theta) \propto \exp\left\{-\Delta_{\text{num}} \sum_{\ell=1}^M \tau_i^{(1)}(s_{\ell-1})\right\} \prod_{\ell=1}^M \left[\tau_i^{(1)}(s_{\ell-1})\Delta_{\text{num}}\right]^{n_\ell^i}. \tag{12}$$

Here $\tau_i^{(1)}(t) = rm_i^{(1)}(\tilde{y}_t, S_t, t)$ is the leading-order approximation to the submission intensity in equation (8); the omitted count-factorial terms do not depend on Θ. Given the preceding public history, submission counts are generated by independent Poisson draws, independent of the latent initial gap and innovation shocks. The likelihood includes all submissions, including multiple submissions within an interval; intervals with no submissions contribute through the exponential term.

For the leaderboard likelihood, we retain one public-gap readout at the end of each grid interval containing at least one submission by either contestant. Denote the retained times and gaps by $t_1 < \cdots < t_N$ and $\hat{y}_{t_k}$, respectively, and let $h_k = t_k - t_{k-1}$, with $t_0 = 0$. Conditional on the preceding history, the readout time t_k, and Θ, the empirical state and observation equations imply

$$\hat{y}_{t_k} \mid \mathcal{F}_{k-1}, t_k, \Theta \sim \mathcal{N}\left(\mu_{k|k-1}, \Omega_{k|k-1}\right),$$

Here $\mathcal{F}_{k-1}$ denotes the submission and public-leaderboard history strictly before t_k. The one-step-ahead predictive moments incorporate initial-state uncertainty, accumulated innovation, and preceding readouts. Appendix B.2 gives their explicit sequential recursion. The leaderboard likelihood is therefore

$$\mathcal{L}_{\text{pub}}(\Theta) = \prod_{k=1}^N \phi\left(\hat{y}_{t_k} \mid \mu_{k|k-1}, \Omega_{k|k-1}\right). \tag{13}$$

After evaluating each predictive density, we use the readout in the latent-state update and, separately, in the perceived-state recursion in equation (11). The moments $(\mu_{k|k}, \mathsf{P}_{k|k})$ belong to the auxiliary filter used for the leaderboard likelihood; the strategic belief pair $(\tilde{y}_t, S_t)$ continues to follow equations (11) and (4). The joint likelihood is proportional to $\mathcal{L}_i(\Theta)\mathcal{L}_j(\Theta)\mathcal{L}_{\text{pub}}(\Theta)$, with all contributions evaluated using the same evolving public history.

We evaluate the likelihood using the leading-order policy from equation (5):

$$m_k^{(1)}(\tilde{y}_t, S_t, t) = \frac{\theta}{c_k\sqrt{2\pi Q_t}} \exp\left(-\frac{\tilde{y}_t^2}{2Q_t}\right), \qquad Q_t = S_t + \sigma^2(T-t). \tag{14}$$

For each parameter vector Θ, equation (4) determines S_t, and equation (11) advances the perceived state and the associated effort paths. The posterior is conditional on both the leading-order policy approximation and the continuous-update approximation.

The leading-order policy also imposes a simple restriction on submission identities. At every grid start,

$$\frac{\tau_i^{(1)}(t)}{\tau_j^{(1)}(t)} = \frac{c_j}{c_i}, \qquad n_\ell^i \mid (n_\ell^i + n_\ell^j = n, \mathcal{F}_{\ell-1}, \Theta) \sim \text{Binomial}\left(n, \frac{c_j}{c_i + c_j}\right),$$

where $\mathcal{F}_{\ell-1}$ is the public history at that grid start. Thus public feedback changes total submission intensity but not the conditional share attributed to either contestant under this approximation. This restriction concerns $m^{(1)}$, not the exact nonlinear policies. Predictive tests should distinguish the timing restriction from this constant-share restriction.

We combine the joint likelihood with priors on the remaining free parameters to obtain the approximate posterior. Prior distributions, hyperparameters, and numerical parameter bounds are reported with each estimation exercise. We sample the resulting approximate posterior primarily using the No-U-Turn Sampler, an adaptive Hamiltonian Monte Carlo method (Hoffman and Gelman 2014).

The following proposition establishes local identification when contestants have different effort costs and the model uses the matched left-point discretization defined below. The proof treats the initial public input as given, whereas the empirical application constructs it from observed data. This initialization procedure is outside the scope of the proposition.

PROPOSITION 2 **(Local Identification under a Matched Grid Approximation)**. *Fix the initial public input before generating observations, with $y_0 \sim \mathcal{N}(\mu_0, S_0)$. For this proposition, use a matched left-point discretization: on each grid interval, hold the leading-order effort policy and submission intensity at their interval-start values and use that same effort value in the state-drift increment. Suppose $\theta > 0$, the grid contains at least three positive time points at which the first submission-triggered readout can occur, and $c_i \neq c_j$. Then the interior vector $\Theta = (c_i, c_j, \sigma, \lambda, \mu_0, r, S_0)$ is locally identified by the resulting finite-grid count and Gaussian-readout law.*

Identification here is a property of the specified family of data distributions, not a guarantee of precise estimation from one contest. The proof uses conditional laws for different possible first-readout times, including histories that may be rare in a finite sample. It therefore remains compatible with the weak cost–submission-scale separation observed below.

The identifying variation has an economic interpretation. At a possible first-readout time s, uncertainty decomposes as $S_0 + \sigma^2 s + 1/(\lambda s)$: initial uncertainty is constant, innovation uncertainty grows with elapsed time, and measurement uncertainty falls with it. Three distinct possible times separate these components in the specified law. Unequal effort costs create a directional score drift; combining that drift with the submission-intensity difference separates effort from the conversion factor r, and hence identifies absolute costs. If costs are equal, the drift difference vanishes and a common rescaling of both costs and r is observationally equivalent. Submission activity alone then cannot reveal how expensive it would be to induce additional effort.

4. Synthetic Experiments

We examine the finite-sample performance of the estimation framework using synthetic contests. Data are generated from the numerically computed equilibrium policy m, while estimation uses its leading-order approximation $m^{(1)}$ in equation (14).

Algorithm 1 Synthetic Data Simulation

Input: $\Delta > 0$, T, θ, c_i, c_j, σ, $\lambda > 0$, r, $\tau^\star$, μ_0, S_0, with $T/\Delta \in \mathbb{N}$

Solve equation (4) for $(S_t)_{t\in[0,T]}$ and the nonlinear HJB system for m_i and m_j.

Draw $y_0 \sim \mathcal{N}(\mu_0, S_0)$.

Sample rate-$\tau^\star$ Poisson candidate sets $\mathcal{C}_i, \mathcal{C}_j$ on $[0, T)$.

Sample $u_s^k \sim \mathcal{U}[0,1]$ for $s \in \mathcal{C}_k$, $k \in \{i, j\}$.

Sample $\Delta W_t \sim \mathcal{N}(0, \Delta)$ and $z_t \sim \mathcal{N}(0,1)$ on the grid; all draws are independent.

Initialize $\ell^i = \ell^j = 1$, $\hat{y}_0 = \tilde{y}_0 = \mu_0$, and $t_{\text{obs}} = 0$.

for $t = 0, \Delta, \ldots, T - \Delta$ **do**

For $k \in \{i, j\}$, evaluate $m_k = m_k(\tilde{y}_t, t)$ and set $\tau_k(t) = r m_k$.

$y_{t+\Delta} \leftarrow y_t + (m_i - m_j)\Delta + \sigma \Delta W_t$ {Discretize (1)}

$\tilde{y}_{t+\Delta} \leftarrow \tilde{y}_t + (m_i - m_j)\Delta + (1 - e^{-\lambda S_t \Delta})(\hat{y}_t - \tilde{y}_t)$ {(11), using m}

$\hat{y}_{t+\Delta} \leftarrow \hat{y}_t$

for each player $k \in \{i, j\}$ and candidate $s \in \mathcal{C}_k \cap [t, t+\Delta)$, chronologically **do**

if $u_s^k < \tau_k(t)/\tau^\star$ **then**

$\hat{t}_{\ell^k}^k \leftarrow s$; $\ell^k \leftarrow \ell^k + 1$

end if

end for

if at least one submission was accepted in this step **then**

$\hat{y}_{t+\Delta} \leftarrow y_{t+\Delta} + z_t / \sqrt{\lambda(t + \Delta - t_{\text{obs}})}$; $t_{\text{obs}} \leftarrow t + \Delta$ {Readout (9)}

end if

end for

Output: (S_t), (y_t), $(\tilde{y}_t)$, $(\hat{y}_t)$, (m_i, m_j), and accepted submission times $\{\hat{t}_k^{i(j)}\}$

The first step of Algorithm 1 computes the variance path S_t and the effort policies m_i, m_j needed to simulate the contest. We obtain S_t from equation (4) and use this path to solve the nonlinear HJB system numerically, with state spacing 0.2 and time step 1/48 day. To check sensitivity to the grid, we halve both spacings; the maximum difference between the resulting effort policies is below 0.43% of peak effort on a common evaluation grid. We then simulate the contest on an hourly grid, $\Delta = 1/24$ day, using thinning (Lewis and Shedler 1979) with $\tau^\star$ bounding both submission

intensities. Accepted submissions retain their event times, whereas the public gap is refreshed only at the end of hours containing submissions and carried forward otherwise. Readout-noise variance depends on the elapsed time between public updates, not individual submissions.

Our baseline simulation is a 90-day contest with $c_i = 1.2$, $c_j = 1.5$, $\theta = 1$, $\sigma = 2$, $\lambda = 1$, $\mu_0 = 0$, $r = 15$, and $S_0 = 4$, with time measured in days. The initial variance is twice its steady-state value, $\sigma/\sqrt{\lambda} = 2$, and the incentive indices are $w_i \approx 0.208$ and $w_j \approx 0.167$. The baseline sample contains 28 submissions by player i and 18 by player j. Figure 4 displays this sample, which we also use for the estimation comparison below. A paired steady-state simulation with $S_0 = 2$, holding all other parameters, random-number streams, and simulation rules unchanged, produces 30 and 19 submissions, respectively.

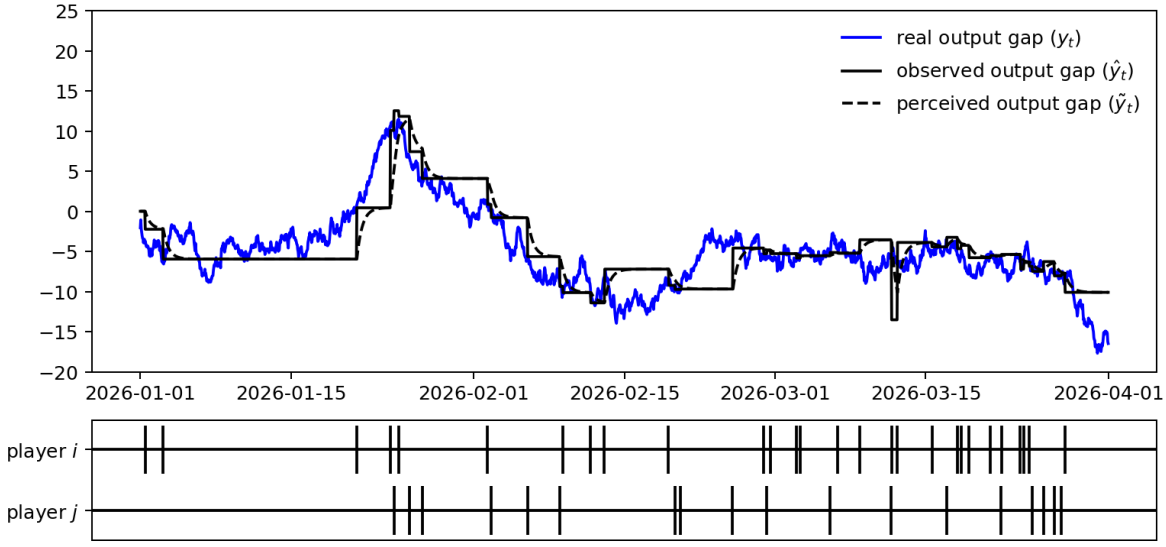


Figure 4 Synthetic Contest Data over a Three-Month Horizon

Note: The upper panel shows the latent gap y_t, carried-forward public gap $\hat{y}_t$, and perceived gap $\tilde{y}_t$; the lower panel marks submissions by players i and j. This is one 90-day realization under the numerically computed equilibrium policy m with time measured in days, $T = 90$, $c_i = 1.2$, $c_j = 1.5$, $\theta = 1$, $\sigma = 2$, $\lambda = 1$, $\mu_0 = 0$, $r = 15$, and $S_0 = 4$. Positive gaps favor player i; negative gaps favor j. Calendar dates on the horizontal axis index the 90 simulated days. Each mark is a submission, giving 28 for i and 18 for j. Public readouts refresh at the end of submission hours and are carried forward otherwise. The perceived gap uses the interval-start public value, so it need not coincide with the new readout. This path illustrates noisy learning, not an average treatment effect of feedback.

We directly compare the leading-order effort policy with the numerically computed nonlinear equilibrium in Figure 5. Panels (a)–(c) vary only $\theta = 1, 2, 4$, giving $w_i \approx 0.21, 0.42, 0.83$. Panels (d)–(f) instead change only feedback precision to $\lambda = 0.1$ or 10, or the opponent's cost to $c_j = 0.6$; all three retain $\theta = 1$ and $S_0 = 4$. The curves show player i's effort at $t = T/2$, while the error statistic E_i covers the time–state evaluation grid. Errors increase from 2.51% to 10.18% as the prize rises, remain below 2.6% under the precision changes, and reach 12.32% with the lower opponent cost.

Thus, accuracy also depends on the strategic environment, even when w_i is unchanged. These controlled comparisons do not establish accuracy throughout the empirical sample or at optimized counterfactual designs.

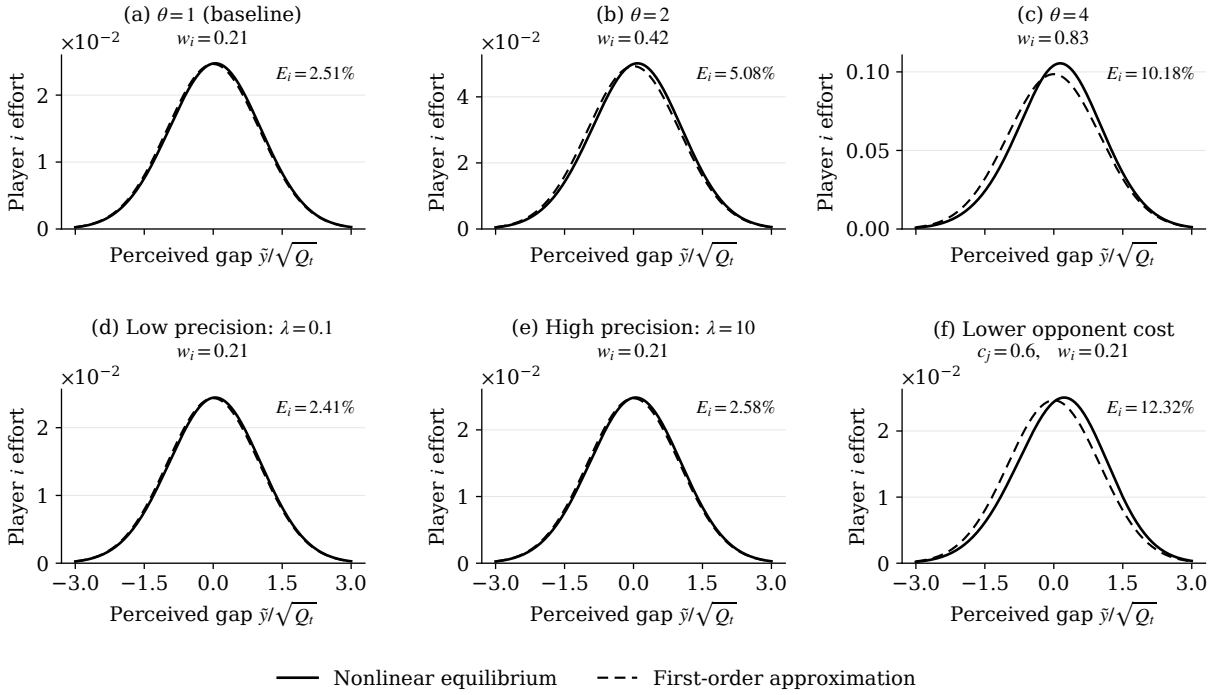


Figure 5 Equilibrium Effort: Nonlinear Numerical Solution and First-Order Approximation

Note: Solid curves show the numerical nonlinear equilibrium and dashed curves the first-order policy at $t = T/2$. Each panel changes only the indicated parameter from the baseline. The horizontal axis is $z = \tilde{y}/\sqrt{Q_t}$. The vertical axis is player i's effort, in model score units per day; $z > 0$ means that i is the perceived leader. The normalized horizontal scale changes with Q_t, so equal z across panels need not mean equal score gaps. Higher prizes increase both players' incentive indices; lowering c_j increases the opponent's ability without changing i's first-order policy at a fixed state. The error E_i uses 601 equally spaced values of $z \in [-3, 3]$ and 378 time points spanning $[0, T]$, with additional points near the deadline. At each time, the largest absolute error is divided by the nonlinear peak over this state range; E_i is its maximum over time. Halving the state and base time spacings changes effort by less than 0.40% under the same time-specific peak normalization in every configuration. The baseline is $T = 90$, $c_i = 1.2$, $c_j = 1.5$, $\theta = 1$, $\sigma = 2$, $\lambda = 1$, $\mu_0 = 0$, $r = 15$, and $S_0 = 4$. Panels (a)–(c) set $\theta = 1, 2, 4$; panels (d) and (e) set $\lambda = 0.1, 10$; panel (f) sets $c_j = 0.6$, with other parameters at baseline.

Table 1 compares two initializations using the same complete baseline sample and jointly estimating r. Panel A imposes $S_0 = \sigma/\sqrt{\lambda}$, so the filtering variance is at its steady state from the outset. Panel B estimates S_0 separately as a nuisance parameter; its posterior is integrated out of the reported parameter summaries. The estimates are similar across initializations, and all reported 95% credible intervals contain the generating values. These results concern a single sample and do not establish the absence of systematic bias.

We compare the models primarily by their marginal likelihoods, computed by bridge sampling (Gronau et al. 2020) under common priors for the shared parameters. The Bayes factor for the non-steady-state model relative to the steady-state model is approximately one. Changing the median of the lognormal prior for S_0 from 2 to 1 or 4, holding its log-scale standard deviation at 0.75, gives Bayes factors between 0.97 and 1.01. Thus, this sample provides little evidence for estimating a separate initial variance, even though it was generated with $S_0 = 4$. The auxiliary BIC favors the steady-state restriction, but boundary likelihood optima make its usual asymptotic interpretation unreliable.

4.1. Parameter Recovery Under Richer Data

Given the limited evidence for estimating a separate S_0 in Table 1, we use the more parsimonious steady-state specification to assess recovery from pooled contests and longer horizons. Data are generated using the nonlinear equilibrium policy m and $S_0 = 4$, while estimation imposes $S_0 = \sigma/\sqrt{\lambda}$ and jointly estimates $(c_i, c_j, \sigma, \lambda, \mu_0, r)$. No initial observations are discarded.

Pooled Contests. We form nested pools of 10 and 20 independent contests under common generating parameters. Each contest draws its own initial gap, innovations, signal noise, and submission events. Estimation initializes a separate latent state for each contest and combines their likelihoods under one shared parameter vector.

In Table 2, expanding the pool from 10 to 20 contests reduces the posterior standard deviations of σ, λ, and μ_0. Uncertainty about c_i, c_j, and r changes little. Strong positive posterior correlations between r and each cost parameter indicate that these samples provide limited separation of effort costs from the submission scale. This is evidence of weak finite-sample separation, not a failure of the stated conditional identification result and not by itself evidence that empirical costs are prior-dominated. However, because submission intensities load on the ratios r/c_k while counterfactual incentive indices depend on absolute costs, the design calculations remain conditional on the cost-scale priors and numerical bounds; Section 6.3 reports targeted sensitivity checks.

Longer Contests. We next use single contests with 180- and 360-day horizons, holding $(\theta, c_i, c_j, \sigma, \lambda, \mu_0, S_0)$ at their baseline values. We adjust the submission scale r to target comparable submission frequencies across horizons. Using equation (8), we calibrate r to match the time-averaged total submission intensity of a 90-day pilot contest with $r = 15$. Calibration uses leading-order effort policies and steady-state initialization.[9] The resulting scales, $r_{180} = 18.6484375$ and $r_{360} = 47.4375$, are used in the simulations with m and $S_0 = 4$; the resulting submission frequencies need not be identical.

[9] On the hourly grid, calibration uses $\frac{r_T \Delta}{T} \sum_{n=0}^{T/\Delta-1} \sum_{k\in\{i,j\}} m_k^{(1)}(\tilde{y}_{n\Delta}, \sigma/\sqrt{\lambda}, n\Delta) \approx 0.6184661122$, with $\Delta = 1/24$ day. The target is computed from a 90-day pilot with $r = 15$, and each pilot perceived-state path uses the candidate r_T. Pilot seeds are 1000 for 90 days and $5000 + T$ for longer horizons, with offsets $0, 1, 2, 3$ for Gaussian shocks, Poisson candidates, uniform marks, and the initial gap, respectively. The absolute tolerance is 5×10^{-5} submissions per day.

Table 1 Bayesian Estimates under Alternative Initializations

Panel A. Steady-State Initialization: $S_0 = \sigma/\sqrt{\lambda}$

Parameters	c_i	c_j	σ	λ	μ_0	r
(true val.)	1.2	1.5	2.0	1.0	0.0	15.0
Posterior Mean	1.168	1.763	2.170	1.562	-0.466	13.878
Posterior SD	0.399	0.631	0.249	0.596	0.900	4.159
Post. RMSD	0.400	0.684	0.302	0.820	1.014	4.307
95% Interval	[0.58, 2.14]	[0.83, 3.34]	[1.74, 2.71]	[0.69, 2.99]	[-2.24, 1.25]	[7.53, 23.61]

Panel B. Non-Steady-State Initialization: S_0 Estimated

Parameters	c_i	c_j	σ	λ	μ_0	r
(true val.)	1.2	1.5	2.0	1.0	0.0	15.0
Posterior Mean	1.149	1.727	2.169	1.560	-0.427	13.677
Posterior SD	0.399	0.622	0.253	0.599	0.940	4.101
Post. RMSD	0.402	0.662	0.304	0.820	1.032	4.309
95% Interval	[0.57, 2.13]	[0.82, 3.21]	[1.74, 2.72]	[0.67, 3.02]	[-2.22, 1.42]	[7.41, 23.46]

Panel C. Model Comparison

	Steady State	Non-Steady State
Estimated parameters	6	7
Log marginal likelihood	-359.124	-359.127
Maximized log likelihood	-352.922	-352.717
BIC ($n = 45$)	728.684	732.080

Note: Both models use the same 46 submissions and 45 hourly public readouts, with no initial observations discarded. The parameters are effort costs c_i, c_j (lower means higher ability), innovation risk σ, feedback precision λ, initial mean gap μ_0, and the effort-to-submission scale r. SD denotes posterior standard deviation. The nonlinear generating model has $T = 90$ days, $\theta = 1$, and $S_0 = 4$; both fitted likelihoods use first-order effort. Higher log marginal likelihood favors a model, whereas lower BIC favors it. The nearly equal marginal likelihoods indicate little evidence for choosing between the two initializations in this sample. Both use $c_i, c_j \sim \text{LogNormal}(\log 1, 0.75^2)$, $\sigma \sim \text{LogNormal}(\log 2, 0.5^2)$, $\lambda \sim \text{LogNormal}(\log 1, 0.75^2)$, $\mu_0 \sim \mathcal{N}(0, 1)$, and $\log r \sim \mathcal{N}(\log 15, 0.35^2)$. Panel B additionally uses $S_0 \sim \text{LogNormal}(\log 2, 0.75^2)$. Priors are normalized on the retained synthetic supports: $c_i, c_j \in (0.1, 5)$, $\sigma \in (0.5, 10)$, $\lambda \in (10^{-6}, 100)$, $\mu_0 \in (-20, 20)$, and $S_0 > 0.01$ when free. Each fit uses four chains with 1,000 warmup and 2,000 retained draws per chain; neither has divergences, maximum $\widehat{R} < 1.002$, and minimum bulk and tail effective sample sizes exceed 4,100 and 3,800. Post. RMSD is the square root of the posterior mean squared distance from the generating parameter, not an RMSE across datasets. Intervals are equal-tailed 95% credible intervals. BIC is $-2\hat{\ell} + k \log n$, using likelihood-only optimization on the same supports and the 45-readout convention for n, not the computational grid size. The likelihood optima approach $c_j = 5$ in both models and $S_0 = 0.01$ in Panel B; BIC is therefore reported only as a heuristic comparison.

In Table 3, the 360-day realization yields smaller posterior standard deviations for σ and λ than the 180-day realization. However, the posterior standard deviation of μ_0 remains close to its prior value of one, whereas pooling independent contests reduces this uncertainty more substantially. These comparisons reflect the realized samples and their different submission scales, not an isolated effect of contest duration.

Table 2 Bayesian Estimates from Pooled Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0	r
(true val.)	1.2	1.5	2.0	1.0	0.0	15.0
Pool of 10 Contests						
Posterior Mean	1.168	1.326	1.994	0.889	-0.195	14.477
Posterior SD	0.355	0.405	0.074	0.118	0.666	4.314
Post. RMSD	0.357	0.441	0.075	0.162	0.694	4.346
Pool of 20 Contests						
Posterior Mean	1.174	1.325	2.031	1.039	0.090	14.492
Posterior SD	0.354	0.402	0.054	0.108	0.542	4.334
Post. RMSD	0.355	0.438	0.062	0.115	0.549	4.364

Note: The nested pools contain 426 submissions (227 by i, 199 by j) and 848 submissions (450 by i, 398 by j), with 418 and 836 distinct hourly readouts. The first 10 contests form the smaller pool. The generating scale is $r = 15$, but r is estimated jointly in both pools. The parameters are effort costs c_i, c_j (lower means higher ability), innovation risk σ, feedback precision λ, initial mean gap μ_0, and the effort-to-submission scale r. SD denotes posterior standard deviation. Each generating contest lasts 90 days with $\theta = 1$ and $S_0 = 4$; estimation uses first-order effort and imposes $S_0 = \sigma/\sqrt{\lambda}$. More independent contests sharpen the risk and precision estimates here but do little to separate absolute costs from r. The two pools are nested, not independent replications of an estimator-comparison experiment. Priors and supports match the steady-state specification in Table 1. Each fit uses four chains with 1,000 warmup and 2,000 retained draws per chain; all $\widehat{R} < 1.002$, bulk and tail effective sample sizes exceed 2,600, and no divergences or tree-depth-limit hits occur. Post. RMSD is the square root of the posterior mean squared distance from the generating parameter vector.

Table 3 Bayesian Estimates from Longer-Horizon Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0	r
Contest of 6 Months						
(true val.)	1.2	1.5	2.0	1.0	0.0	18.648
Posterior Mean	1.165	1.249	2.190	1.444	-0.362	18.230
Posterior SD	0.374	0.404	0.177	0.367	0.959	5.450
Post. RMSD	0.376	0.476	0.260	0.576	1.025	5.466
Contest of 12 Months						
(true val.)	1.2	1.5	2.0	1.0	0.0	47.438
Posterior Mean	1.074	1.445	2.085	1.033	-0.596	45.969
Posterior SD	0.327	0.442	0.096	0.189	0.921	13.654
Post. RMSD	0.350	0.446	0.128	0.191	1.097	13.732

Note: The 180-day contest contains 114 submissions (59 by i, 55 by j) and 109 distinct hourly readouts; the 360-day contest contains 235 submissions (135 by i, 100 by j) and 231 readouts. The parameters are effort costs c_i, c_j (lower means higher ability), innovation risk σ, feedback precision λ, initial mean gap μ_0, and the effort-to-submission scale r. SD denotes posterior standard deviation. Both datasets use nonlinear effort, $\theta = 1$, and $S_0 = 4$; estimation uses first-order effort and steady-state initialization. Each horizon contributes one realization, and the generating r_T also changes. Differences therefore describe these samples, not the causal effect of extending a contest. Both specifications estimate r, with $\log r \sim \mathcal{N}(\log r_T, 0.35^2)$ centered at the respective generating scale. Other priors and supports match Table 1. Each fit uses four chains with 1,000 warmup and 2,000 retained draws per chain; all $\widehat{R} < 1.002$, bulk and tail effective sample sizes exceed 3,100, and no divergences or tree-depth-limit hits occur. Post. RMSD is the square root of the posterior mean squared distance from the generating parameter vector.

5. Empirical Application

This section applies the estimation framework to Kaggle contests, using submission behavior and public-leaderboard histories to estimate innovation risk and feedback precision. Private-leaderboard

diagnostics assess their empirical interpretation, and a subsequent design exercise searches the nonlinear game for cost reductions that preserve baseline expected quality.

Following the parsimonious specification examined in Section 4, we impose the steady-state initialization $S_0 = \sigma/\sqrt{\lambda}$ and jointly estimate $(c_i, c_j, \sigma, \lambda, \mu_0, r)$ for each contest.[10] Estimation uses the leading-order policy in equation (14), so precision affects both observation noise and effort incentives through $Q_t(\lambda)$. Private-leaderboard paths, reconstructed from the private scores of historical submissions after the contest ends, are excluded from the likelihood and used to assess the estimates of λ and σ. Because reconstructed private-score rankings enter focal-pair selection, we interpret the evidence as diagnostic validation conditional on the selected pair, rather than as validation on a fully independent sample. Figure 6 summarizes the data-model structure and validation design.

5.1. Preparing Data for Structural Estimation

Kaggle contests differ in prize structures, participation patterns, and scoring metrics. Our contest selection follows three criteria. The resulting sample contains 73 contests concluded before April 15, 2025 that satisfy the eligibility filters on leaderboard availability, public-private score information, and focal-player activity.[11]

We first restrict attention to Kaggle competitions offering one, two, or three monetary prizes denominated in U.S. dollars. Competitions with a single prize naturally conform to the winner-take-all incentive structure. For competitions with two or three prize tiers, we use the difference between the first- and second-place cash prizes as the effective prize θ in the two-player approximation. This proxies the incentive for top rivalry, rather than the exact payoff difference for the selected focal pair.[12]

The second criterion requires an active focal pair with substantially overlapping participation. We select the pair from contestants ranked by their last recorded private scores. In the source records, each contestant must submit at least five times over an activity span of at least ten days, and the pair's activity windows must overlap for more than ten days. We interpret the two-player model as an approximation of rivalry among active, high-performing contestants, rather than a representation of the full participant population.

The third criterion restricts the sample to bounded, higher-is-better scoring metrics, including accuracy, AUC, average precision, Kappa, and correlation measures. To place these scores on a

[10] The empirical priors are $c_i, c_j \sim \text{LogNormal}(\log 1, 0.75^2)$, $\sigma \sim \text{LogNormal}(\log 2, 0.5^2)$, $\lambda \sim \text{LogNormal}(\log 100, 2.25^2)$, $r \sim \text{LogNormal}(\log 10, 0.5^2)$, and $\mu_0 \sim \mathcal{N}(\hat{y}_{t_1}, 1)$. Free positive parameters have a numerical lower bound of 0.01 and no upper bound. Centering the prior for μ_0 on the first observed public gap is an empirical-Bayes initialization, on which the posterior is conditional.

[11] The Meta-Kaggle database is updated daily, with new competition data added after the contests conclude.

[12] Noncash awards and separately conditioned supplementary awards, such as travel reimbursements and software-specific prizes, are excluded. Contest-specific prize conventions are reported in Appendix C.

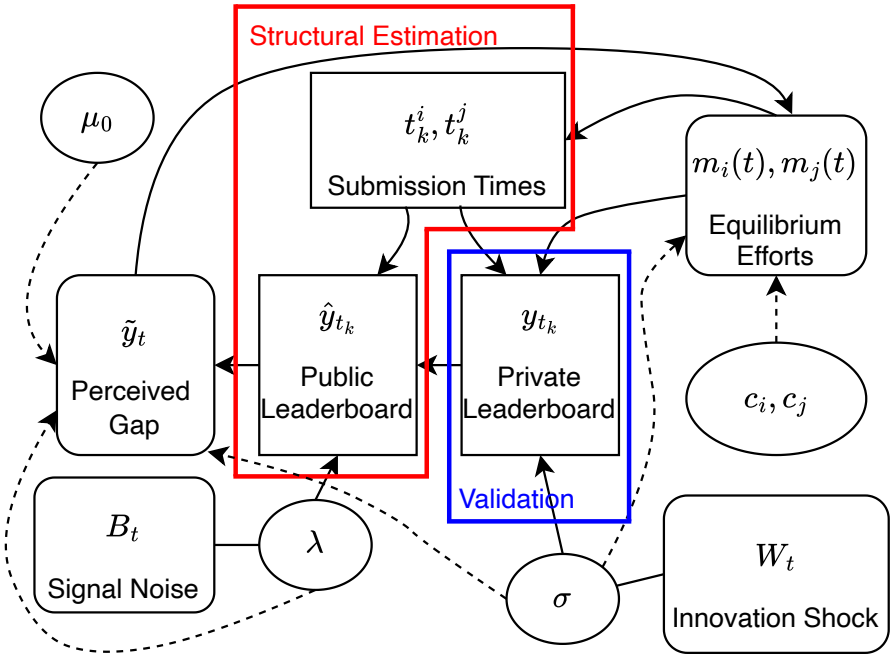


Figure 6 Meta-Kaggle Dataset and the Model Structure

Note: This figure shows the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles represent observed data, rounded rectangles represent latent states, and ellipses represent contest parameters. Solid lines depict temporal relationships among time series variables and state variables, while dashed arrows show the influence of contest parameters on the data-generating process. The private gap y_t is unavailable to contestants during the contest and is used for post-estimation diagnostics. Private-path values are excluded from the estimation likelihood, although reconstructed private-score rankings are used to select the focal pair. Here y_t, $\hat{y}_t$, and $\tilde{y}_t$ denote the latent private gap, public-gap proxy, and contestants' perceived gap. The parameters c_i, c_j determine effort costs, σ innovation risk, λ signal precision, and r the conversion from effort to submission intensity. Arrows describe the maintained model, not separately identified causal effects.

common transformed scale, we first map scores with range $[-1,1]$ to $[0,1]$ by $(s+1)/2$ and then apply a logit transformation:[13]

$$x_t = a \cdot \log\left(\frac{p_t}{1-p_t}\right), \quad 0 < p_t < 1 \tag{15}$$

where p_t is the clipped individual leaderboard score, x_t is the corresponding transformed score, and the scale parameter is fixed at $a = 1$ across contests. Within the clipping bounds, the transformation preserves score ordering and magnifies score differences near the endpoints of the unit interval. For the focal pair (i,j), we then construct the model variables as transformed-score gaps: $\hat{y}_t = \hat{x}_t^i - \hat{x}_t^j$ for the public leaderboard and $y_t = x_t^i - x_t^j$ for the private leaderboard.

Submission records provide calendar dates rather than within-day times. Preprocessing retains all submissions within the observation window and assigns reproducible random times within each

[13] Preprocessing clips the rescaled scores to $[10^{-4}, 0.9999]$, affecting 54 of 26,830 public/private score entries (about 0.2%).

date before constructing hourly submission counts and carried-forward score paths. These times are imputed, not observed; the estimates condition on this reconstruction. Appendix C reports implementation details and time-imputation sensitivity results.

The empirical horizon T is the length of the constructed hourly data window, including the initialization window, rather than the exact interval between the official contest opening and deadline. On this grid, we record each contestant's submission count and carry forward their latest transformed score between submissions.[14] For each hour containing at least one submission by either focal contestant, we retain one public-gap readout at the interval endpoint, following Section 3. Because submissions are asynchronous, only the submitting contestant's score is updated, while the opponent's most recent score is carried forward. The observed public gap may therefore compare scores submitted at different times and only approximates the contemporaneous latent performance gap. Our observation equation treats this gap as an approximate readout of the latent competitive state; this approximation is most credible for focal pairs with frequent submissions and substantial overlap in active periods.

5.2. Private-Leaderboard Primitive Validation

This subsection assesses whether the estimated feedback precision λ and innovation risk σ are consistent with private-leaderboard outcomes. As background, Table 4 summarizes public–private rank agreement across the 73 contests without using structural estimates. Although overall rank correlations are high, overlap among top-ranked teams is incomplete, indicating that public standings do not fully reveal private rankings.

Table 4 Public–Private Leaderboard Agreement

Statistic	Mean	Median	95% CI for Mean
Public–private rank correlation	0.937	0.984	[0.902, 0.963]
Top-2 overlap rate	65.1%	50.0%	[56.8%, 73.3%]
Top-5 overlap rate	66.6%	80.0%	[59.7%, 73.2%]

Note: The sample consists of the same 73 Kaggle contests used in the structural analysis. For each contest, we use the official public and private leaderboard ranks in Meta-Kaggle, excluding benchmark teams and teams missing either rank. Public–private rank correlation is the Spearman correlation across all retained teams within a contest. The top-k overlap rate is $|\mathcal{T}_k^{pub} \cap \mathcal{T}_k^{pri}|/k$, where $\mathcal{T}_k^{pub}$ and $\mathcal{T}_k^{pri}$ are the sets of teams ranked in the public and private top k, respectively; the ordering of teams within the top-k sets need not coincide. Each statistic is first calculated separately by contest, and the table reports the equally weighted mean and median across the 73 contests. The 95% confidence intervals pertain to the across-contest means and are percentile intervals obtained from 10,000 nonparametric bootstrap resamples at the contest level. Higher values indicate greater agreement. A top-2 overlap of 50% means that one of the public top two also belongs to the private top two. High overall rank correlation can coexist with substantial uncertainty near the prize threshold; these statistics alone do not distinguish overfitting from other sources of public–private disagreement.

[14] The earliest joint public gap is backfilled into the initialization hours, which supply recursion inputs but contribute neither submission-count exposure nor validation outcomes.

For each of the 73 contests, we estimate the model using the full observation window. For the time-split diagnostics, we reestimate it using only the first 75% of the contest-phase window, retaining the full-sample horizon. We report posterior medians for parameter estimates and use joint posterior draws for credible intervals and precision counterfactuals. All 146 fits have no divergent transitions or tree-depth saturations, rank-normalized $\widehat{R} < 1.01$, and bulk and tail effective sample sizes above 400. Appendix C reports the full-sample parameter estimates.

We next examine whether contests with higher estimated λ exhibit less public–private disagreement and those with higher estimated σ exhibit greater private-gap volatility. Appendix C provides the regression details.

5.2.1. Private-Leaderboard Diagnostic for Signal Precision λ. We assess the estimated feedback precision $\hat{\lambda}_\ell$ by comparing it with a benchmark constructed from public–private leaderboard discrepancies. Let $\ell \in \{1, \ldots, L\}$ index contests. Let $\mathcal{E}_\ell$ denote the ordered set of submission-triggered leaderboard readouts for the focal pair, and let $N^\ell = |\mathcal{E}_\ell|$. Let $h_k^\ell = t_k^\ell - t_{k-1}^\ell$ denote the elapsed time associated with readout t_k^ℓ. Using the Gaussian discrepancy approximation $\hat{y}_{t_k}^\ell - y_{t_k}^\ell \sim \mathcal{N}(0, 1/(\lambda_\ell h_k^\ell))$ for $t_k^\ell \in \mathcal{E}_\ell$, we construct

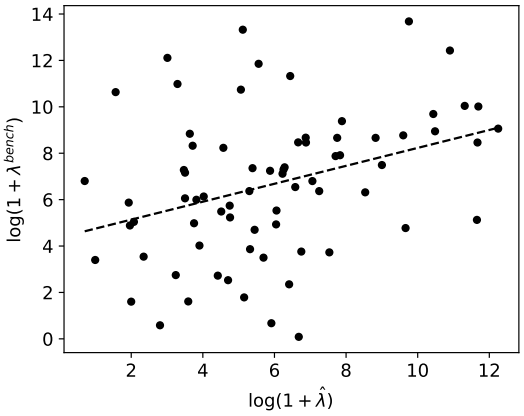
$$\lambda_\ell^{\text{bench}} = \frac{N^\ell}{\sum_{t_k^\ell \in \mathcal{E}_\ell} h_k^\ell (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2}.$$

The benchmark therefore uses only submission-triggered leaderboard readouts rather than repeated hourly carry-forward observations.[15] This working Gaussian law does not follow from the score transformation, and asynchronous updates may induce dependence across readouts. The benchmark therefore checks the observation-precision interpretation of λ, not its behavioral effect through $Q_t(\lambda)$ and effort. Figure 7 reports a regression slope of 0.386 and a Spearman correlation of $\rho = 0.364$ (Spearman $p = 0.002$). Because the two measures share public-data inputs, their association is a directional consistency check rather than an independent holdout validation.

5.2.2. Public-Private Leaderboard Disagreement Diagnostics. Whereas the preceding diagnostic compares two precision measures, we now examine whether higher estimated precision is associated with less public–private disagreement. For contest ℓ, we measure disagreement by the mean absolute gap between the two leaderboard paths:

$$\mathcal{D}_\ell^{path} = \frac{1}{|\mathcal{H}_\ell|} \sum_{h \in \mathcal{H}_\ell} |y_{\ell h} - \hat{y}_{\ell h}|,$$

[15] As a robustness comparison to the irregular-interval estimator above, we also consider the fixed-grid approximation $\lambda_{\ell,\Delta}^{\text{bench}} = N^\ell / \left\{ \Delta \sum_{t_k^\ell \in \mathcal{E}_\ell} (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2 \right\}$. Despite this difference, both constructions yield the same qualitative conclusion of a positive but modest cross-contest association.



	coef.	stderr	95% CI
const.	4.365***	0.809	[2.780, 5.950]
$\log(1+\hat{\lambda})$	0.386***	0.112	[0.167, 0.606]
Obs.		73	
R^2		0.121	
Adj. R^2		0.109	

(1) Significance levels: *** $p<0.001$, ** $p<0.01$, * $p<0.05$. (2) OLS residual diagnostics: Omnibus = 0.601, Jarque–Bera = 0.374, Skewness = 0.175, Kurtosis = 3.024. (3) Condition Number = 15.726.

Figure 7 $\lambda_\ell^{\text{bench}}$ **vs. Bayesian Estimate** $\hat{\lambda}_\ell$

Note. Each point represents one of 73 contests. The approximate benchmark $\lambda_\ell^{\text{bench}}$ uses elapsed-time-weighted squared public–private discrepancies at submission-triggered readouts; $\hat{\lambda}_\ell$ is the posterior median structural estimate. The dashed line regresses $\log(1+\lambda_\ell^{\text{bench}})$ on $\log(1+\hat{\lambda}_\ell)$. The horizontal axis is the log-transformed structural estimate and the vertical axis the log-transformed benchmark; larger values indicate more precise feedback. A positive slope means that the two measures tend to rank contests similarly, not that they agree in levels or identify a causal effect of disclosure. Regression standard errors and confidence intervals use HC1 heteroskedasticity-robust covariance estimates.

where $\mathcal{H}_\ell$ contains the hourly observation points in the contest-phase path, excluding the backfilled initialization window. Higher λ reduces signal noise at a given readout interval, motivating a negative association with $\mathcal{D}_\ell^{path}$; update frequency and score characteristics also affect observed disagreement.

Because $\hat{\lambda}_\ell$ and $\mathcal{D}_\ell^{path}$ use public data from the same period, their association may partly reflect shared path variation. We therefore also relate the early estimate $\hat{\lambda}_\ell^{early}$ to disagreement in the final quarter. Let $\mathcal{H}_\ell^{late}$ contain the hours in the final 25% of the contest-phase window and define $\mathcal{D}_\ell^{late} = |\mathcal{H}_\ell^{late}|^{-1} \sum_{h\in\mathcal{H}_\ell^{late}} |y_{\ell h} - \hat{y}_{\ell h}|$. Initialization hours are excluded when locating the cutoff.

We regress $\log(1+\mathcal{D}_\ell^{path})$ on $\log(1+\hat{\lambda}_\ell)$ for the full-path comparison and $\log(1+\mathcal{D}_\ell^{late})$ on $\log(1+\hat{\lambda}_\ell^{early})$ for the time-split comparison. Table 5 reports slopes of -0.013 ($p=0.006$) for the full path and -0.015 ($p=0.025$) for the time split. Higher early precision is associated with less disagreement in the final quarter, with $R^2=0.029$.

5.2.3. Private-Leaderboard Consistency Benchmark for Innovation Risk σ. We assess the estimated innovation risk $\hat{\sigma}_\ell$ using private-gap increments adjusted for estimated effort drift. Let $(t_k^\ell, y_k^\ell)_{k=0}^{n_\ell}$ denote the private-leaderboard gap evaluated at the ordered update times used for this benchmark, with the first retained readout serving as the initial observation. No increment is constructed from the artificial backfill. Motivated by a Gaussian increment approximation to

Table 5 Signal Precision and Public-Private Path Disagreement

Specification	coef.	stderr	95% CI	R^2	Spearman ρ
Full path	-0.013**	0.005	[-0.022, -0.004]	0.064	-0.346**
Time split	-0.015*	0.007	[-0.028, -0.002]	0.029	-0.343**
Obs.			73		

Note: The full-path row regresses $\log(1+\mathcal{D}_\ell^{path})$ on $\log(1+\hat{\lambda}_\ell)$, where $\mathcal{D}_\ell^{path}$ is mean absolute public–private gap disagreement. Negative coefficients mean that higher estimated precision is associated with smaller public–private discrepancies. These are cross-contest associations on a $\log(1+x)$ scale, not elasticities or the effect of changing a contest's disclosure rule. Inference treats the posterior medians as given. The time-split row relates the same measure in the final 25% to precision estimated from the first 75% of contest-phase data, retaining the full-sample horizon. Initialization hours are excluded from the discrepancy measures. Both regressions include an intercept and use 73 contests and posterior median estimates. Spearman ρ reports the corresponding rank association. Regression standard errors and confidence intervals use HC1 heteroskedasticity-robust covariance estimates. (1) Significance markers: *** $p<0.001$, ** $p<0.01$, * $p<0.05$. (2) OLS residual diagnostics (Full path; Time split): Omnibus = 66.882; 90.471, Jarque–Bera = 364.820; 1131.650, Skewness = 2.872; 3.831, Kurtosis = 12.324; 20.701. (3) Condition Number (Full path; Time split): 15.726; 15.742.

equation (1), we subtract the posterior-estimated effort drift $\hat{\mu}_k^\ell = \int_{t_{k-1}^\ell}^{t_k^\ell} [\hat{m}_i^\ell(s) - \hat{m}_j^\ell(s)]\, ds$ from each increment and scale the squared residual by the elapsed time. The resulting benchmark is

$$\sigma_\ell^{\text{bench}} = \left(\frac{1}{n_\ell} \sum_{k=1}^{n_\ell} \frac{\left[y_k^\ell - y_{k-1}^\ell - \hat{\mu}_k^\ell \right]^2}{t_k^\ell - t_{k-1}^\ell} \right)^{1/2}$$

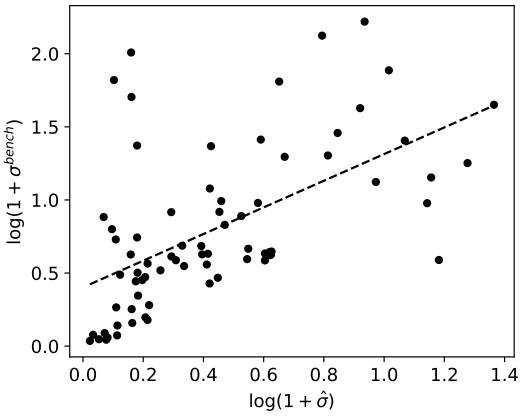
Appendix B.4 derives this expression. The drift adjustment $\hat{\mu}_k^\ell$ is evaluated at the posterior median parameters. Together with asynchronous score updates, this makes $\sigma_\ell^{\text{bench}}$ a model-assisted consistency benchmark, rather than an exact maximum likelihood estimator of latent diffusion volatility.

As for λ, we examine the cross-contest relationship between $\log(1+\sigma_\ell^{\text{bench}})$ and $\log(1+\hat{\sigma}_\ell)$. Figure 8 reports a regression slope of 0.912 and a Spearman correlation of $\rho = 0.597$ (Spearman $p<0.001$).

5.2.4. Private-Leaderboard Volatility Diagnostics. We examine whether contests with higher estimated innovation risk exhibit greater private-path volatility. Unlike the preceding benchmark, this diagnostic does not use an estimated effort-drift adjustment. For contest ℓ, let $\Delta y_{\ell h} = y_{\ell h} - y_{\ell,h-1}$ denote the hourly private-gap increment and define $\mathcal{V}_\ell^{path} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_\ell,\ h>1\}$. Hours are indexed within the contest-phase path, excluding artificial initialization increments.

Carrying scores forward produces zero increments between updates and jumps when new scores arrive. Thus, $\mathcal{V}_\ell^{path}$ reflects both submitted-score changes and update frequency. It provides a less model-dependent check on $\hat{\sigma}_\ell$, rather than a direct measure of latent diffusion volatility.

We also apply the same time split used for signal precision. For the final-quarter hours $\mathcal{H}_\ell^{late}$, define $\mathcal{V}_\ell^{late} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_\ell^{late}\}$. The full-path regression relates $\log(1+\mathcal{V}_\ell^{path})$ to $\log(1+\hat{\sigma}_\ell)$; the time-split regression relates $\log(1+\mathcal{V}_\ell^{late})$ to $\log(1+\hat{\sigma}_\ell^{early})$.



	coef.	stderr	95% CI
const.	0.402***	0.094	[0.218, 0.586]
$\log(1+\hat{\sigma})$	0.912***	0.173	[0.573, 1.250]
Obs.		73	
R^2		0.320	
Adj. R^2		0.310	

(1) Significance levels: *** $p<0.001$, ** $p<0.01$, * $p<0.05$. (2) OLS residual diagnostics: Omnibus = 21.321, Jarque–Bera = 28.003, Skewness = 1.302, Kurtosis = 4.557. (3) Condition Number = 3.585.

Figure 8 σ^{bench} **vs. Bayesian Estimate** $\hat{\sigma}$

Note. Each point represents one of 73 contests. The model-assisted benchmark $\sigma_\ell^{\text{bench}}$ uses private-gap increments between readouts, adjusted for elapsed time and effort drift evaluated at posterior median parameters; $\hat{\sigma}_\ell$ is the posterior median structural estimate. The dashed line regresses $\log(1+\sigma_\ell^{\text{bench}})$ on $\log(1+\hat{\sigma}_\ell)$. The horizontal axis is the log-transformed structural estimate and the vertical axis the log-transformed benchmark; larger values indicate greater innovation risk. A positive slope is directional agreement. Because the benchmark uses estimated drift and asynchronous private scores, it is not an independent observation of latent volatility. Regression standard errors and confidence intervals use HC1 heteroskedasticity-robust covariance estimates.

Table 6 reports slopes of 0.243 ($p<0.001$) for the full path and 0.265 ($p=0.006$) for the time split. The positive time-split association also appears without the benchmark's drift adjustment. These regressions are descriptive and do not control for early public-gap volatility, submission counts, or scoring-metric fixed effects; we also do not report event-time outcomes that avoid hourly carry-forward padding. The regressions use posterior medians with HC1 standard errors and do not propagate joint posterior uncertainty or intraday timestamp-imputation variation. Accordingly, the time-split associations do not establish incremental validation beyond these observable or persistent contest-level features.

The preceding diagnostics support the empirical interpretations of λ and σ, but do not establish the accuracy of the leading-order effort approximation. The incentive indices $w_k=\theta/(\sigma^2 c_k)$ are not uniformly small across the sample, limiting the applicability of Theorem 1 as a justification for this approximation. Figure 5 directly compares effort policies around the synthetic baseline, while Appendix C.2 reports a targeted parameter-recovery check using data generated under nonlinear effort. The counterfactuals below solve the nonlinear game numerically, conditional on parameter draws estimated with the leading-order policy. They remain subject to approximation error in estimation.

5.3. Counterfactual Analysis of Contest Design

We examine whether joint adjustments to prizes, deadlines, and feedback precision can preserve expected solution quality at lower cost. Let $\overline{\mathcal{B}}_\ell(\theta,T,\lambda)$ denote the numerically computed posterior

Table 6 Private-Leaderboard Volatility Diagnostics

Specification	coef.	stderr	95% CI	R^2	Spearman ρ
Full path	0.243***	0.049	[0.147, 0.339]	0.309	0.674***
Time split	0.265**	0.095	[0.078, 0.452]	0.275	0.386***
Obs.			73		

Note: The full-path row regresses $\log(1+\mathcal{V}_\ell^{path})$ on $\log(1+\hat{\sigma}_\ell)$, where $\mathcal{V}_\ell^{path}$ is the sample standard deviation of hourly private-gap increments within the contest-phase path. Positive slopes indicate that larger estimated innovation risk accompanies greater private-path variation. Hourly increments include zeros between submissions, so the outcome also reflects update frequency. Coefficients on the $\log(1+x)$ scale are not elasticities; the confidence intervals treat posterior medians as given. The time-split row uses increments ending in the final 25% and innovation risk estimated from the first 75%, retaining the full-sample horizon. Artificial initialization increments are excluded. Both regressions include an intercept and use 73 contests and posterior median estimates. Spearman ρ reports the corresponding rank association. Regression standard errors and confidence intervals use HC1 heteroskedasticity-robust covariance estimates. (1) Significance levels: *** $p<0.001$, ** $p<0.01$, * $p<0.05$. (2) OLS residual diagnostics (Full path; Time split): Omnibus = 50.904; 13.440, Jarque–Bera = 180.803; 15.401, Skewness = 2.220; 0.879, Kurtosis = 9.304; 4.405. (3) Condition Number (Full path; Time split): 3.585; 3.502.

mean of expected best-solution quality in the nonlinear game for contest ℓ, and let $\overline{\mathcal{B}}_\ell^{obs}$ denote its baseline value at the observed design. To avoid specifying the organizer's monetary valuation of quality, we minimize prize and delay costs subject to maintaining baseline quality:

$$\inf_{\theta,T,\lambda}\{\theta+D_\ell(T)\} \quad \text{subject to} \quad \overline{\mathcal{B}}_\ell(\theta,T,\lambda)\geq\overline{\mathcal{B}}_\ell^{obs}. \tag{16}$$

Within each draw, costs and the initial gap distribution remain fixed, with $S_0=\sigma/\sqrt{\lambda^{obs}}$; feedback changes affect subsequent learning. The constraint preserves posterior mean expected quality, rather than quality in every draw or realization. Appendix C.1 details the calculation and a precision-only benchmark.

Following the numerical analysis in Section 2.1, we retain quadratic delay costs, $D_\ell(T)=\delta\theta_\ell^{obs}(T/T_\ell^{obs})^2$, and vary $\delta>0$ across scenarios. This parameter measures delay cost at the observed deadline relative to the observed prize. Prizes are nonnegative. We search eight deadlines between 0.1 and 2 times the observed duration and a finite menu of positive precisions, together with full disclosure and retention of the observed feedback rule. The reported designs are the best numerically supported candidates found in this search; continuous global optimality is not established. Table 7 reports the results across the 73 contests.

Table 7 shows how the cost of delay changes the value of time relative to prize expenditure. When delay is inexpensive ($\delta=0.1$), the median selected duration rises by 50% and the median prize falls by 74.39%; the median total cost reduction is 50.39%. More time allows effort and innovation to accumulate, making a smaller prize compatible with the quality target in the selected designs. At $\delta=100$, shortening duration saves much more delay cost: the median duration falls by 25% while the median prize rises by 40.91%, with a median total cost reduction of 42.75%. Larger prizes can

Table 7 Quality-Constrained Nonlinear Design Comparisons

Panel A: Feedback choice (number of contests)

δ	Full disclosure $\lambda = \infty$	Less disclosure $\lambda < \infty$	Precision irrelevant (zero prize)
0.1	33	20	20
0.25	33	27	13
0.5	25	38	10
1	16	48	9
2	18	55	0
5	15	58	0
10	14	59	0
25	14	59	0
100	12	61	0

Panel B: Design changes and cost reductions (medians)

δ	ΔT (%)	$\Delta\theta$ (%)	Total cost reduction (%)	Additional reduction versus searched full disclosure (%)
0.1	50.00	-74.39	50.39	[1.64, 2.16]
0.25	25.00	-58.41	25.15	[4.75, 4.83]
0.5	25.00	-36.92	9.72	[4.52, 4.85]
1	0.00	-0.20	1.56	[13.58, 13.71]
2	0.00	0.00	0.14	[21.93, 22.27]
5	0.00	0.00	5.39	[29.47, 29.74]
10	0.00	0.00	5.23	[32.69, 32.76]
25	-25.00	40.91	23.95	[34.64, 34.67]
100	-25.00	40.91	42.75	[35.66, 35.67]

Note: Each scenario covers 73 contests; Panel A classifies the selected numerically supported designs, with less disclosure including retention of observed feedback. Total cost is prize plus delay cost. "Total cost reduction" is $100(1 - C^*_\ell / C^{obs}_\ell)$, where C^*_ℓ is the selected design's cost and $C^{obs}_\ell = (1+\delta)\theta^{obs}_\ell$. The last column instead reports $100(1 - C^*_\ell / C^{FD}_\ell)$, where C^{FD}_ℓ is the best supported searched full-disclosure cost at the same quality target. Thus, the two savings columns have different denominators. ΔT and $\Delta\theta$ are percentage changes from the observed deadline and prize. Negative values mean shorter duration or a smaller prize. Here T^{obs} is the constructed estimation-window length, not the official contest duration. Component medians need not refer to the same contest and cannot be combined to reconstruct the median cost reduction. All Panel B medians use 73 contests except the last column, whose subgroup is positive-prize finite-feedback designs. One contest has an unresolved full-disclosure benchmark; brackets give the range of the subgroup median when this comparison is left unrestricted, not a confidence interval. Other comparisons use the best supported full-disclosure design found on the same deadline grid. Quality calculations solve the nonlinear game conditional on the original approximate Bayesian posterior. Selected changes pass paired numerical quality diagnostics; these do not guarantee realized quality or give rigorous continuous error bounds. Unresolved alternative designs remain unexcluded. The parameter δ sets $D_\ell(T^{obs}_\ell) = \delta\theta^{obs}_\ell$.

therefore be part of a cost-saving design when earlier delivery is sufficiently valuable. These are separate cross-contest medians, not components of a single representative design.

At $\delta = 1$, the selected designs use full disclosure in 16 contests, finite feedback in 48, and zero prizes in 9; the median total saving is 1.56%. The last column isolates the extra saving from

allowing finite feedback relative to the searched full-disclosure comparator, among contests selecting a positive prize and finite feedback. It does not measure the benefit of changing precision alone, because prize and duration can change in both designs. At $\delta = 2$, for example, the median total saving is only 0.14%, even though finite feedback is selected in 55 contests. Frequent selection of a feedback regime need not imply a large benefit relative to the observed design.

The direction of the prize–duration trade-off also follows from the objective. Within a fixed feasible candidate set, a higher δ weakly lowers the cost-minimizing duration and weakly raises the associated prize.[16] No analogous ordering of precision follows, because different feedback rules can support different prize–duration combinations. A short horizon leaves less time for innovation to overturn a perceived lead, which can increase the value of preserving competitive uncertainty; a long horizon can substitute for prize expenditure through effort accumulation and selection from innovation. Organizers should therefore evaluate all three instruments together.

The calculations condition on the approximate parameter posterior, a fixed focal pair, and a finite search; unresolved alternatives remain open and large prize changes are extrapolations. Delay costs are assumed, not estimated. The precision-only comparison in Appendix C.1 separates what can be gained when prize and duration are held fixed.

6. Robustness

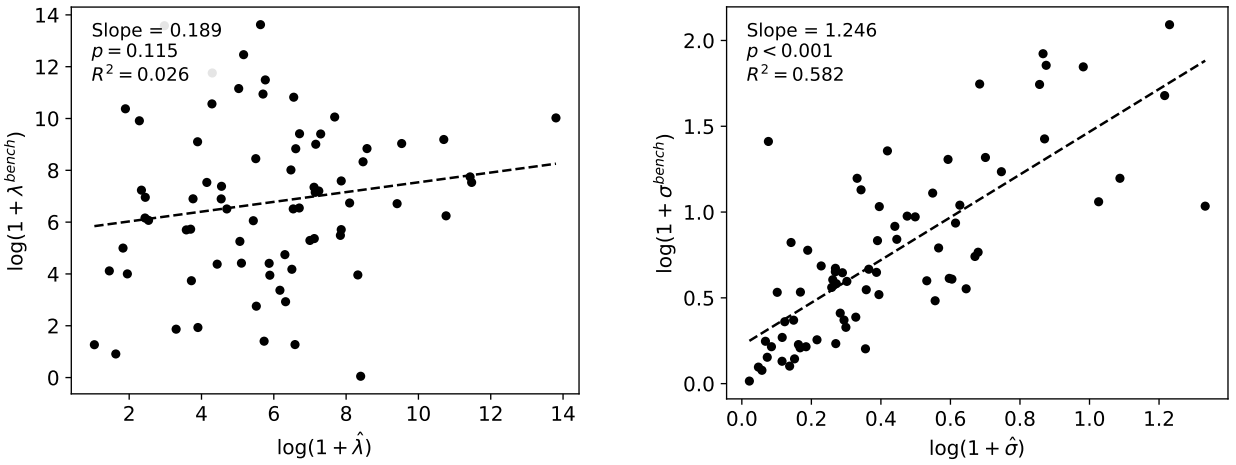
We assess whether the empirical findings are sensitive to the choice of focal contestants and the score calibration. The first exercise repeats the benchmark comparisons using alternative focal pairs; the second examines parameter stability under alternative score scales within the maintained Gaussian framework. Additional checks examine the sensitivity of the joint design comparisons to cost specifications and the predictive value of the equilibrium submission rule.

6.1. Alternative Focal Pairs

To assess whether the benchmark associations depend on the selected rivalry, we exclude baseline contestant i and search for an alternative pair among the remaining top 20 contestants, ranked by their last recorded private score. Each pair satisfies the activity and overlap requirements in Section 5.1. All 73 contests admit an alternative pair; prize conventions, the model horizon, and preprocessing rules are retained.

Figure 9 shows that the innovation-risk association remains positive and statistically significant, with a slope of 1.246 ($p < 0.001$). The signal-precision association is also positive, with a slope of 0.189, but is not statistically significant ($p = 0.115$). The evidence is therefore stronger for innovation risk than for feedback precision under alternative pair selection. Appendix C.3 reports the detailed statistics and changes in usable observation windows.

[16] Write normalized candidate cost as $p + \delta d$, where $p = \theta/\theta^{obs}$ and $d = (T/T^{obs})^2$. Optimality at $\delta_2 > \delta_1 > 0$ gives $(\delta_2 - \delta_1)(d_2 - d_1) \le 0$ and $p_2 - p_1 \ge \delta_1(d_1 - d_2) \ge 0$. This comparison assumes the same quality-feasible set at both delay costs.



(a) Signal precision (b) Innovation risk

Figure 9 Private-Leaderboard Validation with Alternative Focal Pairs

Note: Each panel contains 73 contests and uses the actual intervals between retained readouts. Panel (a) uses the elapsed-time-weighted public–private discrepancy benchmark for λ; panel (b) uses the square root of the mean squared drift-corrected private-gap increment per unit elapsed time for σ. The plotted regressions relate $\log(1+\text{benchmark})$ to $\log(1+\text{posterior median estimate})$; dashed lines are OLS fits with intercepts. HC1 heteroskedasticity-robust standard errors are used for inference. Private-path values are excluded from the estimation likelihood, although reconstructed private-score rankings are used to select the focal pairs. Each horizontal axis is the structural estimate and each vertical axis its benchmark, both on the stated log scale. Alternative pairs exclude baseline player i and satisfy the same activity criteria. The precision slope is 0.189 ($p = 0.115$); the innovation-risk slope is 1.246 ($p < 0.001$). Thus the precision association is less conclusive under this change of rivalry.

6.2. Sensitivity to Score Calibration

We next examine whether the parameter estimates are sensitive to the scale of the transformed scores. We calibrate the score-scale parameter a in equation (15) to a standard normal reference under L_1 and L_2 losses, then re-estimate all 73 contests using the baseline numerical prior specifications in the transformed units.

The cross-contest rankings of feedback precision and innovation risk remain stable under alternative score calibrations. Relative to the baseline, the Spearman correlations under the L_1 and L_2 calibrations are 0.992 and 0.999 for λ, and 0.945 and 0.996 for σ. This stability concerns parameter rankings; it does not establish robustness of the counterfactual designs. Nor does rescaling test Gaussianity, since it leaves standardized distributional shape unchanged. Appendix C.3 gives the calibration details and results. These common score-scale checks do not control for cross-contest dispersion, evaluation-metric type, or submission frequency, any of which may affect both structural estimates and private-path diagnostics. Thus, the scale exercise does not isolate the validation associations from persistent contest-level differences.

6.3. Additional Checks

Cost-prior and bound checks support a cost-saving adjustment direction in four selected scenarios, with a smaller prize cut needed in one case. They do not establish full-sample robustness or stable

optimal designs. Contest 8219's reported saving is only 0.001204%, so its economic significance is negligible.

Across both time splits, the equilibrium model has higher submission-count predictive scores than a filtered kernel with free gap and deadline coefficients, but lower scores than a constant-rate Poisson benchmark. Its advantage over a deadline-dependent Poisson benchmark is inconclusive at the reported 95% level. These results support the equilibrium restrictions relative to one filtered extension, without establishing general predictive superiority. Tables 13 and 14 in Appendix C.4 report the checks; the GitHub repository contains supporting implementation materials.

For a manager, this distinction matters: a model that connects design choices to incentives need not be the best short-run submission forecaster. The flexible filtered benchmark adds parameters and can fit noise as well as behavior; the comparison is consistent with a useful restriction, but does not identify why it helps. The constant-rate model's advantage cautions against treating every fluctuation in submission activity as evidence of strategic effort. Policy comparisons therefore draw on the maintained economic structure, rather than on demonstrated predictive dominance.

7. Conclusion

The main design lesson is that feedback changes where incentives are strongest. Precise information focuses effort on close races; noisy information can preserve incentives when contestants would otherwise regard the outcome as settled. Prize size and time remaining determine whether preserving that uncertainty is valuable to the organizer. Because the objective is the quality of the best terminal solution, both induced effort and innovation matter. The model therefore supports joint evaluation of prize, deadline, and feedback, rather than a universal recommendation for more or less disclosure.

This paper develops a dynamic contest model and Bayesian structural estimation framework linking noisy feedback, latent performance, and effort. Under Proposition 1's conditions, finite precision can strictly outperform full disclosure. Across 73 Kaggle contests, innovation-risk estimates are positively associated with private-path volatility, and precision estimates with public–private agreement, including in time-split diagnostics. These descriptive associations condition on ex post pair selection. Submission predictions favor equilibrium over the flexible filtered kernel, favor the constant-rate benchmark over equilibrium, and are inconclusive against the deadline-dependent benchmark.

Several limitations qualify the empirical findings. Latest-submission scores may differ from prize-relevant best-so-far scores and contestants' common information. Carry-forward gaps may reuse stale discrepancies, whereas the likelihood assumes fresh errors and inverse-elapsed-time measurement variance. Validation regressions omit controls for early public-gap volatility, submission

counts, and metric effects; use hourly rather than event-time outcomes; and do not propagate posterior or timestamp-imputation uncertainty. Persistent contest-level differences may therefore contribute to the associations.

Nonlinear design comparisons remain conditional on the approximate parameter posterior and finite numerical search. Synthetic experiments show weak cost–submission-scale separation; sensitivity checks cover only selected cases. Asymmetric empirical designs do not isolate Proposition 1's symmetric mechanism. The identification result excludes data-dependent initialization. Numerical regime boundaries remain provisional: 1,179 of 14,553 points are unresolved, including 926 with potentially competitive failed candidates. Future work should assess measurement assumptions and improve numerical coverage.

Appendix A: Proofs

Proof of Theorem 1 Fix a finite $\lambda > 0$. Put $a_t = \lambda S_t^2$, $\alpha_k = \sigma^2 w_k$, and $\alpha_* = \max_k \alpha_k$. Write $U^k = V^k/\theta$, $\mathfrak{p} = U^i_y$, and $\mathfrak{n} = -U^j_y$. In reverse time $\tau = T - t$, the differentiated HJB equations are

$$\mathfrak{p}_\tau = \tfrac{a_{T-\tau}}{2}\mathfrak{p}_{yy} + \partial_y(\alpha_i \mathfrak{p}^2/2 - \alpha_j \mathfrak{p}\mathfrak{n}), \qquad \mathfrak{n}_\tau = \tfrac{a_{T-\tau}}{2}\mathfrak{n}_{yy} + \partial_y(\alpha_i \mathfrak{p}\mathfrak{n} - \alpha_j \mathfrak{n}^2/2),$$

with $\mathfrak{p}(y,0) = \mathfrak{n}(y,0) = G_{S_T}(y)$, where G_q is the centered Gaussian density of variance q.

Existence. We apply the integral-equation version of the abstract Cauchy–Kowalevski theorem in Nishida (1978, Chapter III, Remark 2.3). Use the analytic norms

$$\|f\|_\varpi = (2\pi)^{-1} \int_{\mathbb{R}} e^{\varpi|\xi|} |\widehat{f}(\xi)|\, d\xi, \qquad \widehat{f}(\xi) = \int_{\mathbb{R}} e^{-i\xi y} f(y)\, dy.$$

For $0 < \varpi' < \varpi$, these satisfy $\|fg\|_\varpi \le \|f\|_\varpi \|g\|_\varpi$ and $\|f_y\|_{\varpi'} \le \|f\|_\varpi/(\varpi - \varpi')$. Let $P_{t,s}$ be Gaussian convolution of variance $\int_t^s a_u\, du$. It is a contraction in every such norm. For $\mathbf{h} = (\mathfrak{p}, \mathfrak{n})$, let $\mathscr{N}(\mathbf{h})$ denote the two nonlinear terms above and use the sum norm for pairs. The gradient equations have the integral form

$$\mathbf{h}(\tau) = P_{T-\tau,T}\mathbf{h}(0) + \int_0^\tau P_{T-\tau,T-s}\mathscr{N}(\mathbf{h}(s))\, ds. \tag{17}$$

On a radius-R ball,

$$\|\mathscr{N}(\mathbf{h}) - \mathscr{N}(\widehat{\mathbf{h}})\|_{\varpi'} \le \frac{C\alpha_* R}{\varpi - \varpi'} \|\mathbf{h} - \widehat{\mathbf{h}}\|_\varpi, \qquad \|\mathscr{N}(\mathbf{h})\|_{\varpi'} \le \frac{C\alpha_* R^2}{\varpi - \varpi'}.$$

Fix $\varpi_0 > \varpi_1 > 0$ and $R > 2\|\mathbf{h}(0)\|_{\varpi_0}$; the Gaussian data make this choice independent of α_*. After rescaling time by $\upsilon = \alpha_* \tau$, the estimates above have constants independent of α_*, since $\alpha_k/\alpha_* \le 1$ and the propagators remain contractions. The cited theorem thus gives an existence interval $0 \le \upsilon < \upsilon_0$ and a bound in the ϖ_1 norm, both independent of α_*. Because $\alpha_* \le \sigma^2(w_i + w_j)$, taking $w_i + w_j$ sufficiently small ensures $\alpha_* T < \upsilon_0$. The solution then exists on $[0,T]$ with a fixed positive analytic radius. Cauchy's estimate on any smaller strip gives uniform bounds on all spatial derivatives used below.

Equilibrium verification. Returning to $t = T - \tau$, with $\mathfrak{b} = \alpha_i \mathfrak{p} - \alpha_j \mathfrak{n}$, the gradients satisfy

$$\mathfrak{p}_t + \tfrac{a_t}{2}\mathfrak{p}_{yy} + \mathfrak{b}\mathfrak{p}_y - \alpha_j \mathfrak{n}_y \mathfrak{p} = 0, \qquad \mathfrak{n}_t + \tfrac{a_t}{2}\mathfrak{n}_{yy} + \mathfrak{b}\mathfrak{n}_y + \alpha_i \mathfrak{p}_y \mathfrak{n} = 0.$$

Their bounded coefficients and strictly positive Gaussian terminal data give $\mathfrak{p}, \mathfrak{n} > 0$ by Feynman–Kac. Bounded drift and reaction coefficients also preserve Gaussian tail bounds. Integrating the reverse-time conservative equations over $\mathbb{R}$ therefore produces no boundary terms and preserves $\int \mathfrak{p}\, dy = \int \mathfrak{n}\, dy = 1$. Consequently, $U^i(y,t) = \int_{-\infty}^y \mathfrak{p}(z,t)\, dz$ and $U^j(y,t) = \int_y^\infty \mathfrak{n}(z,t)\, dz$ are bounded values with the required terminal and spatial limits. Integrating the gradient equations fixes the possible constants of integration by these spatial limits and gives the normalized HJB system $U^k_t + (a_t/2)U^k_{yy} + F_k(U) = 0$, where

$$F_i(U) = \alpha_i (U^i_y)^2/2 + \alpha_j U^i_y U^j_y, \qquad F_j(U) = \alpha_j (U^j_y)^2/2 + \alpha_i U^i_y U^j_y.$$

The positive policies $m_i = \alpha_i \mathfrak{p}$ and $m_j = \alpha_j \mathfrak{n}$ attain the constrained HJB maxima. For any admissible deviation q_k, Itô's formula and completion of the square give

$$\mathbb{E}_{y,t}\left[\Pi_k(\tilde{y}_T) - \int_t^T \frac{c_k q_{k,s}^2}{2}\, ds\right] = V^k(y,t) - \frac{c_k}{2}\mathbb{E}_{y,t}\int_t^T [q_{k,s} - m_k(\tilde{y}_s, s)]^2\, ds. \tag{18}$$

The stochastic integral has zero expectation by bounded derivatives and admissibility. The last term in (18) is nonpositive and vanishes for $q_k = m_k$. Hence the policies are mutual best responses and form an interior Markov-perfect equilibrium.

Leading-order effort. Direct differentiation, using $Q_t' = -a_t$ and $Q_T = S_T$, verifies $U_i^{(0)} = \Phi(y/\sqrt{Q_t})$, $U_j^{(0)} = 1 - U_i^{(0)}$ as the zero-incentive HJB solution. The exact values satisfy

$$U^k(t) = U_k^{(0)}(t) + \int_t^T P_{t,s} F_k(U(s))\, ds. \tag{19}$$

Indeed, $Q_t - Q_s = \int_t^s a_u\, du$, so propagating the terminal density G_{S_T} from T to t gives G_{Q_t}. On a smaller fixed analytic strip, the uniform existence bound controls $\mathscr{N}(\mathbf{h})$ by $C\alpha_*$. Heat contraction and integration over an interval of length at most T therefore give

$$\sup_{y,t}\{|\mathfrak{p}(y,t) - G_{Q_t}(y)| + |\mathfrak{n}(y,t) - G_{Q_t}(y)|\} \le CT\alpha_* = O(w_i + w_j).$$

Since $m_k = \alpha_k \mathfrak{p}_k = \sigma^2 w_k \mathfrak{p}_k$, multiplying by $\sigma^2 w_k$ gives the stated expansion. Finally, $Q_t' = -a_t < 0$, $Q_T = S_T$, and $Q_0 = S_0 + \sigma^2 T$, so $S_T \le Q_t \le S_0 + \sigma^2 T$; the Gaussian factor is therefore uniformly bounded. This proves (5) uniformly in state and time. □

Proof of Proposition 1 The utility assumption implies $u'(B) \le Ce^{-\zeta B}$ for all sufficiently large B, so u is bounded above and $Bu'(B) \to 0$. Write $\alpha = \theta/c$, $b_0 = \sqrt{S_0/(2\pi)}$, and $b(T) = \sqrt{(S_0 + \sigma^2 T)/(2\pi)}$; put $Q_t = S_t + \sigma^2(T - t)$.

Part (i): finite feedback strictly dominates full disclosure. Fix $\bar{\lambda}, \bar{\alpha} > 0$. The analytic existence argument in the proof of Theorem 1 is uniform for $0 < \sigma \le \bar{\sigma}$ and sufficiently small T. Indeed, $S_t \ge S_0/(1 + \bar{\lambda} S_0 t)$ keeps the Gaussian terminal data bounded on a common analytic strip. With $\alpha_i = \alpha_j = \bar{\alpha}$, the gradient integral equation gives $m_k(y,t) = \bar{\alpha} G_{Q_t}(y) + O(T)$ uniformly in state and time.

Under zero effort, $\tilde{y}_t$ is centered Gaussian with variance $Q_0 - Q_t$. Gaussian convolution therefore gives $\mathbb{E}[G_{Q_t}(\tilde{y}_t)] = G_{Q_0}(0)$. Moreover, the leading terms of m_i and m_j coincide, so $m_i - m_j = O(T)$; integrating this drift over a horizon of length T changes the perceived-gap path by $O(T^2)$. The Lipschitz Gaussian kernels give the same order of error after taking expectations. Substitution in (27) first gives

$$\mathcal{B}(0, c\bar{\alpha}, T, \bar{\lambda}; S_0) = b(T) + \bar{\alpha} G_{Q_0}(0) T + o(T).$$

Since $G_{Q_0}(0) = 1/\sqrt{2\pi(S_0 + \sigma^2 T)}$, $b'(0) = \sigma^2/\sqrt{8\pi S_0}$, and $D'(0) = 0$, Taylor expansion gives, uniformly in $0 < \sigma \le \bar{\sigma}$,

$$u(\mathcal{B}) - D(T) = u(b_0) + \frac{u'(b_0)(\bar{\alpha} + \sigma^2/2)}{\sqrt{2\pi S_0}} T + o(T).$$

The displayed coefficient is bounded below by $u'(b_0)\bar{\alpha}/\sqrt{2\pi S_0} > 0$. Uniformity of the remainder lets us choose a single short $\bar{T} > 0$ and $\varepsilon_0 > 0$, independent of $\sigma \in (0, \bar{\sigma}]$, such that the payoff before deducting the prize is at least $u(b_0) + \varepsilon_0$. The prize equals $\bar{\theta} = c\bar{\alpha}$. Hence, for $c < c_F := \varepsilon_0/(4\bar{\alpha})$, this finite-precision candidate has net payoff greater than $u(b_0) + 3\varepsilon_0/4$.

Use the full-disclosure benchmark in Appendix B.1, with $w = \alpha/\sigma^2$.

Couple the equilibrium scores with their zero-effort counterparts using the same Brownian paths. The maximum function is one-Lipschitz in the Euclidean norm, and $(\int_0^T m_k dt)^2 \le T\int_0^T m_k^2 dt$. Conditional Cauchy–Schwarz and (26) therefore give, for $Y_0 \sim \mathcal{N}(0, S_0)$,

$$\mathcal{B}(0,\theta,T,\infty;S_0) - b(T) \le \sqrt{2\alpha T}\,\mathbb{E}\sqrt{1 - f(d(Y_0,0))}.$$

For $d>0$ and $x = wd/2$,

$$\frac{wf'(d)K(d)}{K'(d)} = 1 + (x\tanh x - 1)\operatorname{sech}^2 x \le 2,$$

using $\cosh^2 x \ge 1 + x^2 \ge 2x$. Because $f(1)=1$ and $K, K' > 0$ on $(0,1]$, the ratio bound implies

$$1 - f(d) = \int_d^1 f'(s)ds \le \frac{2}{w}\int_d^1 \frac{K'(s)}{K(s)}ds = \frac{2}{w}\log\frac{K(1)}{K(d)}.$$

Equation (25) and symmetry in y therefore yield

$$1 - f(d(y,0)) \le -\frac{2}{w}\log\left|2\Phi\left(\frac{y}{\sigma\sqrt{T}}\right) - 1\right|.$$

The constant $C := 2\int_{\mathbb{R}} \sqrt{-\log|2\Phi(z)-1|}\,dz$ is finite: its integrand is $O(\sqrt{|\log|z||})$ near zero and has Gaussian decay in the tails. Since $w = \alpha/\sigma^2$, the preceding bounds imply

$$\mathcal{B} - b(T) \le 2\sigma\sqrt{T}\,\mathbb{E}\sqrt{-\log|2\Phi(Y_0/(\sigma\sqrt{T})) - 1|}.$$

Changing variables $Y_0 = \sigma\sqrt{T}z$ and bounding its Gaussian density by $1/\sqrt{2\pi S_0}$ gives

$$\mathcal{B}(0,\theta,T,\infty;S_0) \le b(T) + \frac{C\sigma^2 T}{\sqrt{2\pi S_0}}. \tag{20}$$

This also holds at $\theta = 0$ or $T = 0$.

Concavity of the square root gives $b(T) \le b_0 + \sigma^2 T/(2\sqrt{2\pi S_0})$. Concavity of u, followed by (20) and omission of the nonpositive term $-\theta$, gives, for every full-disclosure prize and deadline,

$$u(\mathcal{B}) - \theta - D(T) \le u(b_0) + \mathsf{a}_{\text{out}}\sigma^2 T - D(T), \qquad \mathsf{a}_{\text{out}} = \frac{(C+1/2)u'(b_0)}{\sqrt{2\pi S_0}}.$$

Since $D'(T)/T$ is nondecreasing and converges to $D''(0)$ as $T \downarrow 0$, $D'(T) \ge D''(0)T$. Integrating from zero gives $D(T) \ge D''(0)T^2/2$, so completing the square yields

$$\sup_{T\ge 0}\{\mathsf{a}_{\text{out}}\sigma^2 T - D(T)\} \le \frac{\mathsf{a}_{\text{out}}^2\sigma^4}{2D''(0)}.$$

Choose $\sigma_F \le \min\{\bar\sigma, 1\}$ so that the last bound is strictly below $\varepsilon_0/2$. Every full-disclosure payoff is then below $u(b_0) + \varepsilon_0/2$, whereas the constructed candidate is above $u(b_0) + 3\varepsilon_0/4$. Thus the candidate strictly beats every full-disclosure design for $0 < c < c_F$ and $0 < \sigma < \sigma_F$.

When $\theta = 0$ or $T = 0$, the payoff is precision-independent, so the same payoff is attained by setting $\lambda = \infty$; these boundary designs are already included in the full-disclosure comparison. Hence no joint optimizer can have zero prize, zero deadline, or full disclosure. Every joint optimizer therefore lies in $(0,\infty)^3$. This proves part (i) without asserting that a joint optimizer exists.

Part (ii): positive-prize full disclosure. Take (c, σ, T_0, b, V_0) from Lemma 3; $c < c_e$ and $\sigma > 1 \ge \sigma_F$. Let $u_\infty := \lim_{B\to\infty} u(B) < \infty$. Any full-disclosure design with payoff at least V_0 satisfies $\theta + D(T) \le u_\infty - V_0$. Since

D is strictly increasing and unbounded, such designs lie in a compact prize–deadline rectangle. The explicit policies are continuous before $T-\varepsilon$, and (26) controls the remaining score change by $O(\sqrt{\varepsilon})$, uniformly on bounded α, including zero prize and zero deadline. It follows by first taking parameter limits on $[0, T-\varepsilon]$ and then sending $\varepsilon \downarrow 0$ that the objective extends continuously to this rectangle. It therefore attains its maximum (Berge 1963, Chapter 6).

Rescaling (30) at $T = T_0$ gives

$$\partial_\theta \{u(\mathcal{B}) - \theta - D(T_0)\}\big|_{0+} = \frac{T_0 u'(b)}{c\sqrt{2\pi(S_0 + \sigma^2 T_0)}} - 1 = \frac{c_e}{c} - 1 > 0. \tag{21}$$

The positive right derivative means that a sufficiently small prize at T_0 yields payoff strictly above V_0; hence the attained full-disclosure maximum exceeds V_0. Lemma 3 excludes every positive-prize finite-feedback design; zero-prize and zero-deadline designs yield at most V_0 and $u(b_0)$. Consequently this is a joint maximum, and every joint maximizer satisfies $\theta^* > 0$, $T^* > 0$, and $\lambda^* = \infty$.

Part (iii): high-cost prize exit. Fix $\sigma > 0$. Let T_0 maximize $u(b(T)) - D(T)$ and write $V_0 = u(b(T_0)) - D(T_0)$, $\Gamma = \lim_{B\to\infty} u(B) - u(b_0) > 0$. Choose $\bar{T}$ from $D(\bar{T}) = \Gamma$. Every design with payoff at least V_0 has $\theta + D(T) \le \Gamma$: indeed, $u(\mathcal{B}) - \theta - D(T) \ge V_0 \ge u(b_0)$ and $u(\mathcal{B}) \le \lim_{B\to\infty} u(B)$. Hence $T \le \bar{T}$ and $\alpha \le \Gamma/c \to 0$ uniformly. We bound the output gain directly, without ranking feedback precision.

For $\lambda S_0^2 < .72\sigma^2$, $S_T \ge S_0$ and $\alpha T/S_T \le \alpha\bar{T}/S_0 \to 0$. Thus (34) applies uniformly for arbitrarily small positive precision. For $\lambda S_0^2 \ge .72\sigma^2$, use (33) with $w = \alpha/\sigma^2 \to 0$. For full disclosure, (26) and $\log\cosh z \le z^2/2$ give

$$1 - f(d) \le \frac{2}{w}\log\cosh(w/2) \le \frac{w}{4}, \qquad \mathbb{E}\int_0^T (m_i^2 + m_j^2)\, dt \le \frac{\alpha^2}{2\sigma^2}.$$

Coupling with the zero-effort scores and applying Cauchy–Schwarz yields $\mathcal{B} - b(T) \le \alpha\sqrt{T}/(\sqrt{2}\sigma)$. In the first precision region, (34) and $S_T \ge S_0$ bound both its linear and quadratic terms by a constant times α, uniformly for $T \le \bar{T}$. In the other region the competitive-output bound does the same, while the displayed full-disclosure estimate covers the endpoint $\lambda = \infty$. Consequently there is $C_\sigma < \infty$, independent of c, T, λ, such that, for all sufficiently small $\alpha > 0$,

$$\mathcal{B} - b(T) \le C_\sigma \alpha, \qquad 0 \le T \le \bar{T}, \quad 0 < \lambda \le \infty.$$

Because T_0 maximizes $u(b(T)) - D(T)$, concavity and $u'(b(T)) \le u'(b_0)$ give, for sufficiently large c,

$$u(\mathcal{B}) - c\alpha - D(T) \le V_0 + \alpha\,[u'(b_0)C_\sigma - c] < V_0.$$

No positive-prize design can therefore compete. The feasible design $(\theta, T) = (0, T_0)$ attains V_0. Moreover, the right derivative of $u(b(T)) - D(T)$ at zero equals $u'(b_0)\sigma^2/(2\sqrt{2\pi S_0}) > 0$, so $T_0 > 0$; all precision choices give this same payoff. This proves part (iii). □

Proof of Proposition 2 This proof uses the matched left-point convention stated in the proposition: on grid interval q, the drift increment is $\Delta_{\text{num}}[m_i^{(1)}(s_{q-1}) - m_j^{(1)}(s_{q-1})]$, and the Poisson intensities are held at the same interval-start values. Thus the cumulative drift and cumulative intensity difference use the same grid sum. Hold the prescribed initial public input fixed under both parameter vectors, with the initial latent law in equation (10). Suppose two interior vectors Θ and Θ' induce the same joint distribution of submission

counts and public readouts. Identification compares all possible data histories, not different observations along a single realized history.

Let E_ℓ be the event that all earlier counts are zero and $n_\ell^i + n_\ell^j > 0$, so the first readout occurs at s_ℓ. Before this readout, the recursion carries forward the fixed initial public input and the intensities are deterministic conditional on the parameters. Finite, positive intensities make each preceding zero-count probability and the current nonzero-total-count probability strictly positive. Their finite product gives $\Pr_\Theta(E_\ell) > 0$ and $\Pr_{\Theta'}(E_\ell) > 0$.

Conditional on all earlier counts being zero, contestant $k \in \{i, j\}$ has zero-count probability $\exp\{-\Delta_{\text{num}} \tau_k^{(1)}(s_{\ell-1})\}$ in interval ℓ. Equality of the joint laws implies equality of these conditional probabilities, so both intensity sequences coincide along the no-previous-submission history. Count draws along this history are independent of the initial latent uncertainty and innovation shocks; conditioning on E_ℓ therefore leaves their joint distribution unchanged.

To see the first implication explicitly, equality of the conditional zero-count probabilities and injectivity of the exponential map give $\tau_k^{(1)}(s_{\ell-1}; \Theta) = \tau_k^{(1)}(s_{\ell-1}; \Theta')$ recursively for each k and ℓ. The additional condition that the total count in interval ℓ is positive depends only on the Poisson draws at that interval. Because its intensity is already fixed along this history, it does not tilt the independent Gaussian state and measurement shocks.

Along this history, define the cumulative intensity difference

$$\mathscr{I}(s_\ell) := \Delta_{\text{num}} \sum_{q=1}^{\ell} \left[\tau_i^{(1)}(s_{q-1}) - \tau_j^{(1)}(s_{q-1}) \right]. \tag{22}$$

The identified intensity sequences make $\mathscr{I}(s_\ell)$ identical under Θ and Θ'. Under the matched left-point discretization, the accumulated drift on E_ℓ is exactly $\mathscr{I}(s_\ell)/r$, since drift and submission intensities use the same interval-start effort values. The independent Gaussian initial uncertainty and innovation give latent variance $S_0 + \sigma^2 s_\ell$. Since $h_1 = s_\ell$, the fresh independent measurement error adds variance $1/(\lambda s_\ell)$. Thus

$$\hat{y}_{s_\ell} \mid E_\ell, \Theta \sim \mathcal{N}\left(\mu_0 + \frac{\mathscr{I}(s_\ell)}{r}, S_0 + \sigma^2 s_\ell + \frac{1}{\lambda s_\ell} \right). \tag{23}$$

Since E_ℓ has positive probability under both vectors, equality of the joint laws implies equality of these conditional Gaussian distributions. No future submission times are conditioned on.

Equality of the first-readout variances under Θ and Θ', multiplied by s_ℓ, gives

$$(\sigma^2 - \sigma'^2) s_\ell^2 + (S_0 - S_0') s_\ell + \left(\lambda^{-1} - \lambda'^{-1} \right) = 0.$$

For $\ell = 1, 2, 3$, the three distinct positive times s_ℓ are roots of the same polynomial of degree at most two. The polynomial is therefore identically zero. Its constant, linear, and quadratic coefficients give $\lambda^{-1} = \lambda'^{-1}$, $S_0 = S_0'$, and $\sigma^2 = \sigma'^2$, respectively. Interior positivity then yields $\lambda = \lambda'$ and $\sigma = \sigma'$.

Equality of the first-readout means gives

$$\mu_0 + \frac{\mathscr{I}(s_\ell)}{r} = \mu_0' + \frac{\mathscr{I}(s_\ell)}{r'}.$$

At s_1, equation (14) has a common strictly positive kernel for the two contestants and differs only through $1/c_k$. Since $r, \theta > 0$ and $c_i \neq c_j$, this implies

$$\mathscr{I}(s_2) - \mathscr{I}(s_1) = \Delta_{\text{num}} \left[\tau_i^{(1)}(s_1) - \tau_j^{(1)}(s_1) \right] \neq 0.$$

Subtracting the mean equalities at these two times gives

$$[\mathscr{I}(s_2) - \mathscr{I}(s_1)] \left(r^{-1} - r'^{-1} \right) = 0,$$

so positivity of r, r' gives $r = r'$. Substitution into either mean equality then gives $\mu_0 = \mu_0'$. Finally, the identified initial intensities and $\tilde{y}_0 = \mu_0$ recover both costs directly:

$$c_k = \frac{r\theta}{\tau_k^{(1)}(0)\sqrt{2\pi(S_0 + \sigma^2 T)}} \exp\left\{ -\frac{\mu_0^2}{2(S_0 + \sigma^2 T)} \right\}, \qquad k \in \{i, j\}.$$

Thus $\Theta = \Theta'$, establishing local identification. At $c_i = c_j$, the two leading-order efforts coincide, so their difference and hence $\mathscr{I}(s_\ell)$ are zero along every public history. A common rescaling of (r, c_i, c_j) leaves each ratio r/c_k, and therefore both count intensities, unchanged; because the effort difference remains zero, it also leaves every readout prediction unchanged. The full vector is therefore not identified at equality. □

Appendix B: Auxiliary Results

B.1. Lemmas for Joint Design

This section establishes the equilibrium and output comparisons used in Proposition 1. We first record the full-disclosure benchmark and the output identities, then establish positive-precision equilibrium existence and compare precision at weak incentives. Finally, we bound competing prize–deadline choices to obtain the global payoff comparison supporting part (ii).

Full-disclosure benchmark. For full disclosure, specialize the strategies and values in Ryvkin (2022, Propositions 1 and 3, equations (18) and (23)) to $c_i = c_j = c$ and $V = \theta > 0$. Put $\alpha = \theta/c$ and $w = \alpha/\sigma^2$. The normalized values $U_k = V^k/\theta$ satisfy $U_i + U_j = f(d)$ and $U_i - U_j = d$, where

$$f(d) = 1 + \frac{2}{w} \log \frac{\cosh(wd/2)}{\cosh(w/2)}, \qquad K(d) = \frac{d + \sinh(wd)/w}{2}, \tag{24}$$

and $d(y,t)$ is determined by

$$K(d(y,t)) = K(1) \left[2\Phi\left(\frac{y}{\sigma\sqrt{T-t}} \right) - 1 \right]. \tag{25}$$

Ryvkin's change of variables can be written as $\mathsf{r}_{\mathrm{R}} = \tanh(wd/2)$. Since the total prize is θ, summing the cited value functions gives

$$\mathbb{E}_{y,t} \int_t^T (m_i^2 + m_j^2)\, ds = 2\alpha[1 - f(d(y,t))]. \tag{26}$$

For finite precision, put $Q_t = S_t + \sigma^2(T-t)$ and $\beta(y,q) = \mathbb{E}|\mathcal{N}(y,q)|/2$. Since $\beta_y = \Phi(y/\sqrt{q}) - 1/2$, $\beta_q = \beta_{yy}/2$, and $Q_t' = -\lambda S_t^2$, the drift from $\beta_q Q_t'$ cancels the quadratic-variation term from $d\tilde{y}_t = (m_i - m_j)dt + \sqrt{\lambda} S_t\, d\widehat{B}_t$. Also, $\max\{x_T^i, x_T^j\} = (x_T^i + x_T^j)/2 + |y_T|/2$ and $\mathbb{E}[|y_T|/2 \mid I_T] = \beta(\tilde{y}_T, S_T)$. Applying Itô's formula to the mean score and to $\beta(\tilde{y}_t, Q_t)$, then using $\beta(0, Q_0) = \sqrt{(S_0 + \sigma^2 T)/(2\pi)}$, gives

$$\mathcal{B} = b(T) + \mathbb{E} \int_0^T \left\{ \frac{m_i + m_j}{2} + (m_i - m_j) \left[\Phi(\tilde{y}_t/\sqrt{Q_t}) - \frac{1}{2} \right] \right\} dt, \qquad b(T) = \sqrt{\frac{S_0 + \sigma^2 T}{2\pi}}. \tag{27}$$

The Gaussian flux estimate used below is

$$G_q(y) - \int_0^q (|G_{q-v}'| * G_v^2)(y)\, dv = \frac{2}{\pi} \int_0^\infty \frac{G_q(\sqrt{2}|y|\cosh h)}{2\cosh^2 h - 1}\, dh \geq 0. \tag{28}$$

The identity follows by taking the Laplace transform in q.

LEMMA 1 **(Positive-precision equilibrium).** *For $\theta, T, \sigma, S_0, c_i, c_j > 0$ and $0 < \lambda < \infty$, the normalized HJB system has a unique bounded classical solution with bounded spatial gradients. Its feedback policies form an interior Markov-perfect equilibrium.*

Here normalization divides each value function by the prize θ. The lemma ensures that every fixed finite-precision design used in the analysis has a well-defined equilibrium in the stated regularity class, even when incentive indices are large. "Interior" means that each contestant exerts strictly positive effort at every finite perceived gap before the deadline. It neither guarantees accuracy of the first-order policy nor asserts uniqueness among equilibria outside this bounded classical Markov class.

Proof of Lemma 1 Use $U^k = V^k/\theta$, $a_t = \lambda S_t^2$, and $\alpha_k = \theta/c_k$ as in the proof of Theorem 1. For fixed positive finite precision, a_t is smooth and bounded above and away from zero on $[0, T]$. The terminal functions $g_i(y) = \Phi(y/\sqrt{S_T})$ and $g_j = 1 - g_i$ are smooth and bounded, with bounded derivatives.

We apply Theorem 2.8 of Jackson (2023) to $U_t^k + (a_t/2)U_{yy}^k + F_k(U_y) = 0$ on $[0, T] \times \mathbb{R}$. For $z = (z_i, z_j)$, the required structure follows from

$$F_k(z) = z_k(\alpha_i z_i + \alpha_j z_j) - \frac{\alpha_k}{2} z_k^2, \qquad k \in \{i, j\}.$$

Thus the common-drift term in the theorem's decomposition is $\mathfrak{b}_0(z) = \alpha_i z_i + \alpha_j z_j$, while $\mathfrak{b}^k(z) = -\alpha_k z_k^2/2$. The driver is smooth and satisfies $|F_k(z)| \le C|z_k||z|$, which verifies the quadratic-growth and local Lipschitz conditions. For the a-priori boundedness condition, take the coordinate vectors $\{\pm Le_i, \pm Le_j\}$ with $L \ge \max\{\alpha_i, \alpha_j\}$. They positively span $\mathbb{R}^2$. For $+Le_k$ the required upper bound is immediate because $\mathfrak{b}^k \le 0$; for $-Le_k$ it follows from

$$(-Le_k)^T b(z) = \frac{L\alpha_k}{2} z_k^2 \le \frac{L^2}{2} z_k^2 = \frac{1}{2}|(-Le_k)^T z|^2.$$

The scalar diffusion coefficient is uniformly elliptic and smooth in time, while F is time independent and the Gaussian terminal functions are C^{2,β_H} for every $\beta_H \in (0, 1)$. These observations verify the remaining regularity hypotheses of the cited classical-solution theorem. The cited theorem therefore gives a unique bounded classical solution with bounded spatial gradients.

Parabolic regularity gives bounded second spatial derivatives. As in the proof of Theorem 1, Feynman–Kac with terminal data $U_y^i = -U_y^j = G_{S_T} > 0$ gives $U_y^i > 0$ and $U_y^j < 0$. The policies $m_i = \alpha_i U_y^i$ and $m_j = -\alpha_j U_y^j$ therefore form an interior Markov-perfect equilibrium by (18). □

LEMMA 2 **(Full disclosure at weak incentives).** *Suppose $c_i = c_j = c > 0$. There is $w_* > 0$ such that, whenever $S_0/(\sigma^2 T) \in [99/2, 198]$ and $0 < \theta/(c\sigma^2) < w_*$, the selected symmetric equilibria satisfy*

$$\mathcal{B}(0, \theta, T, \infty; S_0) > \mathcal{B}(0, \theta, T, \lambda; S_0) \qquad \text{for every } 0 < \lambda < \infty. \tag{29}$$

This comparison holds prize and deadline fixed and ranks expected best-solution quality. The ratio $S_0/(\sigma^2 T)$ measures initial uncertainty relative to innovation variance accumulated over the whole contest; the stated interval is a sufficient condition used in the proof. The weak-incentive threshold works for every finite precision, including arbitrarily informative feedback. Thus the result is stronger than a comparison with any fixed finite menu. At zero prize, all precision choices give the same output; their first-order output gains also coincide. The strict ranking comes from higher-order equilibrium responses, so the leading-order effort formula alone cannot establish it. This weak-incentive result does not contradict Proposition 1(i), where low costs and low innovation risk need not produce weak incentives.

Proof of Lemma 2 Measure time in units of T and scores in units of $\sigma\sqrt{T}$. Write $w=\theta/(c\sigma^2)$, $s=S_0/(\sigma^2T)$, $\Lambda=\lambda\sigma^2T^2$, and $q=1+s$. In these units, $S_t'=1-\Lambda S_t^2$ and $Q_t=S_t+1-t$. Denote output by $\widehat{\mathcal{B}}(w,s,\Lambda)$. All estimates below are uniform for $s\in[99/2,198]$. Equation (27) gives the common first variation $G_q(0)$ at zero prize. For full disclosure, the explicit equilibrium gives the same coefficient:

$$\widehat{\mathcal{B}}(w,s,\infty)=\sqrt{q/(2\pi)}+wG_q(0)+O(w^2). \tag{30}$$

Bounded precisions. For every finite $\Lambda_{\max}$, the positive lower bound on S_T makes the analytic construction in Theorem 1 uniform on $[99/2,198]\times(0,\Lambda_{\max}]$. The coefficients and expansion extend continuously to the limiting endpoint $\Lambda=0$, which is used only to obtain a uniform comparison for arbitrarily small positive precision. Indeed, the analytic-norm proof uses only contraction of the Gaussian propagators and a uniform analytic bound on the terminal density; it does not divide by the diffusion coefficient. Its contraction radius and remainder constant are therefore uniform for $\Lambda\in[0,\Lambda_{\max}]$. The Riccati solution and every coefficient in the expansion depend continuously on (s,Λ) on this compact set. The gradient mild equations and the output identity give, uniformly,

$$\widehat{\mathcal{B}}(w,s,\infty)-\widehat{\mathcal{B}}(w,s,\Lambda)=w^2\Delta_2(s,\Lambda)+O(w^3).$$

The full-disclosure formulas (24)–(25) are analytic in w at zero. With $\mathsf{V}_t=s+t-S_t$, Gaussian integration gives $\Delta_2(s,\Lambda)=b_2(s,\infty)-b_2(s,\Lambda)$, where

$$b_2(s,\Lambda)=\int_0^1\int_t^1\mathscr{K}_q(\mathsf{V}_t,\mathsf{V}_r)\,dr\,dt,\qquad \mathscr{K}_q(A,B)=\frac{A}{\pi\sqrt{2\pi}\sqrt{q-B}(q+B)\sqrt{q^2+qB-2A^2}}.$$

For $\Lambda>0$, $\varrho_t=\mathsf{V}_t/(s+t)$ increases strictly from zero and stays below one: its derivative has the opposite sign to $E_t=(s+t)S_t'-S_t<0$, since either $S_t'\le 0$ or $E_0=-\Lambda s^3<0$ and $E_t'=-2\Lambda(s+t)S_tS_t'<0$. The kernel is strictly increasing in its first argument because

$$\partial_A\log\mathscr{K}_q(A,B)=\frac{1}{A}+\frac{2A}{q^2+qB-2A^2}>0.$$

It also increases along radial rays: for $x=B/q$ and $z=A/q$,

$$\frac{A\mathscr{K}_A+B\mathscr{K}_B}{\mathscr{K}_q}=\frac{1+x^3+xz^2-3x^2z^2}{(1-x^2)(1+x-2z^2)}>0,\qquad 0\le z\le x<1.$$

The numerator is affine in z^2, so its minimum is either $1+x^3$ or $(1-x)(3x^3+x^2+x+1)$, both positive. Since $0<\varrho_t<\varrho_r<1$ for $0<t<r<1$, first increasing the first argument and then moving along the radial ray gives

$$\mathscr{K}_q(\mathsf{V}_t,\mathsf{V}_r)<\mathscr{K}_q(\varrho_r(s+t),\varrho_r(s+r))<\mathscr{K}_q(s+t,s+r).$$

At $\Lambda=0$, the Riccati equation gives $S_t=s+t$, hence $\mathsf{V}_t=0$ and $b_2=0$. The endpoint bound $C[(1-t)(1-r)]^{-1/2}$ gives continuity by dominated convergence. For $\Lambda>0$ the strict kernel inequality above gives $b_2(s,\Lambda)<b_2(s,\infty)$, while the same conclusion at $\Lambda=0$ follows from positivity of the full-disclosure integral. Thus Δ_2 is positive and continuous on $[99/2,198]\times[0,\Lambda_{\max}]$, so it has a strictly positive minimum. Choosing w small enough for the uniform $O(w^3)$ remainder proves the comparison for $0<\Lambda\le\Lambda_{\max}$.

Unbounded precisions. Put $\eta=\Lambda^{-1/2}<1/4$. Use an auxiliary contest with constant posterior variance η and initially disclosed mean variance $s-\eta$; total initial uncertainty remains s. We first identify a positive leading

term in the full-disclosure output advantage. With $\tau = 1-t$, set $h = \tau+\eta$, $v = \log(h/\eta)$, and $z = y/\sqrt{h}$. The full-disclosure profiles $R_w, \mathsf{D}_w, \mathsf{b}_w$ satisfy

$$\mathscr{L}_0 R_w + w\{3(R'_w)^2 - (\mathsf{D}'_w)^2\}/4 = 0, \qquad \mathscr{L}_0 \mathsf{D}_w + (w/2)R'_w \mathsf{D}'_w = 0,$$

$$(\mathscr{L}_0 - 1/2)\mathsf{b}_w + wR'_w \mathsf{b}'_w + (w/2)\mathsf{D}'_w = 0, \qquad \mathscr{L}_0 = (\partial_{zz} + z\partial_z)/2.$$

Here $R_0 = 1$, $\mathsf{D}_0 = 2\Phi - 1$, $\mathsf{b}_0 = \beta = G_1 + z(\Phi - 1/2)$, and $R_w = 1 - w\Phi(1-\Phi) + O(w^2)$ by the explicit equilibrium.

The profile error $E = (\varphi, \psi, \chi)$ satisfies $E_v = \mathcal{A}_w E + \mathscr{N}_w(E)$, where $\mathcal{A}_w$ is the linearization, $\mathscr{N}_w$ the quadratic remainder, and $E(0) = (1 - R_w, 2\Phi - 1 - \mathsf{D}_w, \beta - \mathsf{b}_w)$. Explicitly,

$$\mathscr{N}_w(E) = w\left(\frac{3\varphi'^2 - \psi'^2}{4}, \frac{\varphi'\psi'}{2}, \varphi'\chi'\right),$$

while the player block of $\mathcal{A}_w$ is

$$\begin{pmatrix} \mathscr{L}_0\varphi + w(3R'_w\varphi' - \mathsf{D}'_w\psi')/2 \\ \mathscr{L}_0\psi + w(\mathsf{D}'_w\varphi' + R'_w\psi')/2 \end{pmatrix},$$

and its organizer block is

$$(\mathscr{L}_0 - 1/2 + wR'_w\partial_z)\chi + w(\varphi'\mathsf{b}'_w + \psi'/2).$$

On the even–odd–even subspace of $\mathscr{H} = L^2(e^{z^2/2}dz)^3$, put $\mathscr{X} = \mathrm{Dom}((-\mathcal{A}_0)^{7/8})$. Writing $g = 2\Phi - 1$, the explicit profiles give $1 - R_w = O(w(1-g^2))$ and $g - \mathsf{D}_w = O(w^2(1-g^2))$, with uniform polynomial-times-Gaussian bounds on their first two derivatives. For the organizer, set $k_w = \mathsf{b}_w - \beta - wG_1$. Subtracting $\beta + wG_1$ from its stationary equation gives

$$(\mathscr{L}_0 - 1/2 + wR'_w\partial_z)k_w = -wR'_w(\beta' + wG'_1) - \frac{w}{2}(\mathsf{D}'_w - 2G_1).$$

The right side is $O(w^2)$ in the scalar weighted L^2 space, since $R'_w = O(w)$ and $\mathsf{D}'_w = 2G_1 + O(w^2)$. The bounded inverse of $\mathscr{L}_0 - 1/2$ and its Neumann perturbation give

$$\|E(0,w)\|_{\mathscr{X}} \le Cw, \qquad \|k_w\|_{\mathrm{Dom}(\mathscr{L}_0)} \le Cw^2. \tag{31}$$

The scalar domain carries its weighted graph norm. For scalar organizer functions, $\|f\|_{\mathscr{X}}$ means $\|(0,0,f)\|_{\mathscr{X}}$, and similarly for $\mathscr{H}$. Only the corrections belong to these weighted spaces; β and b_w grow linearly. The explicit profiles and organizer resolvent also give analytic coefficients for $\mathcal{A}_w$ near $w = 0$.

At $w = 0$ these blocks are Ornstein–Uhlenbeck operators. Their Hermite spectrum has the simple leading eigenpair $(-1/2, (G_1, 0, 0))$; parity places every other eigenvalue at or below -1. Moreover,

$$\|(\mathcal{A}_w - \mathcal{A}_0)f\|_{\mathscr{H}} \le C|w|\|(-\mathcal{A}_0)^{1/2}f\|_{\mathscr{H}}.$$

Spectral perturbation (Kato 1980, Chapter VII, Sections 1–2) and sectorial estimates (Lorenzi et al. 2005, Proposition 3.2.2 and Chapter 5) therefore preserve one simple real eigenvalue $\nu(w)$ with $|\nu(w) + 1/2| < 1/20$ and give decay rate $7/10$ on the complementary subspace. If $\mathscr{P}_w$ is its spectral projection, the corresponding uniform semigroup estimates are

$$\|e^{v\mathcal{A}_w}\|_{\mathscr{H}\to\mathscr{X}} \le C(1 + v^{-7/8})e^{-9v/20}, \qquad \|e^{v\mathcal{A}_w}(I - \mathscr{P}_w)\|_{\mathscr{H}\to\mathscr{X}} \le C(1 + v^{-7/8})e^{-7v/10}.$$

Sobolev embedding gives $\|\mathscr{N}_w(E)\|_{\mathscr{H}} \le C|w|\|E\|^2_{\mathscr{X}}$; since $7/8 < 1$, the singular factor is integrable in Duhamel's formula. Using (31), Duhamel's map is a contraction on a radius-Cw ball under the norm $\sup_{v\ge 0} e^{2v/5}\|E(v)\|_{\mathscr{X}}$ for small $w > 0$, with Lipschitz constant $O(w^2)$. Its fixed point satisfies $\|E(v)\|_{\mathscr{X}} \le Cwe^{-2v/5}$.

Normalize the slow right eigenvector e_w and its left functional $\mathfrak{l}_w$ by $\mathfrak{l}_w(e_w) = 1$. Projection of Duhamel's formula gives the exact amplitude

$$\mathfrak{a}(w) = \mathfrak{l}_w E(0) + \int_0^\infty e^{-\nu(w)r}\mathfrak{l}_w\mathscr{N}_w(E(r))\,dr.$$

The integral is $O(w^3)$ because $Cw^3e^{-(4/5+\nu(w))r}$ is integrable, whereas $R_w = 1 - w\Phi(1-\Phi) + O(w^2)$ and $\int \Phi(1-\Phi)dz = 1/\sqrt{\pi}$ give $\mathfrak{l}_w E(0) = w/\sqrt{\pi} + O(w^2)$. Hence $\mathfrak{a}(w) = w/\sqrt{\pi} + O(w^2) > 0$ for small $w > 0$. The designer component ξ_w of the eigenvector, normalized to $(G_1, 0, 0)$ at $w = 0$, is $-wG_1\beta + O(w^2)$: its first-order equation is $-\mathscr{L}_0\xi_1 = G_1'(\Phi - 1/2)$, verified directly by $\mathscr{L}_0(G_1\beta) = G_1'(\Phi - 1/2)$. Put $E_{\mathrm{f}}(v) = E(v) - \mathfrak{a}(w)e^{\nu(w)v}e_w$. Projection of Duhamel's formula gives

$$\mathscr{P}_w E(v) = e^{\nu(w)v}e_w\left\{\mathfrak{a}(w) - \int_v^\infty e^{-\nu(w)r}\mathfrak{l}_w\mathscr{N}_w(E(r))\,dr\right\}.$$

The complementary semigroup estimate bounds $(I - \mathscr{P}_w)E$ by $Cwe^{-7v/10}$, while the displayed tail is $O_{\mathscr{X}}(w^3e^{-4v/5})$. Thus $\|E_{\mathrm{f}}(v)\|_{\mathscr{X}} \le Cwe^{-7v/10}$. Let $\mathcal{A}^{\circ}_w = \mathscr{L}_0 - 1/2 + wR_w'\partial_z$ on even functions and $\chi_{\mathrm{rem}}(v) = \chi(v) - \mathfrak{a}(w)e^{\nu(w)v}\xi_w + we^{-v}G_1$. The organizer equation and eigenvector equation imply

$$\partial_v\chi_{\mathrm{rem}} = \mathcal{A}^{\circ}_w\chi_{\mathrm{rem}} + w\{\mathfrak{b}_w'(E_{\mathrm{f}})_1' + (E_{\mathrm{f}})_2'/2\} + w\varphi'\chi' - w^2e^{-v}R_w'G_1'.$$

Here $\chi_{\mathrm{rem}}(0) = -k_w - \mathfrak{a}(w)\xi_w = O_{\mathscr{X}}(w^2)$. The three forcing terms are $O_{\mathscr{H}}(w^2e^{-7v/10})$, $O_{\mathscr{H}}(w^3e^{-4v/5})$, and $O_{\mathscr{H}}(w^3e^{-v})$. Since $\mathcal{A}^{\circ}_w - (\mathscr{L}_0 - 1/2) = O(w^2)\partial_z$, its semigroup satisfies, for even scalar f,

$$\|e^{v\mathcal{A}^{\circ}_w}f\|_{\mathscr{X}} \le C(1 + v^{-7/8})e^{-9v/10}\|f\|_{\mathscr{H}}.$$

Duhamel's formula therefore yields

$$\chi(v) = \mathfrak{a}(w)e^{\nu(w)v}\xi_w - we^{-v}G_1 + \chi_{\mathrm{rem}}(v), \qquad \|\chi_{\mathrm{rem}}(v)\|_{\mathscr{X}} \le Cw^2e^{-7v/10}. \tag{32}$$

At the initial date, $h = 1 + \eta$ and $e^{-v} = \eta/h$. For $1 \le h \le 1 + \eta$, let $Y_h \sim \mathcal{N}(0, (q-h)/h)$, with q fixed. The identities $\sqrt{h}\,\mathbb{E}\beta(Y_h) = \sqrt{q/(2\pi)}$ and $\sqrt{h}\,\mathbb{E}G_1(Y_h) = hG_q(0)$ cancel all first-order terms directly:

$$\begin{aligned}\widehat{\mathcal{B}}(w, s, \infty) - \widehat{\mathcal{B}}_{\mathrm{aux}}(w, s, \eta) = &-\sqrt{h}\,\mathfrak{a}(w)e^{\nu(w)v}\mathbb{E}\xi_w(Y_h)\\ &+ \mathbb{E}k_w(Y_1) - \sqrt{h}\,\mathbb{E}k_w(Y_h) - \sqrt{h}\,\mathbb{E}\chi_{\mathrm{rem}}(v, Y_h), \qquad h = 1 + \eta.\end{aligned}$$

Gaussian differentiation gives

$$\frac{d}{dh}\left\{\sqrt{h}\,\mathbb{E}k_w(Y_h)\right\} = \frac{1}{2\sqrt{h}}\mathbb{E}[(1 - 2\mathscr{L}_0)k_w(Y_h)] = O(w^2),$$

using (31); the Gaussian bounds justify interchanging differentiation and expectation. Thus the background correction is $O(w^2\eta)$. Moreover, $-\xi_w = wG_1\beta + O_{\mathscr{X}}(w^2)$, $\mathfrak{a}(w) \ge w/(2\sqrt{\pi})$, and $\mathbb{E}[G_1(Y_h)\beta(Y_h)]$ has a positive uniform lower bound. Since $|\nu(w) + 1/2| < 1/20$ and $h < 5/4$, $e^{\nu(w)v} \ge (5/4)^{-11/20}\eta^{11/20}$. Together with (32), this gives

$$\widehat{\mathcal{B}}(w, s, \infty) - \widehat{\mathcal{B}}_{\mathrm{aux}}(w, s, \eta) \ge w^2\{d\eta^{11/20} - C(\eta^{7/10} + \eta)\}.$$

It remains to compare the auxiliary contest with the original path. In the common variance clock Q, their incentive weights are $\gamma_o = \eta^2/S^2_{t(Q)}$ on $[S_1, q]$, zero below S_1, and $\gamma_a = \mathbf{1}_{[\eta,1+\eta]}$. Both lie in $[0,1]$, have integral one, and the Riccati solution gives $\int |\gamma_o - \gamma_a|\, dQ < 2\eta$. More precisely, $S_t = \eta(1+\chi_S e^{-2t/\eta})/(1-\chi_S e^{-2t/\eta})$, where $\chi_S = (s-\eta)/(s+\eta) \in (0,1)$. Thus $\varepsilon_\eta := S_1 - \eta \le C\eta e^{-2/\eta}$, and $|\gamma_o - \gamma_a| \le 4e^{-1/\eta}$ below $Q = 1/2$ except on $[\eta, S_1]$.

Extend both systems below their terminal variances by zero incentive weight. Their gradients start from the common heat profile $\mathfrak{p} = \mathfrak{n} = G_Q$; in the mild equations, w is replaced by $w\gamma(Q)$. The Gaussian bilinear bound in (28) makes these equations contractions for sufficiently small $w > 0$, uniformly in η. For conditional output J, set $\mathscr{Z} = \partial_y(J - \beta)$, where $\beta(y,Q) = \mathbb{E}|\mathcal{N}(y,Q)|/2$. Its mild equation is obtained from

$$J_Q = \tfrac{1}{2} J_{yy} + w\gamma\{(\mathfrak{p} - \mathfrak{n})J_y + (\mathfrak{p} + \mathfrak{n})/2\}.$$

The same bound and Gaussian smoothing give $|\mathfrak{p} - G_Q| + |\mathfrak{n} - G_Q| \le CwG_Q$, $\|\mathscr{Z}\|_\infty \le Cw$, and bounded spatial derivatives on $1/2 \le Q \le q$. Let $\mathfrak{d}$ be the sum of the two gradient differences in the norm $\sup_{Q,y} |f(Q,y)|/G_Q(y)$ and the difference of $\mathscr{Z}$ in the supremum norm. Subtract the three mild equations. Above $Q = 1/2$, differentiation of the bounded source profiles bounds the weight-error terms by $C|w|\eta$. Below that level, the bilinear bound controls the uniform weight error. On the exceptional interval, Gaussian multiplication gives

$$\frac{(|G'_{Q-v}| * G_v^2)(y)}{G_Q(y)} \le C\left\{\frac{1}{v} + \frac{1}{\sqrt{v(Q-v)}}\right\}, \qquad 0 < v < Q.$$

Integration over an interval of length ε_η with $v \ge \eta$ bounds its contribution by $C|w|\sqrt{\varepsilon_\eta/\eta}$. The same bound for the output gradient follows from $\|\partial_y P_v\|_{L^\infty \to L^\infty} \le Cv^{-1/2}$. The remaining terms are Lipschitz in the three unknowns, so

$$\mathfrak{d} \le C|w|\{\eta + e^{-1/\eta} + \sqrt{\varepsilon_\eta/\eta}\} + C|w|\mathfrak{d} \le C|w|\eta + C|w|\mathfrak{d}.$$

Take $Cw < 1/2$, so $\mathfrak{d} \le Cw\eta$. To compare output, remove the common first-order source G_Q: $\mathcal{R}_\gamma = (\mathfrak{p} + \mathfrak{n})/2 - G_Q + (\mathfrak{p} - \mathfrak{n})(\beta_y + \mathscr{Z})$ satisfies $|\mathcal{R}_\gamma| \le CwG_Q$ and $|\mathcal{R}_o - \mathcal{R}_a| \le C\mathfrak{d}G_Q$. With P_v the heat semigroup, $P_{q-Q}G_Q = G_q$ and $\int_0^q \gamma\, dQ = 1$ give the exact root identity

$$J_\gamma(q,0) = \beta(0,q) + wG_q(0) + w\int_0^q \gamma(Q)[P_{q-Q}\mathcal{R}_\gamma(Q)](0)\, dQ.$$

The common first-order term cancels, hence

$$|J_o(q,0) - J_a(q,0)| \le Cw\left(w\int_0^q |\gamma_o - \gamma_a|\, dQ + \mathfrak{d}\right) \le Cw^2\eta.$$

Because $11/20 < 7/10 < 1$, the positive term decays more slowly than both error terms. Choose $\eta_* > 0$ so that $d\eta^{11/20} > C(\eta^{7/10} + \eta)$ for $0 < \eta < \eta_*$. Combine this tail bound with the compact comparison for $\Lambda_{\max} = \eta_*^{-2}$ and take the smaller prize threshold. Rescaling proves (29). □

LEMMA 3 **(A full-disclosure payoff comparison)**. *Suppose $c_i = c_j = c$, u satisfies the curvature condition in Proposition 1, and S_0 is sufficiently large. There exist $\sigma > 1$ and $0 < c < c_e$, where $T_0 > 0$ maximizes $u(\sqrt{(S_0 + \sigma^2 T)/(2\pi)}) - D(T)$ and*

$$c_e = \frac{T_0 u'(\sqrt{(S_0 + \sigma^2 T_0)/(2\pi)})}{\sqrt{2\pi(S_0 + \sigma^2 T_0)}}.$$

Let V_∞ be the supremum of the organizer's payoff over full-disclosure designs at these parameters. Every positive-prize, positive-deadline finite-feedback design has payoff strictly below V_∞.

The deadline T_0 is optimal when no prize is paid. At that deadline, c_e is the effort-cost threshold at which the marginal benefit of introducing a small prize equals its marginal expenditure: the payoff derivative at zero prize is $c_e/c-1$. The construction takes costs slightly below this threshold, so a positive prize is worthwhile but strong incentives are too expensive relative to their extra quality benefit. Lemma 2 then ranks precision for the competitive designs. The present lemma extends that comparison to all competing prize–deadline choices, supplying the global comparison needed for Proposition 1(ii); it does not describe every high-innovation contest.

Proof of Lemma 3 The utility condition gives $-Bu''(B)/u'(B) \geq \zeta B \to \infty$ and $Bu'(B) \to 0$. We construct a parameter point at which only weak incentives can compete with the zero-prize design, then apply Lemma 2.

Put $\epsilon_* = 1/100$, $b = \sqrt{S_0/\{2\pi(1-\epsilon_*)\}}$, and $\mathfrak{m} = bu'(b)$. Choose

$$T_0 D'(T_0) = \mathfrak{m}\epsilon_*/2, \qquad \sigma^2 T_0 = \epsilon_* S_0/(1-\epsilon_*).$$

The second equality makes $b(T_0) = b$. It also gives

$$T_0 u'(b) b'(T_0) = \frac{\mathfrak{m}\epsilon_*}{2} = T_0 D'(T_0),$$

so T_0 satisfies the zero-prize first-order condition. Because $u(b(T)) - D(T)$ is strictly concave, this condition makes T_0 its unique maximizer. As $S_0 \to \infty$, $\mathfrak{m} \to 0$ by the utility-tail assumption. The first equality and $D''(0) > 0$ then imply $T_0 \to 0$, while the second implies $\sigma^2 = \epsilon_* S_0/\{(1-\epsilon_*)T_0\} \to \infty$. Write $V_0 = u(b) - D(T_0)$, $\alpha = \theta/c$, $w = \alpha/\sigma^2$, and $\vartheta = T/T_0$. Then $c_e\sigma^2 = \mathfrak{m}\epsilon_*$. Monotonicity of $D'(T)/T$ gives

$$D(T) - D(T_0) \geq \mathfrak{m}\epsilon_*(\vartheta^2 - 1)/4, \qquad D(T_0) \leq \mathfrak{m}\epsilon_*/4.$$

Indeed, $D'(s) \geq (s/T_0)D'(T_0)$ for $s \geq T_0$ and the reverse inequality holds for $s \leq T_0$; integrating on the appropriate interval gives the first bound. Integrating $D'(s) \leq (s/T_0)D'(T_0)$ on $[0, T_0]$ gives the second.

We choose a sufficiently large fixed γ below, then increase S_0 so that $-Bu''(B) \geq (\gamma+1)u'(B)$ for $B \geq .99b$ and $D'(T_0)/\{D''(0)T_0\} < 1 + 10^{-5}$. At $c = c_e$, any design with payoff at least V_0 satisfies the budget $\theta + D(T) \leq u_\infty - V_0 \leq \mathfrak{m}/\gamma + \mathfrak{m}\epsilon_*/4$, where $u_\infty = \lim_{B\to\infty} u(B)$ and the tail bound follows by integrating the curvature inequality above b: it gives $u'(B) \leq u'(b)(B/b)^{-(\gamma+1)}$ and hence $u_\infty - u(b) \leq bu'(b)/\gamma = \mathfrak{m}/\gamma$. Since $D(T) \geq D''(0)T^2/2$, this gives

$$w + \frac{D''(0)T_0}{D'(T_0)}\frac{\vartheta^2}{4} \leq \frac{1}{4} + \frac{1}{\gamma\epsilon_*}.$$

Take $\gamma \geq 10^6$; this implies $w + \vartheta^2/4 < .252$, hence $\vartheta < 2$. Moreover, concavity gives $u(b) - u(.99b) \geq .01\mathfrak{m} > D(T_0)$, so a design with $\mathcal{B} \leq .99b$ has payoff strictly below $u(b) - D(T_0) = V_0$. Competitiveness therefore requires $\mathcal{B} > .99b$.

A coarse output bound. We verify, for this entire competitive set,

$$\mathcal{B} \leq b(T) + \alpha T G_{S_0+\sigma^2 T}(0)(1 + 2.9w), \qquad b(T) = \sqrt{(S_0 + \sigma^2 T)/(2\pi)}. \tag{33}$$

Put $Q_t = S_t + \sigma^2(T-t)$, $a_t = \lambda S_t^2$, $U_i = V_i/\theta$, $U_j = V_j/\theta$, $\mathfrak{p} = \partial_y U_i$, $\mathfrak{n} = -\partial_y U_j$, $\mathfrak{H} = \mathfrak{p} + \mathfrak{n}$, and $\mathfrak{A} = \mathfrak{p} - \mathfrak{n}$.

If $\lambda S_0^2 < .72\sigma^2$, the Riccati path increases and $S_T \geq S_0$. Thus

$$\varepsilon_{\text{loc}} := \alpha T/S_T \leq 2w/99 < .01, \qquad Q_0/S_T \leq 101/99.$$

Here we used $\vartheta < 2$ and $S_0/(\sigma^2 T_0) = (1-\epsilon_*)/\epsilon_* = 99$, so $\sigma^2 T/S_0 < 2/99$; the two displayed bounds then follow from $S_T \geq S_0$. Normalize scores by $\sqrt{S_T}$ and reverse time by $v = \alpha(T-t)/S_T$. In the Fourier-moment scale of Theorem 1, the normalized terminal Gaussian has the coefficientwise majorant

$$G_1(0)e^{\omega^2/2} + \frac{1}{\pi}\sum_{j\geq 0}\frac{\omega^{2j+1}}{(2j+1)!!} \leq \Psi(\omega) := .4e^{\omega^2/2} + .32\omega/(1-\omega^2/3), \qquad 0 \leq \omega < \sqrt{3}.$$

Put $\kappa = 7/4$. For nonnegative moment majorants X, Y of $\mathfrak{H}, \mathfrak{A}$, respectively, the weighted sum of their quadratic fluxes satisfies

$$\frac{XY}{4} + \frac{\kappa}{8}(3Y^2 + X^2) \leq \frac{\kappa}{8}(X + \kappa Y)^2,$$

because $\kappa^2 > 3$. Heat contraction and Picard iteration therefore majorize the weighted moment sum $(X + \kappa Y)/2$ by the scalar equation $\mathfrak{h}_v = \kappa \mathfrak{h}\mathfrak{h}_\omega$, $\mathfrak{h}(0,\omega) = \Psi(\omega)$, whose analytic branch satisfies $\mathfrak{h}(v,\omega) = \Psi(\omega + \kappa v \mathfrak{h}(v,\omega))$. In the Picard expansion, subtracting the propagated Gaussian removes the order-zero term. Each individual gradient correction is bounded by half the sum of the moment majorants of $\mathfrak{H} - 2G_Q$ and $\mathfrak{A}$, so its Fourier norm is at most $S_T^{-1/2}\{\mathfrak{h}(v,0) - .4\}$. Direct substitution gives $\mathfrak{h}(v,0) - .4 \leq v/4$ for $v \leq .01$: at the largest argument $\omega = (7/4)(.01)(.4025)$,

$$\frac{.4(\omega^2/2)e^{\omega^2/2} + .32\omega/(1-\omega^2/3)}{.01} < \frac{1}{4}.$$

Thus $d(t) = \alpha(T-t)/(4S_T^{3/2})$ bounds each correction. Since the Fourier-moment norm dominates the pointwise norm, this means

$$|\mathfrak{p} - G_{Q_t}| \leq d(t), \qquad |\mathfrak{n} - G_{Q_t}| \leq d(t),$$

and hence, with $\mathfrak{E} = \mathfrak{H} - 2G_{Q_t}$, $|\mathfrak{E}| \leq 2d(t)$ and $|\mathfrak{A}| \leq 2d(t)$. These are pointwise bounds; the Gaussian-weighted supremum norm used in the next, high-diffusion case is a different norm.

Let $J(t,y)$ be the conditional continuation contribution to $\mathcal{B} - b(T)$. It has terminal value zero and solves

$$J_t + \mathscr{L}_t J + \frac{\alpha}{2}\{\mathfrak{H} + [2\Phi(y/\sqrt{Q_t}) - 1]\mathfrak{A}\} = 0, \qquad \mathscr{L}_t = \frac{a_t}{2}\partial_{yy} + \alpha\mathfrak{A}\partial_y.$$

Consider the unnormalized barrier

$$\overline{J}(t,y) = \alpha(T-t)G_{Q_t}(y) + \int_t^T 2\alpha d(s)\left\{1 + .486\frac{\alpha(T-s)}{Q_s}\right\} ds.$$

It has zero terminal value. Since $\partial_t G_{Q_t} + (a_t/2)\partial_{yy} G_{Q_t} = 0$,

$$(\partial_t + \mathscr{L}_t)\overline{J} = -\alpha G_{Q_t} + \alpha^2 \mathfrak{A}(T-t)G'_{Q_t} - 2\alpha d(t)\left\{1 + .486\frac{\alpha(T-t)}{Q_t}\right\}.$$

The source in the equation for J is $\alpha G_{Q_t} + \alpha\mathfrak{E}/2 + \alpha[2\Phi(y/\sqrt{Q_t}) - 1]\mathfrak{A}/2$. Using $|\mathfrak{E}|, |\mathfrak{A}| \leq 2d$, $|2\Phi - 1| \leq 1$, and $|G'_Q| < .243/Q$, the supersolution residual is at most

$$.486\alpha^2 d(t)\frac{T-t}{Q_t} + 2\alpha d(t) - 2\alpha d(t)\left\{1 + .486\frac{\alpha(T-t)}{Q_t}\right\} = -.486\alpha^2 d(t)\frac{T-t}{Q_t} \leq 0.$$

Itô verification gives the shared low-diffusion bound

$$\mathcal{B} - b(T) \leq \alpha T G_{S_0 + \sigma^2 T}(0) + \frac{\alpha^2 T^2}{4S_T^{3/2}}\left(1 + .486\frac{\alpha T}{S_T}\right). \tag{34}$$

This bound requires only $\lambda S_0^2 < .72\sigma^2$ and $\alpha T/S_T \le .01$. On the present competitive set, the excess output coefficient is $\sqrt{2\pi Q_0/S_T}(\varepsilon_{\mathrm{loc}}/w)(.5+.243\varepsilon_{\mathrm{loc}})/2 < .02$, proving (33) in this case.

If $\lambda S_0^2 \ge .72\sigma^2$, the whole Riccati path has $a_t \ge .72\sigma^2$. Indeed, $a_t' = 2\lambda S_t(\sigma^2 - a_t)$, so a_t moves monotonically toward σ^2 and never falls below $\min\{a_0, \sigma^2\} \ge .72\sigma^2$. In the norm $\|f\| = \sup|f|/G_{Q_t}$, write $\mathfrak{E} = \mathfrak{H} - 2G_Q$ and $\mathcal{T}(f,g) = \int_t^T \partial_y P_{Q_t - Q_s}(f_s g_s)\, ds$. The centered gradient equations are

$$\mathfrak{E} = \tfrac{\alpha}{2}\mathcal{T}(\mathfrak{A}, 2G_Q + \mathfrak{E}), \qquad \mathfrak{A} = \mathfrak{A}_1 + \tfrac{\alpha}{4}\{3\mathcal{T}(\mathfrak{A},\mathfrak{A}) - \mathcal{T}(\mathfrak{E}, 4G_Q + \mathfrak{E})\}.$$

Changing variables from time to Q_s gives $\alpha\|\mathcal{T}(f,g)\| \le K_{\mathrm{b}}\|f\|\|g\|$, where $K_{\mathrm{b}} = \alpha/\min a_t \le w/.72$. Indeed, (28) bounds the convolution integral after this variance substitution. The explicit source $\mathfrak{A}_1 = -\alpha(2\sqrt{\pi})^{-1}\int_t^T Q_s^{-1/2} G'_{Q_t - Q_s/2}\, ds$ has norm at most K_{b}: the same variance substitution evaluates its upper bound as $K_{\mathrm{b}}|2\Phi(y/\sqrt{Q_t}) - 1|G_{Q_t}$. The centered map contracts on $\|\mathfrak{E}\| \le 6w^2$, $\|\mathfrak{A}\| \le 3w$. Indeed, for $0 < w \le .252$, its map-to-radius ratios satisfy

$$\max\left\{\frac{2+6w^2}{4\cdot.72}, \frac{1+w^2(51+36w^2)/4}{3\cdot.72}\right\} < 1,$$

and its Lipschitz row sums are bounded by

$$\frac{w}{.72}\max\left\{\frac{2+3w+6w^2}{2}, \frac{4+18w+12w^2}{4}\right\} < 1.$$

Lemma 1 identifies the fixed point with the equilibrium.

For its transition operator $\mathcal{K}_{t,s}$,

$$\mathcal{K}_{t,s} g \le \cosh^2(.35/2) P_{Q_t - Q_s} g, \qquad g \ge 0.$$

To justify the factor, note first that each player can choose zero effort, so $U_i, U_j \ge 0$, while summing their equilibrium payoffs gives the unit total prize minus nonnegative effort costs and hence $R := U_i + U_j \le 1$. Applying Itô's formula shows that $e^{-K_{\mathrm{b}}U_i}$ and $e^{-K_{\mathrm{b}}U_j}$ are submartingales. Their terminal values are exponentials of complementary probabilities in $[0,1]$, which gives $0 \le 1 - R \le 2\log\cosh(K_{\mathrm{b}}/2)/K_{\mathrm{b}}$. With $\kappa_t = \alpha/a_t$, Girsanov's log likelihood is

$$\kappa_s(R_s - 1) - \kappa_t(R_t - 1) - \int_t^s \kappa_r'(R_r - 1)\, dr - \frac{1}{4}\int_t^s \alpha\kappa_r(\mathfrak{H}_r^2 - \mathfrak{A}_r^2)\, dr, \qquad R = U_i + U_j.$$

Monotonicity of κ, in either direction, bounds this by $2\log\cosh(K_{\mathrm{b}}/2)$; $\mathfrak{H}^2 - \mathfrak{A}^2 = 4\mathfrak{p}\mathfrak{n} \ge 0$ makes the last term nonpositive. Exponentiating and applying the heat semigroup gives the stated $\cosh^2(K_{\mathrm{b}}/2)$ domination. Since $K_{\mathrm{b}} \le w/.72 \le .35$, the displayed constant follows. Bounded gradients and positive diffusion justify the change of measure.

Let $\beta(y,Q) = \mathbb{E}|\mathcal{N}(y,Q)|/2$. Subtracting $\beta + \alpha(T-t)G_{Q_t}$ from conditional best output gives a zero-terminal linear equation with source $\alpha\mathfrak{E}/2 + \alpha\mathfrak{A}\beta_y + \alpha^2(T-t)\mathfrak{A}G'_{Q_t}$. Using $|\beta_y| \le 1/2$, $(T-t)/Q_t \le 1/\sigma^2$, and $|G_Q'| < .243/Q$, its absolute value is at most $\alpha w G_{Q_t}\{3w + 3/2 + .729w\}$. Since $\cosh^2(.35/2)\{3w + 3/2 + .729w\} < 2.9$ for $w \le .252$, the transition bound proves (33). This high-diffusion bound requires only $\lambda S_0^2 \ge .72\sigma^2$ and $w \le .252$, not the prior ratio.

Exclude prizes outside the weak-incentive range. Put $h = 1 + \epsilon_*(\vartheta - 1)$ and $\varrho = \sqrt{h} + \epsilon_*\vartheta(w + 2.9w^2)/\sqrt{h}$. Integrating the utility-curvature inequality gives $u(\mathcal{B}) - u(b) \le (\mathfrak{m}/\gamma)\{1 - (b/\mathcal{B})^\gamma\}$. The construction gives $b(T)/b = \sqrt{h}$, while

$$\frac{\alpha T G_{S_0+\sigma^2 T}(0)}{b} = \frac{\epsilon_*\vartheta w}{\sqrt{h}}.$$

Thus (33) implies $\mathcal{B}/b \le \varrho$. Since the curvature bound is increasing in $\mathcal{B}$, and since $\theta = c_e\alpha = \mathfrak{m}\epsilon_* w$, substitution together with the delay bound yields

$$\frac{u(\mathcal{B}) - \theta - D(T) - V_0}{\mathfrak{m}\epsilon_*} \le \mathscr{F}_\gamma(\vartheta, w) := \frac{1 - \varrho^{-\gamma}}{\gamma\epsilon_*} - w - \frac{\vartheta^2 - 1}{4}. \tag{35}$$

Let $w_* > 0$ be the threshold in Lemma 2 and put $w_0 = \min\{w_*/2, .252\}$. We need $\mathscr{F}_\gamma < 0$ only on the compact set

$$\{(\vartheta, w) \in [0,2] \times [0, .252] : \vartheta \le 1/2 \text{ or } w \ge w_0\},$$

which excludes $(1, 0)$.

On $w + (\vartheta^2 - 1)/4 \le 0$, we have $\vartheta \le 1$ and $w \le (1 - \vartheta^2)/4$. Since ϱ increases in w, $\varrho < 1$ reduces to

$$\vartheta\left\{\frac{1+\vartheta}{4} + \frac{29(1-\vartheta)(1+\vartheta)^2}{160}\right\} < \frac{\sqrt{1 - \epsilon_*(1-\vartheta)}}{1 + \sqrt{1 - \epsilon_*(1-\vartheta)}}, \qquad \vartheta < 1.$$

The left side is at most $.5 - (1-\vartheta)/40$, because the difference is $(1-\vartheta)^2(29\vartheta^2 + 87\vartheta + 76)/160$. The right side exceeds this bound: its difference from $1/2$ is $\epsilon_*(1-\vartheta)/\{2(1 + \sqrt{1 - \epsilon_*(1-\vartheta)})^2\} < (1-\vartheta)/40$. Equality occurs only at $(1, 0)$. The inequality $e^z \ge 1 + z$ makes $\mathscr{F}_\gamma$ nonincreasing in γ. Every point of the displayed compact set eventually has $\mathscr{F}_\gamma < 0$: if $w + (\vartheta^2 - 1)/4 \le 0$, then $\varrho < 1$ and $\mathscr{F}_\gamma \to -\infty$; otherwise $\limsup_{\gamma\to\infty} \mathscr{F}_\gamma < 0$. The sets $\{\mathscr{F}_\gamma < 0\}$ therefore increase with γ and cover the displayed compact set. A finite subcover gives one $\gamma \ge 10^6$ and a strictly negative uniform margin. Fix it before increasing S_0.

Finally take $c < c_e$ sufficiently close to c_e. The competitive budget still implies $w + \vartheta^2/4 < .252$, and replacing c_e by c changes the normalized prize term from $-w$ to $-(c/c_e)w$. Thus the right side of (35) gains only $(1 - c/c_e)w$. The uniform margin excludes $\vartheta \le 1/2$ or $w \ge w_0$ after this small cost perturbation. Every competitive finite-feedback design therefore has $T_0/2 < T < 2T_0$ and $0 < w < w_*$. Lemma 2 strictly improves its payoff by full disclosure. All other designs have payoff below $V_0 \le V_\infty$. □

Numerical illustration. Figure 3 compares the best-performing design families among the evaluated candidates on a grid of 14,553 parameter points. Of these, 1,179 remain unresolved under the numerical reliability checks and are shown separately by dark hatching, rather than assigned to a design family. At 926 of these points, potentially competitive candidates failed to solve reliably and therefore cannot be ruled out. The three design regimes and the unresolved points form a numerical illustration; the curves are smoothed for display using the unresolved-cell intervals, without resolving this classification uncertainty. The failures are not reported by design family, so their distribution across full-disclosure, finite-precision, and zero-prize candidates is unknown. The displayed boundaries should therefore be treated as provisional rather than as robust estimates of the regime frontiers.

B.2. Sequential Leaderboard Likelihood

For the retained readouts defined in Section 3, let

$$\mathfrak{g}_k = \Delta_{\text{num}} \sum_{\ell:\, t_{k-1} < s_\ell \le t_k} \left[m_i^{(1)}(\tilde{y}_{s_{\ell-1}}, S_{s_{\ell-1}}, s_{\ell-1}) - m_j^{(1)}(\tilde{y}_{s_{\ell-1}}, S_{s_{\ell-1}}, s_{\ell-1}) \right],$$

where the retained readouts are grid endpoints. This is the matched left-point approximation to the drift integral, using the same effort values as the count intensities. It is determined by the preceding public history, t_k, and Θ under the maintained carried-forward path. Set $\mathsf{R}_k = (\lambda h_k)^{-1}$ and initialize $\mu_{0|0} = \mu_0$ and $\mathsf{P}_{0|0} = S_0$. Conditional on $\mathcal{F}_{k-1}$, t_k, and Θ, prediction and measurement updating give

$$\mu_{k|k-1} = \mu_{k-1|k-1} + \mathfrak{g}_k, \qquad \mathsf{P}_{k|k-1} = \mathsf{P}_{k-1|k-1} + \sigma^2 h_k,$$
$$\Omega_{k|k-1} = \mathsf{P}_{k|k-1} + \mathsf{R}_k, \qquad \mathsf{K}_k = \mathsf{P}_{k|k-1}/\Omega_{k|k-1},$$
$$\mu_{k|k} = \mu_{k|k-1} + \mathsf{K}_k(\hat{y}_{t_k} - \mu_{k|k-1}), \qquad \mathsf{P}_{k|k} = (1 - \mathsf{K}_k)\mathsf{P}_{k|k-1}.$$

Thus $\mu_{k|k-1}$ and $\Omega_{k|k-1}$ are the predictive moments entering equation (13). After its density contribution is evaluated, $(\mu_{k|k}, \mathsf{P}_{k|k})$ initializes the next prediction.

B.3. Solve S_t in Equation (4)

For $\lambda > 0$, put $\bar{S} = \sigma/\sqrt{\lambda}$ and $\mathsf{s}_t = S_t/\bar{S}$. Equation (4) becomes

$$\dot{\mathsf{s}}_t = \sigma\sqrt{\lambda}\,(1 - \mathsf{s}_t^2).$$

The constant solution is $\mathsf{s}_t = 1$. Otherwise, separation of variables with the initial condition imposed directly gives

$$\int_{\mathsf{s}_0}^{\mathsf{s}_t} \frac{dv}{1 - v^2} = \sigma\sqrt{\lambda}\, t.$$

If $\mathsf{s}_0 < 1$, the antiderivative is $\tanh^{-1}$; if $\mathsf{s}_0 > 1$, it is $\coth^{-1}$. Inverting the corresponding identity yields

$$S_t = \begin{cases} \bar{S} \cdot \tanh\left\{ t \cdot \sigma\sqrt{\lambda} + \tanh^{-1}\left(S_0/\bar{S}\right) \right\} & \text{if } S_0 < \bar{S} \\ \bar{S} & \text{if } S_0 = \bar{S} \\ \bar{S} \cdot \coth\left\{ t \cdot \sigma\sqrt{\lambda} + \coth^{-1}\left(S_0/\bar{S}\right) \right\} & \text{if } S_0 > \bar{S} \end{cases}$$

The vector field points toward $\bar{S}$, so the solution remains in the branch determined by S_0 and converges monotonically to $\bar{S}$. The steady-state variance diverges as $\lambda \downarrow 0$. This is a long-run statement: with S_0 fixed and any fixed finite horizon, the Riccati solution instead converges to $S_0 + \sigma^2 t$ as $\lambda \downarrow 0$, so Q_t converges to the finite constant $S_0 + \sigma^2 T$.

B.4. Consistency Benchmark for σ^2

This appendix derives the private-leaderboard benchmark for σ used in Section 5.2. For a given contest, let $\Delta y_k = y_k - y_{k-1}$ denote the private-leaderboard gap increment between two adjacent update times, and let $h_k = t_k - t_{k-1}$ be the corresponding elapsed time. Effort paths evaluated at the posterior median parameters provide the plug-in drift adjustment $\hat{\mu}_k = \int_{t_{k-1}}^{t_k} [\hat{m}_i(s) - \hat{m}_j(s)]\, ds$. As a working Gaussian approximation, we use $\Delta y_k - \hat{\mu}_k \sim \mathcal{N}(0, \sigma^2 h_k)$, $k = 1, \ldots, n$. This is not an exact conditional distribution given an endogenous

effort path. Treating these increments as approximately independent and holding the plug-in drift fixed, the working log-likelihood for σ^2 is, up to constants,

$$\log \mathcal{L}_\sigma(\sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2}\sum_{k=1}^{n}\log h_k - \frac{1}{2\sigma^2}\sum_{k=1}^{n}\frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}.$$

Differentiating with respect to σ^2 and setting the derivative equal to zero gives

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{k=1}^{n}\frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k} = 0.$$

Therefore,

$$(\sigma^{\text{bench}})^2 = \frac{1}{n}\sum_{k=1}^{n}\frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}. \tag{36}$$

In the empirical application, $\hat{\mu}_k$ is computed from effort paths evaluated at the posterior median parameters, so this benchmark is a plug-in private-leaderboard consistency check rather than a fully model-free estimator. Moreover, asynchronous update timing means that adjacent displayed gaps may combine a new score for one contestant with a carried-forward score for the other. The expression in (36) is therefore an approximate benchmark for the carried-forward gap process, not an exact maximum likelihood estimator for synchronously observed latent diffusion increments.

Appendix C: Statistics

Table 9 reports contest-level estimates and diagnostic benchmarks; large λ^{bench} values can result from very small public–private score discrepancies. "Public Data %" reproduces Meta-Kaggle's reported public-leaderboard test-data share; it is not used in estimation or sample selection, and the recorded zero for contest 3288 does not indicate missing public-score observations.

Data construction. Focal-pair selection searches the top 20 contestants in the reconstructed private-score rankings. Contestants are ordered by their final carried-forward private scores; ties are broken by the order of the tied contestants' final score updates, with the later update ranked first. It first selects the highest-ranked contestant satisfying the individual activity criterion, then searches lower ranks for the first eligible opponent whose activity window overlaps by more than ten days. A pair is retained only if both contestants have at least five submissions in the constructed observation window. For Don't Overfit! (contest 2489), θ is the \$250 prize for the prediction task; the separate variable-identification prize is excluded. The common score transformation in Section 5.1 does not make the different scoring metrics equivalent probabilities; each retains its substantive interpretation.

Within each submission date, preprocessing draws distinct second offsets with seed 1234 and assigns them in submission-ID order, retaining multiple submissions by the same contestant on the same date. Backfilling the earliest joint public gap supplies a data-dependent initial public input, not additional leaderboard observations. The fixed-input identification result in Proposition 2 does not cover the sampling law induced by this initialization.

For eight contests preselected at evenly spaced submission-count ranks, we repeat the time imputation with two alternative seeds. Relative to the baseline, the median absolute change in the posterior median of

λ is 18.5% for full-sample fits and 38.0% for early fits. The 95% posterior intervals overlap their baseline counterparts in all 32 comparisons.

Posterior summaries. We report posterior medians to reduce sensitivity to long right tails, while retaining posterior means and their Monte Carlo errors in the replication outputs. Equal-tailed credible intervals and precision counterfactuals are calculated from the joint posterior draws.

Sampling diagnostics. Of the 397 full, early, and auxiliary fits, two multimodal posteriors use fixed-mixture Metropolis–Hastings refinement under the same likelihood and priors. All fits use four chains and common criteria for rank-normalized convergence and effective sample size; HMC-specific diagnostics apply only to NUTS runs.

Regression inference. All cross-contest regressions in Section 5.2 use posterior median parameter estimates, with HC1 heteroskedasticity-robust standard errors and confidence intervals. Reported inference does not propagate uncertainty from parameter estimation, focal-pair selection, or intraday time imputation.

Approximation diagnostics. Across the 146 full-sample player–contest pairs, the posterior median of $w_k = \theta/(\sigma^2 c_k)$ is below one in 76 pairs; the median of these posterior summaries is 0.909. At the contest level, the posterior median of $\max\{w_i, w_j\}$ is below one in 34 of 73 contests. These statistics describe incentive strength but do not establish approximation accuracy.

C.1. Precision Counterfactual Calculations

We condition on the joint parameter posterior estimated with the leading-order effort policy. Baseline and candidate quality calculations instead solve the original nonlinear game, using its coupled value equations and expected-output equation. Within every draw, $(c_i, c_j, \sigma, \mu_0)$ and the initial variance $S_0^{base} = \sigma/\sqrt{\lambda^{obs}}$ remain fixed across designs; changing feedback affects subsequent learning. Under full disclosure, contestants observe the initial gap, while the organizer averages over the unchanged initial distribution $\mathcal{N}(\mu_0, S_0^{base})$.

For each contest, the precision menu retains observed feedback, adds full disclosure, and uses the four common positive precisions $\lambda_{ref}\{0.001, 0.03, 0.3, 3\}$, where λ_{ref} is the median of 128 joint parameter draws. Each specified precision is common across draws; retaining observed feedback preserves draw-specific uncertainty in λ^{obs}. The lowest searched positive precision is not a no-feedback limit. Deadlines use $T/T^{obs} \in \{0.1, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2\}$. Prizes are searched directly in the nonlinear game, without a linear prize inversion or a monotonicity assumption.

For the joint-design table, paired baseline–candidate quality differences are integrated using 128 coarse-grid draws (32 per chain) and an independent fine-grid correction on 16 draws (four per chain), sampled with replacement from the accepted chains. Both levels solve the nonlinear game. The estimator averages the coarse difference and the fine-minus-coarse difference with equal chain weights. A candidate passes when its estimated quality difference, less a 95% Welch integration diagnostic and the mean numerical diagnostic radius, is nonnegative. The observed design is feasible by identity. Nested-grid and cross-coordinate comparisons supply numerical radii, not rigorous bounds on continuous-equation error. These conditional integration diagnostics are distinct from parameter credible intervals and MCMC error. Failed draws are retained as unresolved; no successful-draw subset substitutes for a complete comparison.

Table 8 uses the same 128 joint draws in a direct refined-grid calculation at observed prize and duration. A common precision is selected only among fully resolved candidates whose mean gain exceeds its numerical diagnostic radius; retaining observed feedback remains available. Small mean gains and close rankings can be affected by integration error. Unresolved candidates remain unranked, including any unresolved full-disclosure alternative.

The design search retains all numerically supported incumbents and locally refines the prize brackets needed by the table to $10^{-4}\theta^{obs}$. Unknown alternatives and unsampled designs are not ruled out. One contest's full-disclosure benchmark remains unresolved after targeted prize probes and a direct refined-grid contrast using all 128 original draws. Its extra saving is left unrestricted in the last-column median range, while that contest remains in all other counts and medians.

Fixed-prize, fixed-duration benchmark. Holding prize and duration at their observed values, Table 8 selects full disclosure in 21 contests, other finite precision in 25, and the lowest searched positive precision in 27. The median model-implied quality gain is 0.086%. Using the 75% threshold for support across draws, 17 contests favor increased precision, 33 favor reduced precision, and 23 have no clear direction.

Table 8 Nonlinear Precision Comparisons at the Observed Prize and Duration

Panel A: Selected searched regime

Regime	Contests	Share (%)
Full disclosure	21	28.8
Other finite precision	25	34.2
Lowest searched precision	27	37.0

Panel B: Model-implied quality gains (%)

Posterior direction group	25th percentile	Median	75th percentile
All contests	0.013	0.086	4.554
Increase precision	0.020	0.037	0.056
Decrease precision	1.018	5.308	10.615
Direction uncertain	0.000	0.021	0.132

Note: Each contest selects the best numerically supported searched precision using 128 joint posterior draws and nonlinear equilibrium output. Other finite precision includes retaining observed feedback; the lowest searched precision is positive and is not a no-feedback limit. At least one alternative remains unresolved in 21 contests, including full disclosure in 5; these alternatives remain unranked. Direction groups compare draw-level preferred precision with observed precision; a direction is named only when more than 75% of draws support it after allowing for numerical diagnostic intervals, and otherwise is uncertain. An unresolved candidate makes that draw's direction uncertain. Reported gains are numerical posterior mean changes; small gains and close rankings can be affected by integration error. Gain is $100(\overline{\mathcal{B}}^{selected} - \overline{\mathcal{B}}^{obs})/\overline{\mathcal{B}}^{obs}$, relative to observed feedback at the same prize and duration. There are 73 contests in total: 17 in the increase group, 33 in the decrease group, and 23 with uncertain direction. Panel A selects a common policy by posterior mean quality; Panel B groups contests by draw-level directional support, so these classifications need not coincide. Percentiles describe variation across contests, not credible intervals. The initial distribution is fixed across designs within each draw.

The small overall median gain shows that changing feedback alone often has limited value at the observed prize and duration. Gains are larger within the group supporting reduced precision, consistent with preserving competition in some settings, but that group is selected using the same model calculations. The 27 selections of the lowest searched positive precision motivate investigation below the search boundary; they do not establish that no feedback is optimal. These conditional comparisons neither identify a universally preferred disclosure rule nor directly test the symmetric mechanism in Proposition 1.

Quality-constrained joint design. We directly search the nonlinear quality constraint in equation (16) over the stated precision menu and deadline grid, with nonnegative prizes. The observed design remains available. Selected changes satisfy the paired numerical quality diagnostic described above; failed alternatives remain unknown. Prize brackets required by the reported designs are locally refined to $10^{-4}\theta^{obs}$, without assuming that quality is globally monotone in prize. The results are supported searched incumbents, not certified continuous optima.

The full-disclosure comparison uses the same initial distributions, baseline quality target, and deadline grid, with prize searched separately. The last column of Table 7 concerns positive-prize finite-feedback designs. Its bracketed medians retain the one unresolved full-disclosure comparison as an unrestricted unknown, rather than omitting that contest.

Delay costs are assumed scenarios; durations at grid edges need not be unconstrained optima, and large prize changes remain extrapolations. Nonlinear output calculations remove the leading-order output approximation from the design exercise, but do not remove approximation error in parameter estimation. Appendix C.4 reports targeted cost-prior and bound sensitivity checks. The asymmetric empirical designs do not isolate the symmetric mechanism of Proposition 1.

Table 9: Contest-Level Structural Estimates and Benchmarks

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Median						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
2445	15	35	0.0500	25	15700.681	0.239	0.197	0.975	0.471	17.508	117.492	0.196
2454	29	21	0.0015	62	852.833	0.861	4.345	0.040	0.058	147.410	42.015	0.862
2464	19	24	0.0095	20	606.828	2.178	-0.682	0.074	0.056	114.472	9.488	2.169
2478	21	23	0.0095	30	31.135	0.524	-0.044	0.204	0.186	42.737	1445.766	1.940
2489	32	58	0.0025	10	114.599	0.601	0.435	0.083	0.043	122.477	310.279	1.292
2549	22	28	0.0050	30	9.449	0.805	-1.825	0.083	0.063	103.432	33.547	3.107
2551	115	52	0.0150	30	36.946	0.115	-0.030	0.302	0.611	40.824	6896.087	1.075
2667	97	143	0.3000	50	35.328	0.399	0.164	0.677	0.447	15.235	4.012	0.729
2749	76	90	0.0300	42	971.451	0.230	0.805	0.253	0.216	37.919	4730.298	0.604
2762	24	41	0.0100	35	6.384	0.197	0.788	0.185	0.107	61.043	3.967	2.943
2860	18	8	0.0100	30	14712.764	0.080	-0.025	0.481	0.806	19.652	6436.776	0.045
3064	25	9	0.0015	25	17233.771	0.075	0.004	0.119	0.341	48.445	874483.872	0.094
3080	46	52	0.0009	25	81893.065	0.178	-0.466	0.061	0.055	126.212	22933.360	0.171
3288	30	20	0.0100	0	794.429	2.261	-1.231	0.062	0.097	97.415	0.090	0.803
3338	73	52	0.0150	30	720.144	0.201	0.329	0.191	0.256	43.884	692.427	0.413
3353	59	38	0.0600	30	8107.518	0.867	-0.138	0.194	0.305	35.375	1791.142	0.870

Table 9 (continued)

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Median						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
3366	30	20	0.0020	50	32.195	0.172	-0.255	0.078	0.123	79.524	425.685	0.871
3507	17	18	0.0050	33	24.638	0.486	0.327	0.128	0.119	63.580	14.607	0.873
3509	15	23	0.0020	30	1867.944	0.829	-0.075	0.098	0.058	98.440	40.560	0.797
3526	20	18	0.0500	20	426.058	0.246	-0.094	0.739	0.809	12.693	251.190	0.324
3641	23	110	0.0150	43	296.069	0.199	0.486	0.641	0.132	32.661	32.094	0.652
3774	126	71	0.0025	50	48.673	0.341	-1.495	0.040	0.073	130.672	54.753	1.500
3800	98	87	0.0300	15	90.274	0.583	2.908	0.115	0.129	65.166	240.703	1.697
3926	121	69	0.0030	25	5085.103	0.117	0.110	0.064	0.114	97.462	551.512	0.304
3928	26	37	0.0068	70	6797.726	0.121	-0.457	0.304	0.223	36.485	5780.185	0.151
3929	143	98	0.0800	50	1.015	0.173	-0.227	0.244	0.370	25.310	898.817	6.452
3960	264	414	0.0800	40	421.322	0.514	-1.207	0.205	0.133	72.494	137.702	0.881
4104	37	19	0.2000	20	117021.122	0.120	-0.022	1.641	1.997	9.269	4729.137	0.076
4366	150	141	0.0800	2	355.937	0.192	-0.251	0.218	0.264	25.415	1397.681	0.558
4407	155	276	0.0400	19	95.846	0.101	0.377	0.443	0.254	36.539	3760.994	1.226
4453	22	27	0.0200	30	518.674	0.565	1.106	0.315	0.254	30.347	1472.125	0.597
4477	11	28	0.0200	50	115191.862	0.085	0.282	1.057	0.614	19.938	167.491	0.060
4488	35	8	0.0200	1	41.715	1.911	0.666	0.072	0.366	50.079	144.770	3.078
4493	46	39	0.0500	40	54.970	0.530	-0.411	0.250	0.295	31.997	460.567	2.926
4657	239	310	0.0400	30	258.181	0.071	-0.146	0.468	0.373	32.837	140654.610	1.418
4699	21	79	0.0500	30	208580.609	0.023	0.087	2.109	2.047	8.539	8603.296	0.036
4986	69	11	0.1000	50	2324.414	0.176	-0.015	0.799	3.369	25.188	5782.765	0.288
5056	50	91	0.0500	33	625.297	0.293	-0.970	0.479	0.276	31.130	83025.549	0.680
5144	53	16	0.2000	20	5.892	1.548	9.940	0.227	0.748	23.099	353.818	8.205
5174	162	163	0.0300	50	157.010	0.691	-0.629	0.103	0.102	77.524	46171.708	1.435
5229	47	57	0.1500	10	81.887	2.586	0.089	0.215	0.177	42.989	14.240	2.498
5261	220	176	0.1000	30	34055.800	0.523	-1.279	0.172	0.219	39.867	16160.346	0.535
5357	99	105	0.0500	30	35766.636	0.230	0.404	0.324	0.302	28.786	7672.799	0.217
5390	220	74	0.0400	30	115.754	0.788	0.344	0.055	0.171	65.713	186.217	1.663
5497	79	40	0.0400	30	165.587	0.217	0.545	0.356	0.661	25.188	611515.682	0.571
6322	158	110	0.1000	66	6.964	0.108	-0.751	0.404	0.550	26.524	153.817	5.171
6927	80	54	0.0400	1	504.258	0.197	-0.082	0.273	0.390	31.341	1230.717	1.102
7115	63	106	0.1000	30	19.300	0.175	0.143	0.956	0.662	21.839	181616.578	4.497
7162	201	32	0.0100	50	40.077	0.954	-1.346	0.029	0.195	107.197	4115.904	2.651
7634	255	122	0.0200	2	32.215	1.329	0.904	0.041	0.088	122.384	1288.305	3.298
7878	7	9	0.0400	29	369.054	1.645	-0.563	0.512	0.403	20.312	0.961	2.073
8076	170	129	0.0600	10	217.682	0.390	-0.930	0.174	0.237	34.543	1564.803	0.988
8078	121	90	0.0400	36	1.702	1.212	2.706	0.082	0.112	78.848	28.935	7.361
8219	42	77	0.0040	22	25.900	2.915	0.021	0.038	0.019	254.447	58937.389	4.212
8396	159	200	0.0050	1	6.124	0.918	6.768	0.043	0.034	179.745	131.304	5.108
9120	301	177	0.1000	20	964.579	0.509	0.162	0.171	0.291	41.689	5828.850	0.748
9949	48	101	0.0500	20	1162.994	0.830	4.429	0.301	0.141	41.348	896.605	0.885
10200	78	77	0.0400	9	2518.354	0.725	-7.283	0.159	0.162	49.989	2733.573	0.813
10684	26	113	0.0400	57	171.999	0.361	-0.361	0.682	0.157	31.826	4.966	0.800
13333	68	67	0.0200	25	119834.456	0.054	-0.069	0.420	0.450	24.280	22323.637	0.048
14242	381	118	0.0300	20	199.159	0.239	-0.013	0.108	0.349	53.191	582.054	0.759
14420	250	43	0.0800	22	15.505	1.510	-3.244	0.064	0.383	58.055	0.794	4.093

Table 9 (continued)

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Median						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
18045	103	181	0.0400	30	109.446	0.573	-2.965	0.224	0.127	50.665	11.530	1.505
19018	294	295	0.0800	30	780.877	0.731	1.172	0.139	0.141	57.144	4758.865	0.947
19991	49	244	0.0400	20	230.794	0.341	-0.118	0.618	0.124	46.446	108.909	0.847
20270	86	65	0.0400	30	203.072	0.481	0.016	0.140	0.191	48.139	46.693	0.984
21669	173	146	0.0100	21	44.613	1.255	-0.983	0.036	0.043	171.563	400.378	2.684
22962	99	61	0.0400	24	2221.663	0.870	-0.874	0.105	0.174	59.898	2633.166	0.910
23249	58	145	0.0100	16	54189.843	0.034	-0.026	1.218	0.865	75.430	249344.779	0.080
23652	85	81	0.0100	20	2655.041	2.136	-0.202	0.037	0.039	187.183	11873.949	1.658
37077	33	199	0.0400	24	1406.717	0.859	0.652	0.470	0.075	50.772	581.433	0.904
38128	48	152	0.0800	42	534.812	0.131	-0.267	2.445	0.919	33.375	1624.612	0.630
38760	13	50	0.0500	33	3.781	1.761	-0.053	0.373	0.090	44.901	41507.819	5.596

Note: Each row is one contest. N_i and N_j are the focal contestants' submission counts in the constructed estimation window. The prize column reports the effective prize θ in units of 100,000 USD: the cash prize for a single-prize contest and the first-minus-second-place cash-prize gap for a multi-prize contest. "Public Data %" reproduces Meta-Kaggle's public-test share and is not used in estimation or sample selection; zero does not denote missing public-score data. Posterior columns are medians estimated from contest-phase public data; λ^{bench} and σ^{bench} are post-contest private-path diagnostic benchmarks, not estimation inputs. The former uses submission-triggered public–private discrepancies and elapsed intervals; the latter is the square root of the mean squared drift-adjusted private-gap increment per unit elapsed time. Very small public–private discrepancies can produce large λ^{bench} values. Scores are on the common logit scale with $a = 1$. The posterior columns report feedback precision λ, innovation risk σ, initial mean gap μ_0, effort costs c_i, c_j, and effort-to-submission scale r. Time is in days. Higher λ means more informative feedback; higher σ means greater innovation risk; lower c_k means higher ability. These medians do not display posterior uncertainty, and differences across scoring metrics need not measure differences in economic value.

C.2. Parameter Recovery under Nonlinear Effort

To assess the effort approximation, we generate data under nonlinear equilibrium policies and estimate parameters with the leading-order likelihood. Ten joint parameter configurations represent high-incentive contests in the original estimates, with four realizations per configuration. All six parameters are re-estimated using the empirical priors in Section 5. The two exploratory realizations used in prior development are excluded. Table 10 reports recovery of feedback precision and innovation risk.

For feedback precision, the 95% credible interval contains the generating value in 14 of the 15 realizations with point-estimation errors above 20%; only one of the 40 datasets has an interval that misses the generating value. For innovation risk, coverage is 33 of 40 datasets (82.5%), below the nominal 95% level. This flags possible undercoverage in the tested configurations; the small, heterogeneous simulation sample does not establish its magnitude more generally or isolate its cause.

The simulation and its likelihood use interval-start public feedback in the perceived-state recursion; the empirical application uses the interval-end public value after computing that interval's effort. Both likelihoods score fresh observations at interval endpoints. The recovery results therefore concern these ten configurations and the simulator-aligned feedback convention, rather than providing an error bound for all empirical estimates.

Table 10 Parameter Recovery from Nonlinear Simulated Contests

Parameter	Mean absolute relative error (%)	Coverage of 95% credible intervals (%)
Signal precision λ	17.0	97.5
Innovation risk σ	12.1	82.5

Note: Each row summarizes 40 simulated datasets at ten fixed parameter configurations. Point estimates are posterior medians. Absolute relative error divides the absolute estimation error by the generating value. Coverage is the fraction of datasets whose equal-tailed 95% posterior interval contains that value. The data are generated under nonlinear-equilibrium effort and estimated with the leading-order likelihood; each of the ten fixed configurations, chosen to represent high-incentive empirical contests, contributes four realizations. Thus, the entries are equally weighted summaries of these 40 datasets, not estimates from a random sample of all contests. Lower relative error means more accurate point recovery; coverage assesses interval calibration. Innovation-risk coverage of 82.5% is below the nominal 95%, and precision coverage of 97.5% does not establish generally calibrated inference.

C.3. Focal-Pair and Score-Scale Robustness

The alternative-pair exercise ranks contestants by their last recorded private score, using submission-ID order to resolve records on the same date and contestant ID to break score ties. It excludes the baseline contestant i and searches the remaining candidates within the top 20 in rank order. This ranking need not coincide with the final leaderboard based on contestants' best scores. Each contestant must have an activity span of at least ten days, the pair must overlap for more than ten days, and each must contribute at least five submissions to the retained count window. Prize conventions, the model horizon, and the within-day time-imputation rule are retained from the baseline. The initialization prefix is recomputed from the alternative pair's first jointly available scores, changing the count-window start in 43 of 73 contests. Thus, the exercise compares alternative pairs with corresponding usable data, rather than holding the effective observation window fixed.

Score-scale calibration uses submitted public scores from the baseline focal pairs, without private scores or repeated carried-forward values. We center their logit-transformed levels by each contest's median for calibration only, assign equal total weight to each contest, and compute pooled weighted quantiles q_j at $p_j = (j - \frac{1}{2})/999$, $j = 1, \ldots, 999$. The L_1 choice minimizes $\sum_j |aq_j - \Phi^{-1}(p_j)|$ over $a > 0$; the L_2 choice minimizes $\sum_j [aq_j - \Phi^{-1}(p_j)]^2$. They give $a = 1.444$ and $a = 0.781$, respectively. Public gaps are multiplied by the selected a for estimation, and private gaps by the same a for diagnostics; the calibration centering does not enter those gaps.

Across the 219 robustness fits (73 alternative pairs and 73 at each score scale), the maximum rank-normalized $\widehat{R}$ is 1.0095, and all bulk and tail effective sample sizes exceed 400. NUTS runs have no divergent transitions or tree-depth saturations. The two multimodal fits use the Metropolis–Hastings refinement described above.

For clarity, a pure change of score units by a factor $a > 0$ has the model-consistent parameter mapping

$$(\sigma, \lambda, \mu_0, S_0, c_i, c_j, r) \longmapsto (a\sigma, \lambda/a^2, a\mu_0, a^2 S_0, c_i/a^2, c_j/a^2, r/a).$$

It multiplies effort and score gaps by a, leaves submission intensities and incentive indices w_k unchanged, and requires transforming the priors as well to preserve the same Bayesian model. Retaining the same numerical

Table 11 Private-Leaderboard Benchmark Regressions under Robustness Specifications

Specification	Slope	Standard error	95% confidence interval	R^2
Signal precision λ				
Baseline ($a=1$)	0.386	0.112	[0.167, 0.606]	0.121
Alternative pair	0.189	0.120	[-0.046, 0.423]	0.026
L_1 scale	0.342	0.114	[0.118, 0.565]	0.102
L_2 scale	0.400	0.113	[0.178, 0.621]	0.126
Innovation risk σ				
Baseline ($a=1$)	0.912	0.173	[0.573, 1.250]	0.320
Alternative pair	1.246	0.168	[0.916, 1.575]	0.582
L_1 scale	0.821	0.176	[0.476, 1.165]	0.268
L_2 scale	0.953	0.179	[0.603, 1.303]	0.324

Note: Each regression uses 73 contests and an intercept. The dependent variable is $\log(1+\lambda^{\text{bench}})$ or $\log(1+\sigma^{\text{bench}})$; the regressor is the corresponding log-transformed posterior median estimate. Estimates and benchmarks use each specification's score units. The alternative-pair signal-precision regression has the largest slope p-value, 0.115. A positive slope means that the structural estimate and private-path benchmark move together across contests; it does not require equality of their levels. The precision benchmark uses elapsed-time-weighted public–private discrepancies, while the innovation benchmark uses drift-adjusted private-gap increments. Inference treats estimated posterior medians as given and does not propagate their uncertainty. Standard errors and confidence intervals use HC1 heteroskedasticity-robust covariance estimates. Baseline rows use the original focal pair and $a=1$; alternative-pair rows use the alternative rivalry, and the L_1 and L_2 rows use the corresponding score-scale calibrations.

Table 12 Score-Scale Sensitivity of Parameter Rankings

Specification	a	Rank correlation: λ	Rank correlation: σ
Baseline	1.000	—	—
L_1	1.444	0.992	0.945
L_2	0.781	0.999	0.996

Note: Each specification covers the same 73 contests. Correlations are Spearman correlations between common-unit posterior medians and baseline posterior medians. Re-estimation retains the baseline numerical priors in transformed units. The common-unit estimates are $a^2\hat{\lambda}_a$ for precision and $\hat{\sigma}_a/a$ for innovation risk. The scale a multiplies logit-transformed scores. L_1 and L_2 calibrations minimize absolute and squared quantile deviations from a standard normal reference, respectively. Correlations near one indicate stable contest rankings, not identical parameter levels, Gaussianity, or stable optimal designs. Dashes denote the baseline reference, whose self-correlation would be one.

priors, as in this exercise, is therefore a sensitivity analysis rather than a pure re-expression of identical inference in different units.

Because the baseline numerical priors and parameter bounds are retained in transformed units, the exercise also captures sensitivity to scale-dependent regularization. Table 11 reports the benchmark regressions. To compare estimates in common score units, convert signal precision to $a^2\hat{\lambda}_a$ and innovation risk to $\hat{\sigma}_a/a$. Table 12 reports their rank correlations with the baseline.

C.4. Additional Checks

Cost-prior sensitivity. We re-estimate four selected contests under two alternative settings: widening the prior standard deviation of the common log-cost component $(\log c_i + \log c_j)/2$ from $0.75/\sqrt{2}$ to 1, retaining the original relative log-cost Gaussian factor, or lowering the cost bound from 0.01 to 0.001, with other specifications unchanged. Quality comparisons use the nonlinear model. Table 13 reports whether a cost-reducing, quality-preserving design remains supported along the original joint adjustment direction.

Table 13 Direction of Joint Design Improvements under Alternative Cost Settings

Contest	δ	Case characteristic	Wider prior	Lower bound
8219	2	Very small cost saving	Supported	Supported
2464	100	Large prize increase	Supported	Supported
5144	1	Lower feedback precision	Supported	Supported
2551	0.1	Longer deadline and lower prize	Supported	Supported†

Note: Case characteristics describe the counterfactual designs selected under the baseline prior. Each row is one contest–delay-cost scenario, where $D(T) = \delta\theta^{obs}(T/T^{obs})^2$. "Supported" means that a design in the original joint adjustment direction has lower prize-plus-delay cost and passes the paired numerical diagnostic for preserving posterior mean expected quality under the alternative fit. It is not a posterior probability or a claim of global optimality. Wider prior increases the common log-cost standard deviation to 1; lower bound reduces the minimum effort-cost parameter to 0.001. The design selected under the baseline prior remains supported in seven comparisons. †For contest 2551 under the lower bound, the prize cut is scaled back from 56.05% to 34.28%; the supported design still lowers total cost by 19.80%. Support concerns the joint adjustment direction, not each variable separately or stability of the optimal design and savings. These selected cases are not a full-sample sensitivity test. Contest 8219's cost reduction is only 0.001204%.

Predictive comparison. We compare sequential, one-hour-ahead joint posterior predictive probabilities of the two contestants' submission counts in all 73 contests. Models are estimated on the training prefix and predict each test hour before updating with that hour's public readout. The primary split uses approximately 75% of the post-initialization window for training; the alternative aligns the split with complete source dates. Both evaluate hourly counts. Scores average log predictive probabilities within contests and then equally across contests. The unrestricted benchmark uses $m_k^U = \theta(c_k\sqrt{2\pi Q_0})^{-1}(Q_t/Q_0)^{-a}\exp[-b\tilde{y}_t^2/(2Q_t)]$, estimating $a \in \mathbb{R}$ and $b > 0$ instead of imposing the first-order equilibrium restrictions $a = 1/2$ and $b = 1$, with submission intensity rm_k^U and the resulting efforts entering the same filtering recursion. This extension retains the constant submission-share restriction.

Table 14 Out-of-Sample Submission-Count Prediction

Benchmark	Primary split	Date-aligned split
Homogeneous Poisson	−0.0604*	−0.0570*
Nonhomogeneous Poisson	+0.1146	+0.1179
Unrestricted filtered activity kernel	+0.0812*	+0.0836*

Note: Entries are equilibrium-model minus benchmark log predictive scores, in nats per contestant-pair hour; positive values favor the equilibrium model. All 73 contests receive equal weight after averaging each contest's one-hour-ahead log probabilities. The primary split trains on approximately the first 75% of the post-initialization window; the date-aligned split keeps source dates intact. Test-hour predictions use earlier public readouts, before observing the current hour. A nat uses the natural logarithm; these differences are not percentage changes in submission counts or solution quality. * indicates that a paired, contest-level 95% bootstrap interval excludes zero, using 500,000 resamples. The homogeneous model assigns a constant intensity to each contestant; the nonhomogeneous model allows intensity to vary with time remaining. The unrestricted kernel freely estimates the deadline and gap coefficients in the same filtering framework, with its effort rule also entering filtered drift. The equilibrium model uses the first-order empirical policy. Results condition on the selected pair, imputed within-date times, and constructed observation window.

Acknowledgments

The authors gratefully acknowledge the valuable research assistance provided by Simian Zhang during the course of this study.

References

Ales L, Cho SH, Körpeoğlu E (2017) Optimal award scheme in innovation tournaments. *Operations Research* 65(3):693–702, URL `http://dx.doi.org/10.1287/opre.2016.1575`.

Ales L, Cho SH, Körpeoğlu E (2021) Innovation tournaments with multiple contributors. *Production and Operations Management* 30(6):1772–1784, URL `http://dx.doi.org/10.1111/poms.13342`.

Aoyagi M (2010) Information feedback in a dynamic tournament. *Games and Economic Behavior* 70(2):242–260, URL `http://dx.doi.org/10.1016/j.geb.2010.01.013`.

Barrau A, Bonnabel S (2017) The invariant extended Kalman filter as a stable observer. *IEEE Transactions on Automatic Control* 62(4):1797–1812, ISSN 0018-9286, 1558-2523, URL `http://dx.doi.org/10.1109/TAC.2016.2594085`.

Bensoussan A (1992) *Stochastic Control of Partially Observable Systems* (Cambridge University Press), 1 edition, ISBN 9780511526503, URL `http://dx.doi.org/10.1017/CBO9780511526503`.

Berge C (1963) *Topological Spaces: Including a Treatment of Multi-valued Functions, Vector Spaces and Convexity* (Edinburgh and London: Oliver and Boyd).

Bergemann D, Heumann T, Morris S, Sorokin C, Winter E (2022) Optimal information disclosure in classic auctions. *American Economic Review: Insights* 4(3):371–388, ISSN 2640-205X, 2640-2068, URL `http://dx.doi.org/10.1257/aeri.20210504`.

Bimpikis K, Ehsani S, Mostagir M (2019) Designing dynamic contests. *Operations Research* 67(2):339–356, ISSN 0030-364X, 1526-5463, URL `http://dx.doi.org/10.1287/opre.2018.1823`.

Blum A, Hardt M (2015) The Ladder: A reliable leaderboard for machine learning competitions. *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, 1006–1014 (PMLR), URL https://proceedings.mlr.press/v37/blum15.html.

Budd C, Harris C, Vickers J (1993) A model of the evolution of duopoly: Does the asymmetry between firms tend to increase or decrease? *The Review of Economic Studies* 60(3):543–573, URL http://dx.doi.org/10.2307/2298124.

Frogerais P, Bellanger JJ, Senhadji L (2012) Various ways to compute the continuous-discrete extended Kalman filter. *IEEE Transactions on Automatic Control* 57(4):1000–1004, ISSN 0018-9286, 1558-2523, URL http://dx.doi.org/10.1109/TAC.2011.2168129.

Gronau QF, Singmann H, Wagenmakers EJ (2020) bridgesampling: An R package for estimating normalizing constants. *Journal of Statistical Software* 92(10), URL http://dx.doi.org/10.18637/jss.v092.i10.

Guerre E, Perrigne I, Vuong Q (2000) Optimal nonparametric estimation of first-price auctions. *Econometrica* 68(3):525–574, ISSN 0012-9682, 1468-0262, URL http://dx.doi.org/10.1111/1468-0262.00123.

Harris C, Vickers J (1987) Racing with uncertainty. *The Review of Economic Studies* 54(1):1–21, ISSN 00346527, 1467937X, URL http://www.jstor.org/stable/2297442.

Hoffman MD, Gelman A (2014) The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research* 15(47):1593–1623, URL http://jmlr.org/papers/v15/hoffman14a.html.

Huang PC, Zhang Z, Erat S (2024) Designing Data Science Contests: The Role of Training vs Test Splits. URL http://dx.doi.org/10.2139/ssrn.4976827.

Huang Y, He M (2021) Structural analysis of Tullock contests with an application to U.S. House of Representatives elections. *International Economic Review* 62(3):1011–1054, ISSN 1468-2354, URL http://dx.doi.org/10.1111/iere.12502.

Hwang SH (2012) Technology of military conflict, military spending, and war. *Journal of Public Economics* 96(1-2):226–236, ISSN 00472727, URL http://dx.doi.org/10.1016/j.jpubeco.2011.09.003.

Jackson J (2023) On quasilinear parabolic systems and FBSDEs of quadratic growth. arXiv preprint arXiv:2206.08334v2, URL https://arxiv.org/abs/2206.08334v2.

Jiang ZZ, Huang Y, Beil DR (2022) The role of feedback in dynamic crowdsourcing contests: A structural empirical analysis. *Management Science* 68(7):4858–4877, ISSN 0025-1909, 1526-5501, URL http://dx.doi.org/10.1287/mnsc.2021.4140.

Kang K (2016) Policy influence and private returns from lobbying in the energy sector. *The Review of Economic Studies* 83(1):269–305, ISSN 0034-6527, 1467-937X, URL http://dx.doi.org/10.1093/restud/rdv029.

Kato T (1980) *Perturbation Theory for Linear Operators* (Berlin: Springer-Verlag), 2 edition, corrected printing.

Khorasani S, Nittala L, Krishnan VV (2023) Screening in multistage contests. *Manufacturing & Service Operations Management* 25(6):2249–2267, ISSN 1523-4614, 1526-5498, URL http://dx.doi.org/10.1287/msom.2021.0378.

Lemus J, Marshall G (2021) Dynamic tournament design: Evidence from prediction contests. *Journal of Political Economy* 129(2):383–420, ISSN 0022-3808, URL http://dx.doi.org/10.1086/711762.

Lewis PAW, Shedler GS (1979) Simulation of nonhomogeneous Poisson processes by thinning. *Naval Research Logistics Quarterly* 26(3):403–413, ISSN 0028-1441, 1931-9193, URL http://dx.doi.org/10.1002/nav.3800260304.

Lorenzi L, Lunardi A, Metafune G, Pallara D (2005) Analytic semigroups and reaction-diffusion problems. URL https://people.dmi.unipr.it/alessandra.lunardi/LectureNotes/I-Sem2005.pdf, internet Seminar 2004–2005.

Mihm J, Schlapp J (2019) Sourcing innovation: On feedback in contests. *Management Science* 65(2):559–576, URL http://dx.doi.org/10.1287/mnsc.2017.2955.

Moldovanu B, Sela A (2001) The Optimal Allocation of Prizes in Contests. *American Economic Review* 91(3):542–558, ISSN 0002-8282, URL http://dx.doi.org/10.1257/aer.91.3.542.

Moscarini G, Smith L (2011) Optimal dynamic contests, URL https://campuspress.yale.edu/moscarini/files/2017/01/contests_current-x5l31k.pdf, working paper.

Nishida T (1978) *Nonlinear Hyperbolic Equations and Related Topics in Fluid Dynamics*. Number 78-02 in Publications Mathématiques d'Orsay (Orsay: Université de Paris-Sud, Département de Mathématique).

Risdal M, Bozsolik T (2022) Meta kaggle. URL http://dx.doi.org/10.34740/KAGGLE/DS/9.

Ryvkin D (2022) To fight or to give up? dynamic contests with a deadline. *Management Science* 68(11):8144–8165, ISSN 0025-1909, 1526-5501, URL http://dx.doi.org/10.1287/mnsc.2021.4206.

Shakhgildyan K (2025) Nonparametric identification and estimation of all-pay auction and contest models. *Review of Economic Design* 29(3):545–583, ISSN 1434-4750, URL http://dx.doi.org/10.1007/s10058-022-00309-3.

Skaperdas S (1996) Contest success functions. *Economic Theory* 7(2):283–290, ISSN 0938-2259, 1432-0479, URL http://dx.doi.org/10.1007/BF01213906.

Yoganarasimhan H (2016) Estimation of beauty contest auctions. *Marketing Science* 35(1):27–54, ISSN 0732-2399, URL http://dx.doi.org/10.1287/mksc.2015.0929.

Zhang S, Singh PV, Ghose A (2019) A structural analysis of the role of superstars in crowdsourcing contests. *Information Systems Research* 30(1):15–33, ISSN 1047-7047, URL http://dx.doi.org/10.1287/isre.2017.0767.